Freedom of Speech, Spirit of Innovation, and Long-term Economic Development: Evidence from the Qing Dynasty of China

Benlai Wang[1], Xuebo Wang[2]

February 2024

Abstract

Expression controls have been a common practice of thought control on citizens in authoritarian regimes. However, how such controls on free speech affect society in the short and long terms remains unclear. The 18th century witnessed severe punishment inflicted upon individuals in Qing China if their speeches were considered offensive to the regime ("the literary inquisition"). By investigating the impact of the literary inquisition on Qing China, we provide novel yet compelling explanations for two fundamental puzzles in Chinese history: (1) Although China was the global leader in science and technology (S&T) over a long time, why did this advantage not lead to an Industrial Revolution (the Needham puzzle)? (2) Why did China decline in the late Qing Dynasty? We first present suggestive evidence that the literary inquisition may have led to China's sharp decline in S&T after 1700. Further difference-in-differences estimates indicate that the literary inquisition reduced the number of scientific and technological innovations by at least 50%–58% in affected regions and negatively affected the long-term land tax revenue in these regions in the Qing Dynasty. These findings provide novel insights into Chinese history, particularly China's decline due to extreme expression controls in the Qing Dynasty.

Keywords: Freedom of speech, spirit of innovation, economic development, literary inquisition

JEL codes: N00, O00, H11

[1] School of Economics, Shanghai University of Finance and Economics.
[2] School of Economics, Shanghai University of Finance and Economics. E-mail: wang.xuebo@mail.shufe.edu.cn.

1. Introduction

In the civilized world, freedom of speech has been accepted as a universal value and a basic human right, protected by the constitutions in most countries. However, even in the modern world, citizens are not fully entitled to the right of free expression in some authoritarian regimes, where speech controls are common and unfavorable remarks against the government are strictly forbidden. How such restrictions on free expression affect society in the short and long terms, specifically, how they impact the spirit of innovation and economic development in the short and long terms, respectively, forms an important empirical issue. Scholars have established that thought controls, such as the Counter-Reformation in European history, may have long-lasting adverse effects on society (e.g., Becker et al., 2021; Blasutto and de la Croix, 2023; Drelichman et al., 2021). However, direct empirical evidence on the effect of expression controls is limited. This study attempts to provide a rigorous empirical analysis on the comprehensive effects of expression controls on society.

The literary inquisition in China's Qing Dynasty (1644–1911) provides an ideal natural experiment to empirically study the aforementioned issue. As defined by Fu (1993), literary inquisition is a "legal punishment for criminal acts committed through speech and written words expressed in various forms, including conversations, letters, essays, poems, pamphlets, books, dramas, novels, and diaries." Literary inquisition had existed in imperial China for a long time, and the earliest case can be traced back to the Zhou Dynasty around 1000 BC. In the earlier period, literary inquisition was usually used by emperors or bigwigs as a tool to eliminate their political opponents; thus, typical literary inquisition cases of the time generally reflected political struggles among political elites and had little to do with commoners (Yang, 1999). By contrast, the literary inquisition in the Qing Dynasty became the rulers' crucial instrument of thought control on citizens to consolidate the Qing regime, and most victims were commoners. From Kangxi to Qianlong reigns of the Qing Dynasty (1711–1788), the emperors implemented the notorious literary inquisition that significantly changed the society of the time in many ways. In this period, if someone's speech was interpreted as offensive to the Qing regime, the offender would be executed by dismembering the body, and the entire family or clan would also be sentenced to death or exiled.

Such extreme controls on expression created a terrifying atmosphere among citizens in the Qing Dynasty and profoundly affected society, particularly in the development of science and technology (S&T) at the time. This period saw intellectuals adopting cautious approaches to communication and writing, as exemplified by Shizheng Liang, an intellectual in the Qianlong reign, who advised, "Never communicate with others through written words, and always promptly destroy any scratch paper with words" (Jin, 2012). This pervasive fear likely hindered effective communication among intellectuals,

reducing their willingness to engage in writing and publishing works on all subjects, including S&T. Given that books were the primary medium for disseminating and sharing ideas on S&T at the time, the literary inquisition could have substantially hampered the progress of S&T.

By investigating the impact of the literary inquisition on Qing China, we attempt to contribute to the explanations for two fundamental puzzles in Chinese history: (1) Although China was the global leader in S&T over a long time, why did this advantage not lead to an Industrial Revolution (the Needham puzzle)? (2) Why did China decline in the late Qing Dynasty?

China had made remarkable achievements in S&T in the premodern period, and it had a considerable lead over the Western world in most of the major areas of S&T over a long period (Lin, 1995). However, China was overtaken by the West in S&T and fell behind of the world in many ways later. Needham first comes up with the following challenging yet fundamental question based on his research on the history of S&T in China: Although China had been more advanced than other civilizations in S&T over a long time, why did not the Industrial Revolution originate in China? This question was later known as the "Needham puzzle" and has been widely discussed but a consensus has yet to be reached.

The literature has noted that the development of modern science, particularly the surge of scientific and technological innovations, served as a pivotal catalyst for the Industrial Revolution (Mokyr, 2002; 2009; McCloskey, 2010). Therefore, the key to explaining the Needham puzzle is to explain China's decline in S&T at the time.

This study argues that the literary inquisition may have stifled the spirit of innovation of the Qing Dynasty society and led to China's sharp decline in S&T. We will show that the time of the implementation of the literary inquisition was also the turning point in China's decline in S&T. The case of Mingshi Dai in 1711 was the first literary inquisition case in the Kangxi period, marking the turning point in the Qing rulers' policy on free expression. Before this case, Emperor Kangxi was tolerant of intellectuals' expression, and free speech was generally allowed. After this case, citizens received strong messages that "inappropriate" speech in any form could spell disaster for the entire family. Mingshi Dai was a famous intellectual at the time. In 1711, his published book titled *nan shan ji* was reported to Emperor Kangxi as containing offensive expressions to the Qing regime. Thereafter, the book was investigated, and Emperor Kangxi was infuriated by a letter included in the book that Dai wrote to a friend. Dai was found guilty and was later executed; over 100 people involved were punished. This sensational literary inquisition case had a huge impact on society, particularly on intellectuals, who were shocked to learn that free speech had become a golden memory. Later, the literary inquisition became the emperors' common instrument of thought control on citizens.

We plot several figures to demonstrate the potential impact of the literary inquisition on innovation in the Qing Dynasty. Figure 1 shows the number of scientific and technological innovations in China and Britain in 1670–1790, in which each point represents the total number of innovations in the country in the past two decades.[1] The dividing line indicates the year of 1711, when Emperor Kangxi initiated the investigation of the first literary inquisition case. Evidently, China had a clear advantage over Britain in innovation before 1711. However, China experienced a dramatic decline in innovation after 1711, whereas Britain achieved considerable growth in innovation. Unsurprisingly, Britain exceeded China in innovation in around 1750–1770, when the Industrial Revolution started in the former. Thereafter, China fell behind Britain in many ways. The implementation of the literary inquisition in China in 1711 appears to be the turning point in the country's decline in S&T, which may not be simply a coincidence.

Figure 2 plots the number of scientific and technological innovations in China and Europe in 1670–1790, which conveys similar information. Innovations in China showed an increasing trend before 1711, and China still dominated Europe in innovation that year. However, China experienced a sharp decline after that, while Europe enjoyed an explosive growth in innovation. Consequently, China fell considerably behind Europe in innovation after 1750. Again, 1711 emerges as the turning point in China's decline and Europe's rise in S&T.

In Figure 3, we compare the number of scientific and technological innovations in the treatment prefectures (those exposed to the literary inquisition) and the control prefectures (those unaffected by it) within China from 1670 to 1790. Given that the literary inquisition cases were primarily concentrated in developed regions with a higher concentration of intellectuals, treatment prefectures generally had a clear advantage over control prefectures in S&T before the literary inquisition. Figure 3 shows that the treatment prefectures had an overwhelming advantage over the control prefectures in the number of innovations before 1711. However, innovation in the treatment prefectures plummeted after 1711, reaching a level similar to that of the control prefectures. Such a significant decline in innovation in the treatment prefectures was probably caused by a certain huge negative shock on innovation in these regions. This study attempts to empirically prove that the literary inquisition was exactly such a shock on innovation that led to China's sharp decline in S&T.

Figures 1–3 present a big picture of the potential impact of the literary inquisition on innovation in the Qing Dynasty of China, based on which we perform our empirical analysis. With a difference-in-differences (DID) strategy, by exploiting the variation of the literary inquisition across regions and

[1] Section 3 discusses the data on innovation in China and Europe in history. The Appendix provides more information on the corresponding data.

over time, we find that the literary inquisition reduced the number of scientific and technological innovations by at least 50%–58% in the treatment prefectures compared with the control ones. Furthermore, it had a negative effect on the long-term land tax revenue, an important indicator of economic development at the time. We also address recent econometric concerns about the reliability of the staggered DID strategy and present a variety of alternative estimators to confirm the robustness of our results.

A natural concern arises that the occurrence of the literary inquisition cases could be correlated with regional characteristics, presenting a typical case of endogenous issue. Figure 3 also indicates that treatment prefectures were generally much more developed than control prefectures. Given that there were more intellectuals in developed regions who were the primary targets of the literary inquisition, these regions were thus more susceptible to this policy. However, Xue (2021) points out that the location and timing of literary inquisition cases were also shaped by idiosyncratic factors. In particular, Chinese languages are subtle and ambiguous, and thus any expression can be interpreted in different ways. Furthermore, there were no general rules in the legal system concerning the criteria of offensive expressions, and the officials' personal discretion mattered to a large extent. Lastly, it was the emperor that made the final sentence of all investigated literary inquisition cases, and such decisions could be extremely subjective and arbitrary. Unsurprisingly, two very similar cases could end up with completely different outcomes. Therefore, these facts generate idiosyncratic variations in the occurrence of the literary inquisition cases across regions and over time.

The DID strategy exactly exploits such idiosyncratic variations in the literary inquisition, thereby facilitating the identification of its effect on innovation. We find that innovation in treatment and control prefectures share a common trend prior to the literary inquisition. However, this common trend disappeared after the literary inquisition, with the number of innovations in the treatment prefectures decreasing sharply compared to that in the control prefectures. Such results present strong evidence of the treatment effect of the literary inquisition on innovation. Furthermore, we adopt the propensity score matching approach to construct a sample in which the treatment and control prefectures were similar prior to the treatment. Reassuringly, our results remain robust for this matched sample.

Our findings indicate that the literary inquisition precipitated China's decline in S&T and, subsequently, its economy. Naturally, an essential question arises: Why could the literary inquisition have such destructive effects on the Qing Dynasty society? We can uncover the potential answers by contrasting the Enlightenment in Europe and the literary inquisition in China in the 1700s.

The Age of Enlightenment, which blossomed in Europe in the 18th century, paved the way for the Industrial Revolution (Mokyr, 2009; Jones, 2017). The Enlightenment, promoted by philosophes who challenged the authority of institutions deeply rooted in society and aimed to reform society with

tolerance, science, and skepticism, was most known for its political achievements. These accomplishments include advocating for such political ideals as freedom, equality, basic human rights, religious tolerance, and democracy (Bristow, 2023). Following the Enlightenment, Europe gradually fostered an institutional environment tolerant of heterodoxy and deviancy, catalyzing the development of a free market for ideas where creative people were highly rewarded for their innovative activities (Mokyr, 2005; 2007). Ultimately, this tolerant institutional environment and idea market led to the technological and institutional changes crucial for the Industrial Revolution.

While the Enlightenment's core ideology was widely accepted in Europe in the 1700s, the Qing government in China was enforcing extreme expression controls to consolidate its authoritarian regime. Under the oppressive climate of the literary inquisition, the formation and development of a tolerant institutional environment and a free market for ideas were impossible, and citizens refrained from engaging in innovative yet potentially risky activities. Thus, the significant difference in the impact of the Enlightenment and the literary inquisition on society may partially account for Europe's rise and China's decline after 1700. The literature has noted that civil liberties and dignities invigorate society, and freedom fosters economic prosperity (McCloskey, 2010; Phelps et al., 2013). Conversely, extreme speech controls can sap a society's vigor and vitality, undermining its capacity for future progress.

This study presents a novel yet compelling explanation for the Needham puzzle. Previous studies have provided several hypotheses to explain the decline of S&T in imperial China. (1) Political Bureaucracy Hypothesis: China's imperial system, particularly the civil service exam system and the criteria of promotion in the bureaucratic system, hindered human capital investment necessary for modern scientific research and led to China's decline in science (Lin, 1995; Cantoni and Yuchtman, 2013; Brandt et al., 2014). (2) Culture Hypothesis: China's traditional Confucian cultural values suppressed the spirit of free inquiry, hindered the formation and development of an open and competitive market for ideas, and thus impeded the development of modern science (Weber, 1915; Landes, 2006; Mokyr, 2009; 2017). (3) Institution Hypothesis: The lack of property rights protection locked imperial China in an agricultural economy and hindered scientific progress (Huang, 1997; Edwards, 2005; Zhang and Gao, 2005; Landes, 2006).

Although the preceding explanations provide important insights into the aforementioned issue, they have evident limitations. Given that China had been the global leader in S&T for an extended period and those claimed adverse factors had always existed in imperial China, why they only became obstacles to innovation after 1700 is not easy to explain. As discussed above, China's sharp decline in S&T after 1700 was more likely due to substantial negative shock on innovation at the time rather than those ever-existing factors in imperial China. Thus, our proposition that extreme expression controls after 1711 led to China's dramatic decline in S&T is more plausible and consistent with the Qing

Dynasty's S&T development trend. Our explanation for the Needham puzzle enriches the literature, providing novel insights into Chinese history, particularly China's decline due to extreme expression controls in the Qing Dynasty.

Our findings also contribute to the growing literature on the comprehensive societal impacts of thought controls in authoritarian regimes. Research has demonstrated that the Counter-Reformation in Europe, following the 15th century and notably in Italy and Spain, adversely impacted knowledge diffusion, human and social capital, and economic development in the affected regions (Becker et al., 2021; Blasutto and de la Croix, 2023; Drelichman et al., 2021). This study will delineate the differences between the Qing literary inquisition and the European Counter-Reformation, highlighting why the former had more severe consequences. Additionally, a recent study by Chen and Yang (2019) suggests that media censorship in modern China could be effective among citizens with a low demand for uncensored information. However, the social costs of such censorship practice remain unclear.

Our study builds upon and diverges from the pioneering research on the literary inquisition in the Qing Dynasty of China. Koyama and Xue (2015) and Xue (2021) have documented its substantial adverse effects on human capital accumulation and social capital, including charity and trust, in affected regions, with some effects persisting into modern China. Our study diverges by focusing on the impacts of the literary inquisition on S&T and economic development in the Qing Dynasty. More importantly, our examination of the literary inquisition's effects provides novel and compelling explanations for the Needham puzzle. Furthermore, our study casts light on present-day China, where the tightening of expression controls over the past decade has culminated in widespread public discontent, epitomized by the White Paper Revolution in November 2022. This historical moment, marked by heightened tensions between state-imposed speech restrictions and the populace's demand for free expression, draws global attention to China's future path. Thus, our findings have broad implications, shedding light on the intricate interplays among civil liberties, social vitality, and economic prosperity in the modern world.

2. Historical Background

2.1 The Literary Inquisition in Qing China

This section briefly introduces the historical background of the literary inquisition from the Kangxi to Qianlong reigns (1661–1796).

The Ming Dynasty collapsed in 1644, and the Manchus invaded Beijing thereafter and established the Qing Dynasty. The Ming-Qing transition was marked by significant unrest, including widespread massacres and heightened tensions between the Manchu conquerors and the Han population

(Wakeman, 1985, Xue, 2021). Such tensions persisted for a long time and became eased only after Emperor Kangxi began to govern the country in 1667.

The investigation of the case of History of the Ming Dynasty in 1661 signified that the tension between the Qing rulers and the anti-Qing Han population reached a peak. In this year, Emperor Shunzhi died, and his son Kangxi inherited the throne at the age of 8. At that time, Kangxi was still too young to govern the country, and four ministers assisted (or substituted) him to make decisions. Tinglong Zhuang, a Han intellectual organized people to compile a book on the history of the Ming Dynasty and published it in 1660. This publication was reported to the central government in 1661 as offensive to the Qing regime. The four ministers in charge of the government perceived this as a serious case and conducted a thorough investigation. The book was interpreted as cherishing and praising the Ming Dynasty and showing immense disrespect toward the Qing regime. As a result, in 1663, more than 70 people associated with the book were executed, and over a thousand were punished (Yan, 2016).

This case was sensational at the time and further intensified the tensions between the Qing rulers and Han intellectuals. It also reflected the Manchus' insecurity of the legitimacy of their regime. While some intellectuals view this as the first literary inquisition case in the Qing Dynasty, it did not signify the onset of tightened expression controls. In fact, no additional cases were investigated in the following 50 years. When Emperor Kangxi fully assumed power in 1667 at 14, he adopted completely different policies toward the Han population, generally allowing free expression.

Historians generally view the case of History of the Ming Dynasty not as the beginning of the literary inquisition in the Qing Dynasty but as part of the broader Qing campaign against anti-Qing activities (Yang, 1999). Compiling a book on the history of the Ming Dynasty during the delicate period of the Ming–Qing transition could easily be interpreted by the Qing rulers as not supporting the new regime. The book even labeled the Manchu armies as robbers and the Qing rulers as foreign enemies (Yu, 1996), adding to its controversial nature. In fact, many people involved in the case had previously engaged in activities of rebelling the Qing and rebuilding the Ming Dynasty. Thus, they deserved the punishments to some extent. Hence, some historians posit that the case was an early Qing effort to suppress potential Han opponents, rather than a literary inquisition per se (Yu, 1996).

Moreover, this incident occurred during a particular period when the four ministers were in power and made the decision. Emperor Kangxi's subsequent reign ended such harsh practices, including the campaign against anti-Qing forces. He implemented conciliation policies aimed at winning the support of the Han population by treating them well and was notably tolerant of intellectuals' free expression (Guo and Lin, 1990; Yang, 1992). Such enlightened policies were successful and are credited with initiating the prosperous "Kangxi–Qianlong Great Ages" (Yan, 2016).

Therefore, the case of History of the Ming Dynasty appears more as an isolated incident rather than the start of a systematic literary inquisition. Its societal impact, if any, was relatively short-lived. Notably, this case is absent in authoritative historical records of the Qing literary inquisition, such as *Qingdai wenziyu dang ji* (Qing Literary Inquisition Archives) (Zhang, 1934) and its updated edition (Wanyan, 2011). In both books, the first recorded literary inquisition case was the case of Mingshi Dai. Thus, these records suggest that the Qing rulers' systematic use of the literary inquisition as a tool to consolidate the authoritarian regime began with the investigation of the Mingshi Dai case in 1711.

When Emperor Kangxi took over the power in 1667, the country was devastated by wars, the economy was in a slump, and people lived in destitution. Many of the Han Chinese, which accounted for 95% of the total population then, were still resistant to the Manchu rulers, posing another potential threat to the Qing regime. To promote the economy and consolidate the regime, Emperor Kangxi adopted the conciliation policy toward the Han population to win their support. In particular, he conducted special examinations *(bo xue hong ru*) to select the Han elites to work for the government. The selected Han intellectuals were generally assigned to important positions. Emperor Kangxi visited famous Han intellectuals to persuade them to serve the Qing regime. Even though several refused to meet him, Emperor Kangxi was not angered and still informed them that they were welcome whenever they were willing to work for the government. His attitude towards the Han, particularly the intellectuals, was appreciated by the Han population and won their support for the Qing regime.

Emperor Kangxi was also tolerant of intellectuals' free expression. In an examination selecting the Han elites, one examinee expressed the idea that Manchus were barbarians, which clearly violated taboos of the Qing regime. However, Emperor Kangxi thought that the statement was not a big deal and let it pass (Yan, 2016). He also introduced a rule that previously published books containing words that violated the taboos of the Qing regime should not be banned and could be republished in their original versions (Yan, 2016). Emperor Kangxi also organized intellectuals to compile a book on the history of the Ming Dynasty, which officially announced that writing books on the Ming Dynasty became legal and virtually overturned the conviction of the previous case of History of the Ming Dynasty. In 1707, an intellectual named Fuxiang Zhang wrote a poem to satirize Emperor Kangxi's extravagant inspection tour to the Chiangnan, where local officials tried their best to please the emperor by spending huge amounts of money to build palaces. Such expressions showed great disrespect to Emperor Kangxi but he did not investigate the case (Yan, 2016).

Emperor Kangxi's conciliation policy significantly eased the tension between the Qing rulers and the Han population and won the support of the majority of the Han Chinese. Moreover, his tolerance for free expression and other enlightened policies injected vitality into society and inspired people to contribute to society with initiatives (Guo and Lin, 1990). Historians argue that Emperor Kangxi's

conciliation policies toward intellectuals were critical in stabilizing society and also boosting the economy, and such policies were the key factors that initiated the "Ages of Prosperity" or the "Kangxi–Qianlong Great Ages" in the Qing Dynasty (Yan, 2016).

The turning point in Emperor Kangxi's policy toward free expression was the case of Mingshi Dai (or the case of *nan shan ji*), which was investigated in 1711. Mingshi Dai was a famous intellectual at the time. In 1711, a government official reported to Emperor Kangxi that a book titled *nan shan ji* written by Mingshi Dai, contained contents that were wildly arrogant and clearly offensive to the Qing regime. Emperor Kangxi examined the articles in the book and was infuriated by one letter that Dai wrote to a friend (Yan, 2016). In the letter, Dai praised a book written by the scholar Xiaobiao Fang, who used to serve the rebel army of Sangui Wu, whom Emperor Kangxi hated considerably. The emperor then assumed that Dai belonged to the same group of people (e.g., Fang) opposed to the Qing regime. Dai also commented in the letter that the team that Kangxi organized to compile the book on the history of the Ming Dynasty was terrible and hopeless. Thus, he had to write a decent treatise on the previous dynasty by himself, further infuriating Emperor Kangxi.

Dai was found guilty and executed in 1713; the descendants of Xiaobiao Fang were also punished severely. The case was sensational at the time and had a huge impact on society, particularly on intellectuals, who received a strong message that personal expressions, even those in private letters, may be investigated by the government and result in severe punishments.

Then a natural question emerges: Given that Kangxi was tolerant of citizens' free expression before 1711, why did he suddenly become sensitive to intellectuals' speech and initiate the literary inquisition in 1711? Some historians argue that the case of Mingshi Dai may only be an accident. For a long time, Emperor Kangxi's sons had been fighting fiercely for the right to inherit the throne, and the crown prince was deprived of the right of succession owing to his evil behavior for a second time in 1711. Emperor Kangxi was extremely angry and sad then, and the scandals about internal struggles within the royal court appealed to the public, possibly further causing political instability. Historians speculate that Kangxi intended to make a sensational case of the literary inquisition to divert the public's attention from the scandals of the royal court (Guo and Lin, 1990). Furthermore, China had reached its zenith of prosperity up to 1711, and there were no external and internal threats to the Qing regime at the time. As founder of this great age of prosperity, Kangxi was also at the peak of his power and authority, and he did not have to treat intellectuals well to win their support anymore and could do whatever he wanted. Therefore, it is not surprising that Emperor Kangxi decided to punish Dai severely when he was infuriated by the latter's offensive speech, considering that the emperor was in a bad mood at the time.

While the case of Mingshi Dai might initially be viewed as an isolated incident, the Qing Dynasty's approach to the literary inquisition underwent significant evolution during the reigns of Emperors Yongzheng and Qianlong. These successors of Emperor Kangxi held him in high regard and were influenced by his legacy. They continued and intensified the literary inquisition. Consequently, the literary inquisition became the Qing rulers' common instrument of thought control on citizens later.

Given that Chinese languages are subtle and ambiguous, any expressions could be interpreted as offensive to the regime (Xue, 2021). As Wang (2002) describes,

"Rash fortune-telling and discussion of military strategy could be offenses, as could poetic works with 'excessive anger' or 'excessive hate,' or even expressions of 'sorrow' regarding specific episodes in history. It was a crime to call oneself a non-collaborator, an expression used to refer to adherents of the former dynasty living under a new one without serving it. Use of taboo words and phrases, or even nonsensical expressions like 'a dog's wild bark' were offenses… Careless use of such words as 'Han,' 'Great Enterprise,' 'Ch'ing (Qing),' 'sun and moon' (the components of the character for 'Ming'), 'barbarian,' 'Ming,' and similar words also could be punishable."

Xue (2021) summarizes the features of the literary inquisition and concludes that it was extremely arbitrary and unreasonable. It was impossible to anticipate what speech or writing may result in a literary inquisition. It was the emperor who made the final decisions on the literary inquisition, and such decisions could be substantially arbitrary and subjective. For instance, two cases that are similar in many ways may end up with entirely different punishments. Furthermore, the punishments were given in public, sending strong messages to citizens that people who dared to oppose the regime would be punished severely.

Xue (2021) also examines all book titles related to the literary inquisition and finds that most books belong to poetry and literary commentary genres, which have nothing to do with politics. Her findings are consistent with historians' views that the culprits of the literary inquisition were not clearly treasonous or even critical of the regime. They may only be the victims of the Qing rulers' arbitrary and unreasonable policy of thought control on citizens.

Although the literary inquisition may appear irrational or unreasonable, using arbitrary punishments signals state power, which could deter future opposition (Xue, 2021). Actually, the literary inquisition was implemented during the Kangxi–Qianlong Great Ages when China was prosperous, and the Qing regime was stable without visible external or internal threats. Moreover, its emperors were among the most powerful in Chinese history. Therefore, the goal of the literary inquisition appears not to purge opponents of the regime, and it may only be a way of building the absolute authority of the emperor and nurturing the consciousness of unconditional obedience to the Qing regime among citizens.

2.2 The Literary Inquisition and Scientific and Technological Innovation in Qing China

Given that the literary inquisition aimed to punish citizens' offensive speech against the Qing regime, people may wonder why innovative activities in politically neutral fields such as physics and mathematics could be affected. Indeed, the literary inquisition was not designed to restrict scientific innovation directly, and few were penalized for their scientific and technological investigations. However, the pervasive and arbitrary nature of the inquisition could exert a substantial deterrent effect on innovative activities in these fields.

As previously discussed, the subtleness and potential ambiguities in the Chinese language meant that any expression could be interpreted as offensive to the Qing regime. Furthermore, given the arbitrary and capricious nature of the literary inquisition, it was impossible for individuals to predict what expressions or writings might lead to punishment. This uncertainty meant that the content of any original work, be it pertained to literature or S&T, risked being interpreted as subversive, placing the authors in jeopardy. Consequently, the literary inquisition fostered a climate of fear among the populace at the time. For instance, during the peak of the literary inquisition under the Qianlong reign, an intellectual named Shizheng Liang advised, "Never communicate with others through written words, and always promptly destroy any scratch paper with words" (Jin, 2012), reflecting the pervasive fear of the literary inquisition. This atmosphere likely hindered effective communication among intellectuals, diminishing their willingness to write and publish works on all subjects, including S&T. Considering that books were the primary medium for disseminating and sharing ideas in S&T at the time, the literary inquisition could have significantly hampered progress in these areas.

Section 5.7 further explores the mechanisms through which the literary inquisition impacted scientific and technological innovation in Qing China. Our analysis indicates that, in the wake of the literary inquisition, intellectuals may have shifted their focus to non-innovative but safer activities, contributing to a stagnation in S&T during this period.

2.3 Literary Inquisition and Censorship

Although literary inquisition appears similar to censorship, the two concepts are different in many ways. Censorship is defined as "the practice of officially examining books, films, etc. and suppressing the unacceptable parts" (Pearsall et al., 2007). This practice is common in authoritarian regimes and aims to prevent the spread of regime-threatening information among citizens. By keeping citizens in ignorance, censorship is considered as the key to the popular support and stability of authoritarian regimes (Ford, 1935; Chen and Yang, 2019). By contrast, instead of pre-examining books or other forms of expression and eliminating unfavorable parts in advance, the literary inquisition re-examines published books and punishes authors severely if their expressions were interpreted as offensive to the

regime. Such a policy aims to deter potential opposition by sending strong messages to citizens that people who dare to oppose to the regime would face severe punishment. In brief, censorship makes citizens supportive of the regime by keeping them in ignorance via information control, while the literary inquisition forces citizens to be obedient to the regime by threatening to kill them otherwise.

Both the literary inquisition and censorship are instruments of thought control on citizens for rulers to consolidate the authoritarian regimes. Nevertheless, the literary inquisition is more unreasonable, making citizens feel insecure and dare not to express their ideas or engage in any potentially risky activities. Such extreme controls on expression created a terrifying atmosphere among citizens in the Qing Dynasty and had profound effects on society.

2.4 The Qing Literary Inquisition and the European Counter-Reformation

People may naturally compare the Qing China's literary inquisition with the European Counter-Reformation and regard them as similar thought control policies. However, they diverged significantly in several aspects and had distinct impacts on China and Europe. While the literary inquisition may have led to China's steep decline in S&T and further its ultimate decline in economy, the Counter-Reformation exerted a relatively limited effect on European S&T and economic development. This section delves into these differences in detail.

The European Counter-Reformation, primarily a response by the Catholic Church to the Protestant Reformation during the 16th and 17th centuries, commenced with the Council of Trent (1545–1563) and largely ended with the conclusion of the European wars of religion in 1648 (Cabello, 2023). The Roman Inquisition, established in 1542, primarily targeted Protestantism, which was defined as heresy in Catholic territories (Dewitte et al., 2023). Operating through local tribunals led by professional inquisitors, the Inquisition enforced adherence to Catholic doctrine by punishing heresy deviating from it. Moreover, the Catholic Church also implemented book censorship to control the production and diffusion of books with heretic ideas (Dewitte et al., 2023).

Three key differences distinguish the Counter-Reformation from the literary inquisition:

1. Severity of Punishments: The punishments during the Counter-Reformation, including admonishments, imprisonment, property confiscation, forced labor, exile, public abjuration, shaming, and flogging (Dewitte et al., 2023), were generally much less severe than those of the literary inquisition. Capital punishment was rare, and accused individuals often had opportunities to defend themselves or could even be exempted from punishment upon renouncing heretical beliefs (Dewitte et al., 2023).
2. Uncertainty of Offenses: The Counter-Reformation, with its clear focus on Protestantism, offered citizens greater certainty regarding punishable offenses. Unlike the literary inquisition, which

created a pervasive atmosphere of fear, the Counter-Reformation did not instill a similar level of terror.

3. Censorship Strategies: The Catholic Church's book censorship, involving publishing lists of forbidden books and inspecting customs and storehouses, mainly targeted bookmen with prohibited volumes, who faced fines and book burning. Authors were not typically punished (Becker, Pino, and Vidal-Robert, 2021). Given that the authors faced little risk of punishment, they could participate in scientific and technological pursuits freely. The flourishing of the Scientific Revolution during this period suggests a limited impact of the censorship on European S&T development.

Furthermore, the divergent state systems in Europe and China may have also contributed to the differing impacts of these policies. Specifically, given the polycentric state system in Europe, there were still many regions out of the reach of the Catholic Inquisition during the Counter-Reformation. Thus, individuals with creative yet heretical ideas could relocate to those regions where such ideas were tolerated and their talents highly rewarded. As shown in the literature, Europe's polycentric state system fostered intellectual pluralism and a competitive market for ideas, instrumental in the Scientific and Industrial Revolutions (Jones, 2003; Mokyr, 2017; Scheidel, 2019; Fernández-Villaverde et al., 2023). In contrast, China's politically unicentric system meant that the literary inquisition overshadowed the entire country, leaving no refuge for talents. For instance, in 1522, Niccolo Buccella, a physician and surgeon from Padua, fled to Romania and then Poland to avoid arrest by the Inquisition. He later became the King's doctor and surgeon, thriving in his new position (Dewitte et al., 2023). Such options were unavailable in Qing China, amplifying the literary inquisition's destructive societal impact.

While the Counter-Reformation had significant adverse effects in regions like Italy and Spain, where the Catholic Church wielded substantial power and the Inquisition was implemented intensely (Becker et al., 2021; Blasutto and de la Croix, 2023; Drelichman et al., 2021, Dewitte et al., 2023; Cabello, 2023), its overall impact on Europe was relatively limited. In particular, the Scientific Revolution blossomed in Europe during the Counter-Reformation, and the Enlightenment ensued, eventually leading to the Industrial Revolution in the 1700s.

In conclusion, the Qing literary inquisition could have led to China's sharp decline in S&T and its ultimate decline in economy, whereas the European Counter-Reformation was a comparatively minor obstacle in the continent's ascent in S&T and economic progress.

3. Discussions: When did China Begin to Decline in S&T?

While Figures 1–3 suggest that China may have begun declining in S&T after 1711, this evidence is far from conclusive. In fact, the literature has not reached a consensus regarding the exact timing of China's S&T decline. From a broader perspective, diverse scholarly views exist concerning the timeframe of China's decline and Europe's rise in S&T. This section seeks to analyze these controversies and unify various perspectives within the literature.

Needham (1981) posits that China maintained its lead over Europe in S&T from the 2nd to the 15th centuries, with Europe overtaking China during its 16th-century Scientific Revolution. Mokyr (1992) notes that China's technological progress slowed down after 1400 and eventually came to a full stop. He observes that "in 1600, their (China's) technological backwardness was evident to most European visitors." However, Needham and Mokyr base their arguments on anecdotal evidence, leaving room for further investigations. A recent project by Huang (2023) traces China's decline in S&T back to the Sui Dynasty (581–619), further intensifying the controversy.

Figures 1–3 are cursory, and it is insufficient for us to draw a conclusion relying on them alone. We thus extend the time horizon of Figure 2 to examine whether a unique time in history signifies a divergence in S&T between China and Europe. Similar to Figure 2, Figure 4 presents the number of scientific and technological innovations in China and Europe during 100–1900. It shows a consistent decline in China's innovations after 600, aligning with Huang's findings. However, this period does not signify the divergence of China's decline and Europe's rise in S&T. Notably, China experienced an innovation surge after 1000,[2] while Europe's innovation levels remained relatively low. Furthermore, consistent with Needham and Mokyr's arguments, China's innovation level was relatively low around 1500. However, China's innovations began to increase after 1500 and paralleled Europe's progress in the following two centuries, indicating that 1500 was not the point of divergence either.

The only discernable divergence point is 1700, characterized by a significant upsurge in Europe's innovations and a steep decline in China's. Despite some recovery in China's innovations after 1800 following the termination of the literary inquisition, China had fallen irretrievably behind Europe in S&T and had no chance to catch up again. Therefore, if a divergence point in S&T exists between China and Europe during 100–1900, it is likely around 1700.

[2] Sheng Bi, an intellectual in the Song Dynasty, invented the movable type printing in 1041, which may have contributed to China's subsequent significant increase in innovation. China later experienced a sharp decrease in innovation after 1300. Historians argue that the feudal autocracy reached a peak when the Ming Dynasty was established in the 1300s, which may have contributed to China's decline in S&T in the following two centuries (Du and Jin, 2003).

People may be concerned whether the data sources for Figure 1–4 are credible or reliable enough to draw a conclusion on this issue. Specifically, people may wonder whether and to what extent the data on scientific and technological innovations in China and Europe are comparable, and whether the definitions of innovations are consistent across the board.

We sourced our data on S&T innovation from Ai and Song (2006) for China and Ito et al. (1984) for Europe. Both books document significant events of scientific and technological activities in their respective histories. We identify the events that reflect important advancements in S&T as innovations. These innovations generally include notable advancements in various disciplines, such as mathematics, physics, printing, agriculture, and medicine.

We compare the scientific and technological event records in the two books and find that they are similar in many ways. Both books chronologically record significant scientific and technological advancements in their respective histories. A typical record includes the essential information on the corresponding innovation, such as the time when the innovation was made, the author of the innovation, the title of the innovation, and a comprehensive description of the innovation itself. Moreover, the recorded innovations in the two books fall into a wide range of categories according to the fields that they belong to. Scientific innovations fall into the following subjects: physics, biology, medicine, mathematics, chemistry, astronomy, geography, and geology. Technological innovations cover the following fields: mechanical engineering, chemical engineering, civil engineering, instrument manufacturing, ironmaking technology, materials engineering, military technology, and agricultural technology. Our detailed comparison of these records reveals no significant differences in the definitions of innovations across the two regions. See more discussions on the data in the Appendix.

Although proving conclusively that the definitions of innovations in the two books are identical is difficult, discrepancies in definitions do not necessarily pose a problem for our analysis. Our primary goal is to identify the timing of the S&T divergence between China and Europe over a historical timeline, rather than comparing the absolute levels of S&T in the two regions at any specific time. Given that the definitions of innovation in both regions are derived from the same book, they are likely to remain consistent over time within each region. Thus, Figure 4 could still reflect the development trends of S&T in Chinese and European history. In other words, China's steep decrease and Europe's significant increase in innovations after 1700 presented in Figure 4 are unlikely driven by the disparate definitions of innovation in the two regions. In summary, provided constant definitions of innovation in both regions over time, Figure 4 remains informative regarding the timing of the S&T divergence between China and Europe.

Moreover, our focus on the quantity of S&T innovations, without considering their quality, may raise concerns about the reliability of our conclusions. If the quality of European S&T innovations

declined substantially after 1700, while that of Chinese innovations improved significantly, then the inferred divergence around 1700 would be questionable. While direct measurement and comparison of S&T innovation quality over time are difficult, the available evidence suggests that the quality of European S&T innovations did not deteriorate after 1700, nor did the quality of Chinese S&T innovations notably improve.

In Europe, the 18th century has witnessed numerous significant scientific and technological breakthroughs. These include James Watt's improvement of the steam engine in 1765, Hargreaves' invention of the Spinning Jenny in 1767, the first application of the steam engine in spinning machinery in 1785, and Lavoisier's introduction of the earliest table of elements in 1789 (Ito et al., 1984). These innovations, particularly the improved steam engine and developments in spinning technology, were instrumental in triggering the Industrial Revolution (Pomeranz, 2000). These key S&T breakthroughs indicate no significant decline in the quality of European innovations during this period.

In China, evidence suggests that the quality of S&T innovations did not significantly improve in the 18th century and may have even declined considerably. Jin (1998) documents the biographies of 77 distinguished scientists from 400 BC to 1900 AD, pivotal in the progression of S&T in Chinese history. The records show a steep decrease in the number of notable scientists in the 18th century, dropping from nine in the 17th century to two in the 18th century. This reduction in the number of distinguished scientists implies a decline in innovations with significant contributions, suggesting that the quality of Chinese S&T innovations in the 18th century likely did not escalate significantly and may have diminished.

Thus, even considering the quality of S&T innovations, the conclusion that China and Europe began diverging in S&T development during the 18th century remains substantiated. This divergence is not solely based on the quantity of innovations but is also supported by the evidence of relative quality within each region.

In conclusion, our findings bridge differing viewpoints in the literature on the Great Divergence, suggesting that the pivotal period for China's decline and Europe's rise in S&T is the 18th century. This conclusion itself represents a significant contribution to the literature.

4. Data

Data on the literary inquisition are from *Qingdai wenziyu dang (zengding ben)* (Qing Literary Inquisition Archives (Updated Edition)) (Wanyan, 2011). The book is a revised edition of *Qingdai wenziyu dang ji* (Qing Literary Inquisition Archives) (Zhang, 1934), which was produced by historians with the Qing Imperial Archives in the 1930s. Compared with the earlier edition, the updated one

published in 2011 includes 25 more literary inquisition cases previously unrecorded. The book records 90 cases of literary inquisition from 1711 to 1788. We exclude a case where the potential culprits were not found and obtain a final sample of 89 cases.

Given that the Qing government implemented the literary inquisition and emperors made the final decisions on all the investigated cases, they should have been meticulously documented in official archives. Thus, *Qingdai wenziyu dang (zengding ben)* stands as the most authoritative records of the Qing literary inquisition. As Zhu (2013) notes, the book compiles its records from primary historical materials sourced directly from the Qing Archives, rendering it the most reliable data source for studying this aspect of Qing history. It provides detailed accounts of each literary inquisition case, including the cause and course of the investigation, the emperor's response, the verdict, and information about the accused. This comprehensive documentation makes *Qingdai wenziyu dang* (*zengding ben*) an indispensable and authoritative historical record of the Qing literary inquisition. In recent studies on this subject, most scholars have relied on this book as their primary data source (e.g., Hu, 2013; Sun and Song, 2018).

Xue (2021) uses *Qingdai wenziyu an* (Qing Literary Inquisition Cases) (Zhang and Du, 1991) as her data source on the literary inquisition, extracting a sample of 86 cases. Our data source has several advantages, and our sample size is also slightly larger than Xue's. Additionally, our data source is more frequently used in Chinese literature. According to the CNKI (China National Knowledge Infrastructure), since its publication in 2011, *Qingdai wenziyu dang (zengding ben)* has been cited 34 times, whereas *Qingdai wenziyu an* has only been cited three times.[3] Reassuringly, the difference between the two data sets is relatively small, and our results are not sensitive to the choice of data source. Our results remain robust whether we conduct our empirical analysis using Xue's sample, our primary source, or a merged sample from both data sets. See more discussions and results in the Appendix.

Following Xue (2021), we examine the effect of the literary inquisition in the culprits' home prefectures, where their identities were lodged and their families and clans resided. In a typical case of literary inquisition, the culprits were executed, and their entire families and even clans were punished severely. Therefore, information on the literary inquisition cases spread rapidly in the culprits' hometowns, and the cases could have acted as a deterrent to residents in these regions. We define the treatment as a prefecture's first exposure to the literary inquisition case: when the hometown of the culprits of the literary inquisition cases turned out to be a prefecture for the first time.

[3] https://kns.cnki.net/kns8s/AdvSearch?classid=YSTT4HG0

Figures 5A and 5B present the time and spatial distributions of the literary inquisition cases in the 18th century. Most literary inquisition cases were investigated after 1750 in the Qianlong reign. Thus, most treatment prefectures were exposed to the literary inquisition in this period. Figures 1–3 show that the number of scientific and technological innovations in China declined significantly after 1711, when the first literary inquisition case was investigated. However, this case concerned only several prefectures at the time. Then a natural question arises: Given that most prefectures were not directly exposed to the first literary inquisition case, why did the number of innovations of the entire country decline sharply after this case?

It is not difficult to explain such facts. As previously discussed, the case of Mingshi Dai was sensational across China at the time, and citizens, particularly intellectuals, received strong messages that free speech was no longer allowed. Consequently, intellectuals throughout the country may no longer be willing to participate in any creative but potentially risky activities. Therefore, the effects of the literary inquisition were not confined to the directly implicated prefectures but also spilled over to other regions that acquired information on the cases.[4] In brief, although the initial literary inquisition only affected several prefectures directly, it still signifies a crucial turning point in China's decline in scientific and technological advancements.

We empirically estimate the effects of the literary inquisition on scientific and technological innovations and long-term economic development in affected regions. Data on innovation are from the *Zhongguo kexue jishu shi: nianbiao juan* (History of Science and Technology in China: Chronological Volumes) (Ai and Song, 2006). This book documents important events of scientific and technological activities in Chinese history, including notable innovations or advancements in various disciplines, such as mathematics, physics, printing, agriculture, and medicine. The record of each innovation contains extensive information, including the author, time, and site where the innovation was made.[5]

We focus on innovation in the Qing Dynasty of China in the 18th century (1700–1800), during which the literary inquisition was implemented and China began to decline in science and technology, and in the middle of which the Industrial Revolution began in Britain.[6] There are 102 recorded scientific and technological innovations in China in this period, among which 93 were made by intellectuals (over 91%), 4 were initiated by the central government, and 5 were introduced by Western missionaries. Undoubtedly, intellectuals were the main contributors of innovation at the time. Unsurprisingly, extreme expression controls on intellectuals could have considerable negative effect on innovation in the Qing Dynasty of China. Given that innovations initiated by the government or

[4] Section 5.6 empirically investigates the potential spillover effects of the literary inquisition on innovation.
[5] We exclude a small percentage of records (approximately 6%) in which the time or location of the innovation is unclear.
[6] We can extend the examined period to 1660–1800 or even longer to 1660–1820, and the estimation results remain similar.

imported from foreign countries were unlikely affected by the literary inquisition, we only include innovations made by intellectuals in our sample, which were likely affected directly by extreme speech controls.[7]

To examine the impact of the literary inquisition on long-term economic development, we employ land tax revenue as a measure of economic performance at the prefectural level. As Wang (1973) points out, the land tax was the most important source of public revenue in the Qing Dynasty. For instance, it accounted for 73.5% of the total tax revenues of the Qing government in 1753. Data on land tax revenue at the prefectural level are sourced from the General Records (*tong zhi*) of various provinces at different times. The General Records provide a record of important events in the province, including land tax revenue across all prefectures. We read through the General Records of all provinces in the Qing Dynasty to collect information on the land tax revenue of all prefectures. Eventually, we construct a decadal panel data set of land tax at the prefectural level for 1680–1850. See the Appendix for additional details on the data.

Our empirical analysis also includes a set of geographic and economic characteristics as controls. In particular, geographic characteristics include the agricultural suitability of different crops and two dummy variables: coast (whether a prefecture is situated on the coast), major river (whether a prefecture includes at least one major river). We also control for the language polarization index, which reflects the degree of cultural integration in one region. Economic and human capital characteristics (prior to the literary inquisition) include the population density in 1630, which reflects the level of economic prosperity at the time; and the number of *jinshi* or "presented scholars" (candidates who passed the national level civil service exam and were selected for high-level government positions) in 1600–1710, which reflects the human capital level prior to the literary inquisition. When we investigate the effect of the literary inquisition on land tax revenue, we also control for additional geographic characteristics, such as land area and fragmentation of basins (basin HHI), which may directly affect agricultural production. Furthermore, given that the tax reform *Tandingrumu* (transitioning from a poll tax to a property tax) during 1723–1728 may also affect land tax revenue, we also control for its potential impact in the regressions.[8] Table 1 presents the data sources and summary statistics for the main variables.

[7] See the Appendix for additional details on innovation data.
[8] See more discussions about the *Tandingrumu* policy in the Appendix.

5. The Effect of the Literary Inquisition on Innovation in Qing China

This section estimates the effect of the literary inquisition on scientific and technological innovation (at the prefectural level) in the Qing Dynasty of China.

5.1 Empirical Strategy

Figure 3 shows that the number of scientific and technological innovations in the control prefectures is relatively stable and remains at a low level over time. However, the number decreases significantly in the treatment prefectures after 1711 (i.e., when the case of Mingshi Dai was investigated), which presents suggestive evidence of the treatment effect of the literary inquisition on innovation. We now provide a rigorous empirical analysis of the effect of the literary inquisition on innovation in affected regions at the time.

We employ a staggered DID strategy to identify the effect of the literary inquisition on innovation. Specifically, we construct a prefectural-level panel in the 1700–1800 period with a five-year time interval between observations, with each observation representing the number of innovations during each five years in a prefecture. The treatment is a prefecture's first exposure to the literary inquisition case.[9] In practice, we estimate the following two-way fixed-effect (TWFE) model:

$$\#Innovation_{pt} = \beta Literary\ Inquisition_{pt} + Treat_p \times After1790_t + \theta X_p \times \gamma_t + \lambda_p + \gamma_t + \delta_{prov} \times \gamma_t + \epsilon_{pt}, \quad (1)$$

where $\#Innovation_{pt}$ is the number of scientific and technological innovations in prefecture p in period t (during the past 5 years); $Literary\ Inquisition_{pt}$ is a dummy indicating whether prefecture p was exposed to the literary inquisition in period t; $Treat_p$ is a dummy indicating whether prefecture p was ever treated, $After1790_t$ is a dummy that equals 1 when observations were after 1790, X_p includes a set of prefectural characteristics, γ_t and λ_p represent time and prefecture fixed effects, respectively; δ_{prov} represents a province-fixed effects; $\delta_{prov} \times \gamma_t$ is a provincial-specific trend, and ϵ_{pt} is an error term. We estimate Equation (1) using ordinary least squares (OLS) and cluster standard errors at the prefectural level.

To improve the precision of our identification, we incorporate an interaction term combining the treatment dummy with the post-1790 dummy into our regression models. This decision stems from the historical context that the last literary inquisition case was investigated in 1788, leading to the cessation

[9] We define that a prefecture was exposed to the literary inquisition in a period if its exposure to the event occurs during the period. Given that in our sample the time intervals between observations are 5 years and a prefecture was exposed to the literary inquisition in a specific year, such a definition may not be completely precise. For instance, if a prefecture was exposed to the literary inquisition in the last year of a period, then it was affected by the event for only less than 1 year during this period. Reassuringly, if we conduct our analysis at the year level to avoid such issues we obtain similar results. See more discussions and results in the Appendix.

of this practice and the halt of further investigations. The termination of the literary inquisition potentially revitalized the innovative spirit within society, possibly triggering a resurgence of innovations within the treatment prefectures. This rebound is illustrated in Figure 4, which depicts a noticeable recovery in the number of innovations across China after 1800, subsequent to the end of the literary inquisition. By introducing the interaction term, we aim to account for the potential rebound in innovation within the treatment prefectures during the decade following the literary inquisition's termination (1790–1800). Our findings remain robust whether this specific control is included or excluded from the regressions. Nevertheless, to accurately estimate the immediate impact of the literary inquisition on innovation and to elucidate the sharp decline in innovation across China in the 1700s during its enforcement, it is imperative to incorporate this control into our regression models.

The staggered DID strategy carries the following underlying assumptions. (1) In the absence of the literary inquisition, the number of scientific and technological innovations in the treatment and control prefectures would have experienced parallel trends. (2) Prefectural-level average treatment effects are homogenous across treated prefectures and over time. If all the assumptions hold, then the coefficient of interest β in Equation (1) identifies the average treatment effect on the treated (ATT) of the literary inquisition on innovation.

The first assumption that the innovation levels in the treatment and control prefectures share a common trend if the literary inquisition were not implemented is likely to hold. As shown in Figure 3, there were considerably more innovations in the treatment prefectures than that in the control prefectures before 1711. However, the gap in innovation between the two groups is relatively stable in the pretreatment period, providing suggestive evidence that the common trend assumption probably holds. In our empirical investigations, we estimate a fully dynamic version of Equation (1) to directly test the common trend assumption. In addition, we further confirm the robustness of our results by obtaining the alternative estimators introduced by De Chaisemartin and d'Haultfoeuille (2020); Borusyak, Jaravel, and Spiess (2021); Callaway and Sant'Anna (2021); and Sun and Abraham (2021).

Recent econometrics literature has shown that the TWFE models for staggered treatment cases can only deliver consistent estimates under strong assumptions that the treatment effects are homogenous across treated groups and over time (Sun and Abraham, 2021; Braghieri et al., 2022). However, if the treatment effects are heterogenous across groups or over time, then we cannot obtain consistent estimates for the ATT from the TWFE model. Thus, we further replicate our results using the robust estimators proposed by De Chaisemartin and d'Haultfoeuille (2020); Borusyak, Jaravel, and Spiess (2021); Callaway and Sant'Anna (2021); and Sun and Abraham (2021), which deliver consistent estimates even if the treatment effects are heterogeneous across groups or over time.

5.2 Baseline Results

We first estimate an event-study version of the TWFE model (Equation (1)) to test the common trend assumption directly and examine the dynamics of the treatment effects of the literary inquisition on innovation. In practice, we estimate the following specification:

$$\#Innovation_{pt} = \sum_{\tau=-(30+),\tau\neq-5}^{20+} \beta_\tau literary\ inquisition_{p\tau} + Treat_p \times After1790_t + \theta X_p \times \gamma_t + \lambda_p + \gamma_t + \delta_{prov} \times \gamma_t + \epsilon_{pt}. \quad (2)$$

Equation (2) is similar to Equation (1), except that the variable $literary\ inquisition_{p\tau}$ is a set of indicator variables that equals 1 if it has been τ years since a prefecture's first exposure to the literary inquisition, where $\tau \in \{-(30+), -30, -25, -20, -15, -10, 0, 5, 10, 15, 20, 25+\}$. We define τ=0 as the period when a prefecture was first exposed to the literary inquisition, and we treat the groups five years before a prefecture's first exposure to the literary inquisition as the reference groups.

Figure 6A demonstrates the event-study figure. Evidently, the coefficients are relatively small and never significant prior to the literary inquisition, and they decrease considerably after the treatment and become significantly negative soon. These results indicate that the treatment and control prefectures may share a common trend in innovation before the treatment. Such a common trend disappears following the literary inquisition, with the treatment prefectures experiencing a sharp decline in innovation. Such results suggest that the literary inquisition may have had a considerable negative effect on innovation in the treatment prefectures.

Given that the fully dynamic version of the TWFE model (i.e., Equation (2)) produces consistent estimates only when the treatment effects are homogeneous across groups and over time (Sun and Abraham 2021; Braghieri et al., 2022), we further present the event-study figures generated by a set of recently developed robust estimators even in the presence of treatment heterogeneity (De Chaisemartin and d'Haultfoeuille, 2020; Borusyak, Jaravel, and Spiess, 2021; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). Figure 6B demonstrates the obtained results. For all robust estimators, the coefficients on the groups prior to the literary inquisition are close to zero and exhibit no discernable pretrends. Furthermore, all coefficients decrease significantly after the treatment and become significantly negative soon. In summary, both the OLS and alternative robust estimators indicate that the common trend assumption holds, and the literary inquisition may have a significant negative effect on innovation in affected regions.

We next estimate Equation (1) and report the estimates of β in Table 2. Columns (1) to (4) of Table 2 show that as we gradually add control variables in the regression, the DID estimators change little and are always significant, indicating that our estimates are largely robust. After controlling for all relevant variables, Column (4) presents that the coefficient is approximately -0.045, which indicates that the literary inquisition reduces the number of innovations within a five-year span in the treatment prefectures by 0.045. Given that the mean of the number of innovations for all prefectures in 1700–1800 is approximately 0.019, such an estimate indicates a substantial negative effect of the literary inquisition on innovation in the treatment prefectures. Specifically, the mean of the number of innovations in the treatment prefectures after treatment is approximately 0.045. This estimate indicates that the corresponding value would be 0.090 if these prefectures were not affected by the literary inquisition. That is, the literary inquisition reduced the number of innovations in the treatment prefectures from the potential value of 0.090 to the actual value of 0.045, indicating a reduction of approximately 50%. Such results indicate that the literary inquisition had a detrimental effect on innovation in the affected regions, aligning with the information presented in Figure 3.

One possible concern is that the literary inquisition may not necessarily reduce the number of innovations. Instead, the recorded innovations merely declined owing to underreporting following the inquisition. Indeed, the underreporting could present a potential pathway through which the literary inquisition affected scientific and technological activities. Specifically, following the inquisition, some intellectuals may have withheld their innovations from the public, leading to a decrease in recorded innovations. Nevertheless, this situation does not pose a problem for interpreting our results. The literature has demonstrated that knowledge diffusion is critical for S&T advancement (Ma, 2021). Therefore, if some innovations were kept secret and not utilized practically, then they could not stimulate scientific and technological progress or catalyze an industrial revolution. In essence, whether the decline in recorded innovations after the literary inquisition was due to a reduction in innovations or a decrease in the public release of innovations does not affect the interpretation of our results.

The Appendix presents four alternative DID estimators that are robust to treatment effect heterogeneity (De Chaisemartin and d'Haultfoeuille, 2020; Borusyak, Jaravel, and Spiess, 2021; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). All the estimates are similar to those presented in Table 2, further confirming the robustness of our results.

5.3 Validity of Our Identification Strategy

A major concern arises from our sample comprising only 93 recorded S&T innovations. Given that we construct a prefectural level panel spanning 1700–1800 with a five-year interval between observations, and that we have more than 200 prefectures in any given interval, most observations must be reporting

zero innovation. Consequently, our results could be disproportionately influenced by events in one or two prefectures or by factors unrelated to the literary inquisition.

For instance, mathematician Zhen Dai (1723–1777) significantly contributed to S&T development in his hometown Huizhou prefecture during his lifetime. Dai died in 1777, and Huizhou was exposed to the literary inquisition in 1779. Therefore, it might have been Dai's advancing age and subsequent death, rather than the literary inquisition, that accounted for the decline in innovation in Huizhou after 1779.[10] In an extreme scenario, if innovation in Huizhou dramatically declined after 1779, while other treatment prefectures exhibited no treatment effects, a significantly negative DID estimate could still be obtained. However, such a result could not be interpreted as valid evidence of the effect of the literary inquisition on innovation in affected regions.

Therefore, we must confirm that our DID estimates are not driven by the impacts of accidental events unrelated to the literary inquisition in a few prefectures. We adopt two strategies to address these concerns.

First, we directly examine whether our DID estimates are driven by any specific prefecture. Following Gross and Sampat (2023), we re-estimate Equation (1), omitting individual prefectures, and illustrate all the obtained DID estimates in Figure 7. Figure 7 shows that all the estimates are significant and close to the baseline estimate (-0.045) in Column 4 of Table 2. Such results confirm that the result is not driven by any single prefecture.

Furthermore, we employ the causal forest method to examine the heterogeneity of treatment effect across prefectures. The causal forest method, a recently developed supervised machine learning technique, is used for predicting heterogeneity in causal treatment effects (Athey and Imbens, 2016; Wager and Athey, 2018; Athey et al., 2019; Britto et al., 2022). This method allows us to estimate the Conditional Average Treatment Effect (CATE) for each prefecture. The CATE is defined as $E(Y_{1i} - Y_{0i}|X_i = x)$, where Y_{1i} and Y_{0i} denote the potential outcomes of interest for the *i*th prefecture when treated and untreated, and X is a vector of observable characteristics (Britto et al., 2022).

We estimate the CATE for each prefecture using all the characteristics controlled in our baseline regressions in Table 2.[11] Figure 8 presents the distribution of the predicted CATE. Consistent with our expectation, the predicted CATEs for all prefectures are negative, with no extreme negative values observed. Such results indicate that our significant negative DID estimates are not driven by outliers. These estimates suggest that innovations in treatment prefectures decrease significantly following their

[10] There are also cases similar to that of Zhen Dai in the control prefectures that were never exposed to the literary inquisition. Theoretically, if such cases are uniformly distributed among the treatment and control prefectures, it would not bias our DID estimates. However, given the limited number of recorded innovations in our sample, we should carefully treat this issue and provide reliable solutions.

[11] See the Appendix for the details of the estimation procedure.

exposure to the literary inquisition, thereby providing strong evidence of the literary inquisition's negative impacts on innovation.

People may still wonder how we could estimate the prefecture level trends and obtain significant DID estimates in cases where many prefectures consistently reported zero innovation. A thorough examination of the data structure reveals a notable pattern: most control prefectures, which were never exposed to the literary inquisition, indeed recorded zero innovation throughout the period. By contrast, many treatment prefectures, those exposed to the literary inquisition, showed a decline in innovation, often from one to zero after their exposure to the event. This observation aligns with the trend depicted in Figure 3. While the innovation levels in control prefectures remained consistently low, treatment prefectures demonstrated a marked decrease in innovation following their exposure to the literary inquisition. Such a differential pattern between control and treatment prefectures is crucial for the validity of our DID estimates and supports our hypothesis about the negative impact of the literary inquisition on innovation.

In summary, despite certain data limitations, our results demonstrate robustness for various specifications. All the results indicate that innovations in treatment prefectures decreased significantly after their exposure to the literary inquisition, providing compelling evidence of its detrimental effect on innovation in affected regions. These findings are highly consistent with the big picture presented in Figures 1–4, suggesting that the literary inquisition may have stifled the spirit of innovation, leading to the Qing Dynasty's sharp decline in S&T.

5.4 Investigating the Impacts of Other Historical Events

Apart from the literary inquisition, other policies or historical events in the 18th century in Qing China may have also negatively impacted scientific and technological innovation. We have accommodated such effects from other potential shocks in our DID strategy, provided that these shocks were not systematically correlated with the incidences of the literary inquisition across regions and over time. We now examine the impacts of other concurrent historical events on innovation and determine whether our findings remain consistent after accounting for these potential confounders.

Such historical events include the expulsion of the Jesuits in the 1720s, the Qing government's implementation of the sea ban policies in the 1700s, the tax reform known as *Tandingrumu* (transitioning from a poll tax to a property tax) during 1723–1728 in the Yongzheng reign, and changes to the civil service examination system. These changes include the assignment of prefectural level quotas for successful candidates during the Yongzheng reign and two revisions to the examination content in 1756 and 1783. In the Appendix, we demonstrate that our estimation results remain robust after controlling for the potential impacts of these events on innovation.

Such results are highly predictable. We acknowledge that other events could have also affected innovation in the Qing Dynasty. However, these events are unlikely systematically correlated with the incidences of the literary inquisition across regions and over time. Theoretically, if whenever a prefecture was first exposed to the literary inquisition, another adverse shock on innovation occurred simultaneously, then our DID estimates of the effects of the literary inquisition on innovation in the treatment prefectures would be confounded by the effects of such concurrent shocks. However, given the large variation in exposure to the literary inquisition across regions and over time, which is also exogenous to some extent, it is highly unlikely that a prefecture would simultaneously experience another adverse shock on innovation each time it was exposed to the literary inquisition. Therefore, if the common trend assumption holds, we do not need to worry too much that other potential confounders could contaminate our DID estimates.

5.5 Further Robustness Check: A Propensity Score Matching Approach

Our analysis so far demonstrates that the literary inquisition appears to be a plausible exogenous shock on innovation in treatment prefectures. These prefectures experienced a significant decrease in the number of innovations after exposure to the literary inquisition, thereby providing strong evidence of the treatment effect. As discussed earlier, the emperor made the final decisions for the literary inquisition cases and such decisions could be arbitrary and subjective. Therefore, the occurrence of literary inquisition cases could be exogenous to some extent. However, given that the literary inquisition cases mainly occurred in developed regions with a higher concentration of intellectuals, who were the primary targets of the literary inquisition, an inherent difference exists between the treatment and control prefectures. Such disparities may evolve over time, leading to potential concerns that the differences in innovation between these two groups might not solely be driven by the literary inquisition.

Given that the treatment and control prefectures differ in many ways, it is difficult to completely exclude the possibilities that other omitted time-varying factors may affect the two groups differently and lead to significantly negative DID estimators. For instance, if a substantial gap in innovation initially existed between the treatment and control prefectures and they began to converge after the literary inquisition, then we would also obtain negative DID estimators. However, these estimators should not be misinterpreted as evidence of the destructive effects of the literary inquisition on innovation in the treatment prefectures.

To alleviate the aforementioned concerns, we further employ matching to reduce heterogeneity in the sample. After matching, systematic differences between treatment and control prefectures prior to the treatment no longer exist, thereby making the DID estimators more likely to capture the

treatment effect of the literary inquisition on innovation. In this scenario, the literary inquisition serves as a plausible natural experiment for identifying the effects of expression controls on innovation. Specifically, if all prefectures in the matched sample were initially similar and some were randomly selected for exposure to the literary inquisition, and if the number of innovations in these treatment prefectures decreases sharply compared with that in the control prefectures after the treatment, then a reasonable conclusion would be that the literary inquisition has a non-negligible negative effect on innovation in the treatment prefectures.

We match the treatment prefectures with the control ones on a range of covariates using propensity score matching.[12] We obtain a balanced sample with no observable differences between the treatment and control prefectures after matching. We now use this matched sample, in which the treatment and control prefectures are similar prior to the treatment, to estimate the effects of the literary inquisition on innovation in the treatment prefectures.

We first perform the event-study analysis and estimate Equation (2) using the OLS with this matched sample. The obtained results are presented in Figure 9A, which visualizes the dynamic effects of the literary inquisition on innovation in the treatment prefectures. Figure 9A shows that prior to the literary inquisition, all the coefficients are close to 0 and insignificant. By contrast, after the literary inquisition, the coefficients sharply decrease and become significantly negative soon. Given the similarity between treatment and control prefectures prior to the literary inquisition, this sharp decrease in the coefficients after the literary inquisition presents compelling evidence of the treatment effect of the literary inquisition on innovation in the treatment prefectures.

Similar to Figure 9B, Figure 9B presents the event-study figures generated by a set of robust estimators in the presence of treatment heterogeneity. As shown in Figure 9B, the coefficients for groups prior to the literary inquisition are close to zero and exhibit no discernable pre-trends. Moreover, all coefficients significantly decrease and remain at the low level after the treatment. Therefore, both the OLS and alternative robust estimators indicate that the common trend assumption holds, and the literary inquisition may have a significant negative effect on innovation in the affected regions.

We next estimate the TWFE model (Equation (1)) using the OLS with the matched sample. Table 3 presents the estimates of β in Equation (1), which generally resemble those obtained from the full sample shown in Table 2, though they are somewhat larger. Columns (1)–(4) of Table 3 show that, with the control variables being gradually added in the regressions, the DID estimators do not change substantially and are always significant at the 5% level. When all control variables are included, the DID estimator is as large as -0.063 (Column (4) of Table 3), which is larger than the corresponding

[12] See the Appendix for the details of the matching procedure.

estimate (−0.045) obtained from the full sample presented in Column (4) of Table 2. Given that the mean of the number of innovations in the treatment prefectures after treatment is approximately 0.046, the corresponding value would be 0.109 had these prefectures not been exposed to the literary inquisition. That is, the literary inquisition reduced the number of innovations in the treatment prefectures from a potential value of 0.109 to an actual value of 0.046, signifying a 57.8% reduction. This estimate is considerably larger than that obtained from the full sample. Considering that the estimates obtained from the matched sample are more likely to capture the treatment effect of the literary inquisition on innovation, such results present compelling evidence that the literary inquisition may have had a substantial negative effect on innovation in the treatment prefectures.

Our results are robust to a large extent, given that the DID estimators obtained from both the full and matched samples are highly consistent. In summary, the literary inquisition could have stifled the spirit of innovation of society, particularly among intellectuals who were the primary drivers of scientific and technological advancements at the time. Consequently, the number of innovations sharply declined in the treatment prefectures following their exposure to the literary inquisition.

5.6 The Spillover Effect of the Literary Inquisition on Innovation

In the preceding empirical analysis, we assume that the literary inquisition only affected innovation in the treatment prefectures, thereby allowing the use of the untreated prefectures as controls. The obtained DID estimators are supposed to capture the treatment effect of the literary inquisition on innovation in those treatment prefectures. However, in reality, all prefectures, including the control ones, could have been affected by the literary inquisition. Note that the Qing rulers implemented the literary inquisition as a deterrent, and it sent strong messages to citizens that people who dared to oppose to the regime would be punished severely. Therefore, the information on the literary inquisition cases could spread throughout the country, and thus a literary inquisition case could have a deterrent effect on innovation at the national level. That is, the literary inquisition may have had a substantial negative impact on innovation in the treatment prefectures and also a spillover effect on innovation in other regions, where people acquired the information on the literary inquisition cases.

We now examine the potential spillover effect of the literary inquisition on innovation in the control prefectures that were not directly exposed to it. Intuitively, people in the control prefectures close to the treatment prefectures were more likely to hear about the literary inquisition cases that occurred in their neighbors. Thus, innovation in such control prefectures may also be affected to some extent. By contrast, given that people in the control prefectures located further away from the treatment prefectures may not be able to acquire information on the literary inquisition cases that occurred far away, innovation in these prefectures may be less affected by those cases. Therefore, we can compare

the estimates of the effect of the literary inquisition on innovation obtained from two subsamples (i.e., treatment and closely located control prefectures and treatment and distant control prefectures) to infer the potential spillover effect.

We divide all control prefectures into two categories based on their proximity to the treatment prefectures: (1) close ones, which were within 60 km from the border of the treatment prefectures; and (2) distant ones, which were beyond 60 km from the border of the treatment prefectures.[13] We then construct two sub-samples to perform the DID analysis: Sub-sample I, which includes all treatment prefectures and closely located control prefectures; and Sub-sample II, which comprises all treatment prefectures and distant control prefectures. Given that the control prefectures that were distant from the treatment prefectures may be generally unaffected by the literary inquisition, they serve as a better control group for the treatment prefectures. By contrast, the control prefectures close to the treatment prefectures may also be affected by the literary inquisition to some extent. Thus, they do not form a perfect control group for the treatment prefectures. Therefore, if a spillover effect of the literary inquisition on innovation exists, we would anticipate the DID estimators obtained from Sub-sample II to be larger (in absolute value) than those obtained from Sub-sample I.

We first estimate Equation (1) with the Sub-samples I and II constructed from the full sample and report the results in Columns (1) and (2) in Table 4. Both estimates are significant, and the estimate obtained from Sub-sample II (-0.061) is larger (in absolute value) than that obtained from Sub-sample I (-0.044). Furthermore, the difference between the two estimates is different from 0(with the *p*-value being approximately 0.21). Thus, the estimates obtained from the full sample present week evidence of the spillover effect.

We then estimate Equation (1) with the Sub-samples I and II constructed from the matched sample, in which the treatment and control prefectures are more comparable. The results are reported in Columns (3) and (4) of Table 4. Both estimates are significant, and the estimate obtained from Sub-sample II (-0.089) is larger (in absolute value) than that obtained from Sub-sample I (-0.064). Furthermore, the difference between the two estimates is different from 0 (with the *p*-value being approximately 0.16), which presents slightly stronger evidence of the spillover effect.

[13] We select a distance of 60 km as the cutoff between the close and distant control prefectures because such a two-day-walk distance might be reasonable for information spread at the time. Our results are not sensitive to the selection of this cutoff distance, and they remain robust if we select 30 km or 90 km as the cutoff (see more results in the Appendix). Alternatively, we can simply select those control prefectures adjacent to the treatment ones (where the distance equals 0) as the close control prefectures and other non-adjacent prefectures as the distant ones. However, the cost of such a practice is that for the matched sample, close control prefectures consist of only a few prefectures. In particular, there are only 71 treatment prefectures and 46 control ones for the matched sample, and only a few control prefectures are adjacent to the treatment prefectures, making it difficult to identify the potential spillover effect for this sample. Thus, we do not impose the strict restriction that the close prefectures must be adjacent to the treatment ones.

To refine our analysis, we delineate close control prefectures as those adjacent to the treatment prefectures, and distant prefectures as those situated beyond 800 km from the treatment prefectures' borders. This clear distinction aims to more accurately identify the potential spillover effects. Accordingly, we further construct two subsamples for our DID analysis: Subsample III, including all treatment prefectures and their adjacent control prefectures, and Subsample IV, consisting of all treatment prefectures and distant control prefectures located over 800 km away. The results, presented in Table A11 in the Appendix, indicate that the estimate obtained from Sub-sample IV is significantly larger (in absolute value) than that obtained from Sub-sample III, with a *p*-value of approximately 0.06. This disparity provides strong evidence of the spillover effect. See more discussions in the Appendix.

If the spillover effect exists, then the DID estimates provide a lower bound for the deterrent effect of the literary inquisition on innovation in the treatment prefectures. Specifically, if both the treatment and control prefectures were affected by the literary inquisition, with the treatment prefectures being affected to a larger extent, then a significant DID estimator indicates that innovation in the treatment prefectures decreased more sharply than that in the control prefectures, which presents strong evidence of the treatment effect. In other words, if the control prefectures were completely unaffected by the literary inquisition, then innovation in the treatment prefectures would decrease to a larger extent compared with that in the control prefectures, and the DID estimators would also be even more negative. Therefore, the existence of the spillover effect makes our results more convincing.

In particular, the existence of the spillover effect implies that the repercussions of the literary inquisition were far-reaching, extending beyond the treatment prefectures into the control ones. Such a spillover effect suggests a widespread chilling effect on innovation due to fear of punishment, stifling the intellectual spirit throughout the country.

5.7 The Mechanism: The Decline of *Shixue* and the Rise of the *Puxue*

We now briefly analyze the potential mechanism through which the literary inquisition could have affected scientific and technological innovation at the time. We previously posit that the intellectuals may have been unwilling to engage in creative yet potentially risky activities following the literary inquisition, thereby leading to S&T stagnation. We present some direct evidence to substantiate this claim.

Prior to the literary inquisition, the academically dominant school among the Chinese intellectuals was the *shixue*. This school of thought emphasized statecraft and advocated for the application of classics to present use. Recognizing the drawbacks of the feudal autocracy, some intellectuals also incorporated anti-autocracy sentiments and equality notions into the core values of *shixue*. Advocates of *shixue* aimed to use their knowledge and talents to stimulate social progress and economic

development. Such ethos prevailed throughout the country before the onslaught of the literary inquisition (Huang, 2003).

However, circumstances dramatically altered following the investigation of the case of Mingshi Dai. Intellectuals were astonished to learn that talking about statecraft could be interpreted as offensive to the regime and induce severe punishments. Moreover, the core values of *shixue*, such as anti-autocracy and equality, stood in stark contrast to the thought control policies that the Qing rulers implemented following the literary inquisition. Consequently, the statecraft ideology embedded in *shixue* dissipated and even perished following the literary inquisition. The study of classics, considered safe and impervious to the literary inquisition, experienced a resurgence (Huang, 2003).

The rise of the *puxue* (or textology) school, which focused extensively on studying and re-examining the Confucian classics, was particularly notable. This discipline comprised three main elements: *jiaozhu* (annotation and collation of classical texts), *bianwei* (identification of errors within these classics), and *jiyi* (collection and compilation of lost classics). *Puxue* developed rapidly after the literary inquisition, peaking during the Qianlong reign when the enforcement of the literary inquisition was also at its zenith (Du and Jin, 2003).

Figure 10 presents the number of *puxue* masters in the Qing Dynasty over time, providing a representation of the evolution of *puxue* in China. Note that prior to the first case of the literary inquisition, the number of *puxue* masters was modest. However, it underwent a sharp increase after 1711 and reached the peak around 1770. Furthermore, the number drastically declined after 1790, coinciding with the cessation of the literary inquisition, and returned to its previous low levels. Such a pattern is consistent with historians' views on the rise and fall of *puxue* in the Qing Dynasty. That is, *puxue* rose quickly due to the literary inquisition and declined soon after its termination (Zheng, 1980; Qian, 1997; Huang, 2003; Jian, 2006; Chen, 2012).

People may wonder why other politically neutral subjects such as mathematics and physics, which appear as impervious to the literary inquisition as *puxue*, declined immediately and only *puxue* developed rapidly following the literary inquisition. The corresponding explanation lies in the unpredictability of the literary inquisition. Note that it was impossible for individuals to anticipate what expressions could induce punishments, and any original works, whether they pertained to literature or science and technology, could be interpreted as offensive to the regime and spell disaster for the authors. By contrast, *puxue*, which studied the Confucian classics, was encouraged and supported by the Qing government even after the literary inquisition, marking it as an absolutely safe pursuit.

In imperial China, the Confucianism served as the cornerstone of the traditional political culture and the ideological foundation of the feudal regime (Jin, 1999). Qing Dynasty emperors also advocated

the learning and studying of Confucian classics among citizens. In 1745, Emperor Qianlong issued a decree to command the ministers to recommend those that were proficient in the Confucian classics, and assign them to important government positions (Chen, 2000). Proficiency in the classics was considered as a virtue and also a qualification for government officials at the time. Given that the emperors explicitly supported and even rewarded the study of the Confucian classics, *puxue* was considered by the intellectuals as the only safe and most rewarding school and developed rapidly after the literary inquisition.

We do not argue that *puxue* was completely useless to society at the time. However, if the majority of intellectuals abandoned themselves to studying the Confucian classics to sidestep the punitive risks associated with more innovative endeavors, the development of modern science and technology was inevitably bound to halt. Therefore, the mechanism through which the literary inquisition affected innovation becomes evident: it catalyzed a shift in the intellectuals' interests from innovative activities with practical applications to less risky yet uncreative pursuits. Hence, if intellectuals, who were the main contributors of innovation in the Qing Dynasty, were no longer willing to participate in innovative activities, then China's decline in science and technology would be inevitable.

6. The Effect of the Literary Inquisition on the Economic Development in Qing China

We have shown that the literary inquisition may have stifled the spirit of innovation among citizens at the time, leading to a significant decrease in the number of innovations in the affected regions. We now extend our investigation to examine the potential impact of the literary inquisition on China's economic development in the Qing Dynasty, particularly in the long term.

Innovation is generally considered the primary engine driving the economic development of a country. Therefore, if the literary inquisition thwarted innovation, then it could ultimately impede the economic development of China in the Qing Dynasty. However, note that the literary inquisition was implemented during the period of the "Kangxi–Qianlong Great Ages" when China enjoyed high economic prosperity. Apparently, the literary inquisition did not immediately destroy China's predominantly agricultural economy at the time. Nonetheless, the literary inquisition could stifle a society's innovative spirit and undermine its capacity for progress, ultimately hampering the long-term economic development of the country.

Our previous discussions have highlighted a significant decline in China's scientific and technological innovation after 1700. Despite this decline, China continued to prosper in this period, with economic downturns only observed after 1800. Apparently, China's decline in S&T did not have an immediate destructive effect on the economy, at least in the short term. However, this decline could

be a crucial determinant in the eventual loss of China's global power status in the late Qing Dynasty. Specifically, if China missed out on the Industrial Revolution due to its decline in S&T, then it was bound to fall behind Europe in economy during the industrial era. Thus, we can generally explain China's ultimate decline in economy in the late Qing Dynasty by proving that the literary inquisition had led to the country's decline in S&T prior to the Industrial Revolution. However, understanding how extreme controls on expression, such as the literary inquisition, impact the long-term economic development of a society remains a fundamental and interesting issue. Thus, we provide empirical evidence of the potential effects of the literary inquisition on the long-term economic development of China in the Qing Dynasty.

We employ the index of land tax revenue at the prefectural level as a measure of each prefecture's economic performance in the Qing Dynasty. Note that land tax was the primary revenue source for the Qing government, and the total land tax revenue reflected the agricultural output level at the time. Therefore, land tax revenue can serve as a reliable indicator of economic development in the Qing Dynasty when agriculture was the dominant industry, contributing a significant portion to the total output.

We construct a panel data set of land tax revenue for the 1680–1850 period. The observation is at the prefectural level with a decadal time interval. The treatment is also defined as a prefecture's first exposure to the literary inquisition. We first estimate an event-study version of the TWFE model to examine the dynamic effects of the literary inquisition on land tax revenue. In practice, we estimate the following equation similar to Equation (2):

$$\ln\left(LandTax_{pt}\right) = \sum_{\tau} \beta_{\tau} literary\ inquisition_{p\tau} + \theta X_p \times \gamma_t + Tandingrumu_{pt} + \lambda_p + \gamma_t + \delta_{prov} \times \gamma_t + \epsilon_{pt}\,, \quad (3)$$

where $ln\,(LandTax_{pt})$ is the logarithm of land tax revenue in prefecture p in time t; $literary\ inquisition_{p\tau}$ is a set of indicator variables that equals 1 if it has been τ years since prefecture p's first exposure to the literary inquisition, where $\tau \in \{-(50+), -50, -40, -30, -20, 0, 10, 20, 30, 40, 50, 60, 70+\}$. Same as before, we define τ=0 as the period when a prefecture was first exposed to the literary inquisition. Moreover, we treat the groups 10 years before a prefecture's first exposure to the literary inquisition as the reference groups. $Tandingrumu_{pt}$ is a dummy variable which equals 1 if prefecture p had implemented the *Tandingrumu* policy in time t, and other variables are similar to these in Equation (2). β_{τ} is the parameter of interest.

Figure 11 demonstrates the event-study results. Prior to the literary inquisition, the estimates of β_τ are close to 0 and not significant. By contrast, after the literary inquisition, the coefficients begin to decrease and become significantly negative 30 years later and continue to decrease thereafter. Such results indicate that the literary inquisition may have a nonnegligible negative effect on the land tax revenue in the treatment prefectures, particularly in the long term.

Table 5 presents the coefficients visualized in Figure 11. From 30 years after the literary inquisition, the estimates become significantly negative and continue to decrease. The significant estimates range from -0.080 to -0.127, suggesting that the literary inquisition reduced the land tax revenue in the treatment prefectures by 8.0% to 12.7%. Such results indicate that the negative effect of the literary inquisition on land tax revenue is modest, particularly in the short term.

In the Appendix, we present the baseline DID estimates of the effect of the literary inquisition on land tax revenue and more robust DID estimators proposed by De Chaisemartin and d'Haultfoeuille (2020); Borusyak, Jaravel, and Spiess (2021); Callaway and Sant'Anna (2021); and Sun and Abraham (2021). All the coefficients are smaller than -0.09, indicating that the overall negative effect of the literary inquisition on the land tax revenue in the treatment prefectures is no more than 9%, which is a modest yet nonnegligible impact.

One potential concern is that land tax may not be a perfect measure of economic performance in Qing China. For instance, some studies also use land tax to measure state capacity because it could also reflect the state's ability to tax (e.g., Tilly, 1990; Kung and Ma, 2014b; Dincecco and Katz, 2016). We acknowledge that finding a perfect measure of economic development in imperial China is challenging. However, given that land tax is strongly correlated with the economic development in Qing China, it is commonly used in the literature as a useful indicator of income level, economic prosperity, and agricultural productivity during that era (e.g., Ma, 2008; Kung and Ma, 2014a; Ma, 2021, 2022; Chen and Ma, 2022). Therefore, despite its limitations, land tax is one of the most reliable indicators of economic development for this historical period.

Although we have presented evidence of the literary inquisition's adverse effect on land tax revenue, the precise mechanism through which it impacted land tax revenue remains unclear. Indeed, the literary inquisition primarily functioned as a policy to control free expression, with no apparent connection to agricultural production.

However, the key to understanding this causality lies in the profound societal impact of the literary inquisition. By instilling a pervasive climate of fear, this policy could immediately diminish citizens' incentive to innovate, leading to a steep decline in S&T advancements in the short term, as suggested by Figures 1–3. While the literary inquisition might not have directly destroyed China's then-

agricultural economy in the short term, its long-term societal consequences were far more insidious. It could gradually erode societal energy and enthusiasm, thereby demotivating people and diminishing their willingness to contribute to society with their talents and initiatives. Over time, this gradual but steady decline in social vitality and vigor could lead to a broader economic decline, including in the agricultural sector.

Hence, the literary inquisition's immediate and pronounced effects on innovation stood in contrast with its gradual yet profound impacts on the overall economy. An analogy of a frog being slowly boiled in water vividly represents the literary inquisition's insidious and escalating impact on the overall economy, with the full extent of the damage only becoming evident over an extended period. This pattern is corroborated by the trends shown in Figure 11, which depict the literary inquisition's delayed but enduring impact on the economic development in Qing China.

Given that the literary inquisition only had an indirect effect on agricultural production, we do not expect this effect to be as large as that on innovation in the affected region. The destructive effect of the literary inquisition on innovation could be a potential mechanism through which it affected land tax revenue. That is, the decline in S&T may have hindered agricultural development, particularly in the long term.

7. Conclusions

The "Kangxi–Qianlong Great Ages" has been widely regarded as one of the pinnacles of imperial China, marked by economic prosperity and global prominence. However, beneath the surface of peace and prosperity, there may have been hidden yet severe crises. In particular, the Qing regime implemented stringent thought controls on its citizens through the literary inquisition, creating an atmosphere of fear at the time and destroying society's foundation for progress. Although the literary inquisition did not immediately devastate the agricultural economy, the extreme expression controls could have drained society's vitality and ingenuity, undermining its capacity for future progress. Our study indicates that the literary inquisition may have stifled the spirit of innovation of society, leading to China's sharp decline in S&T in the short term and ultimately inducing its decline in economy in the long term. Such findings, seemingly surprising yet reasonable and thought-provoking, provide important insights and prompt us to reevaluate the "Kangxi–Qianlong Great Ages" in the broader context of Chinese history. Hence, the following question must be answered: Was this perceived "Great Age" merely an illusion or perhaps the dying gasp of the totalitarian regime in imperial China?

Although our focus has been on the destructive impact of extreme expression controls on the spirit of innovation, we have to recognize that such controls can harm society in various other ways. For

instance, by imposing strict restrictions on free expression, the government can effectively silence criticisms, removing the incentive to improve services to meet the citizens' needs. Consequently, the quality of governance may stagnate, and tensions between the government and its citizens may escalate over time. For authoritarian regimes, suppressing complaints seems easier than exerting effort to appease citizens. Hence, it is not surprising that restrictions on free speech are prevalent in such systems.

However, we should interpret the findings of this study with caution. Although expression controls in authoritarian regimes are widespread, extreme practices such as the literary inquisition seen in Qing China are rare and have faded in modern times. Current controls on speech are typically milder, aimed at promoting regime support by keeping citizens in ignorance. Although mild expression controls may also affect society negatively and lead to undesirable outcomes, their impacts may not be equivalent to those of extreme policies such as the literary inquisition. How such milder controls on free expression impact society in the short and long terms, whether and to what extent they undermine the societal foundations for future progress, remains unanswered questions warranting further investigations.

References

Ai, Suzhen, and Zhenghai Song. 2006. *Zhongguo kexue jishu shi: nianbiao juan (History of Science and Technology in China: Chronological Volumes)*. Beijing: Science Press.

Athey, Susan, and Guido Imbens. 2016. "Recursive Partitioning for Heterogeneous Causal Effects." *Proceedings of the National Academy of Sciences* 113 (27): 7353–60.

Athey, Susan, Julie Tibshirani, and Stefan Wager. 2019. "Generalized Random Forests." *The Annals of Statistics* 47 (2): 1148–78.

Bai, Ying, and Ruixue Jia. 2016. "Elite Recruitment and Political Stability: The Impact of the Abolition of China's Civil Service Exam." *Econometrica* 84 (2): 677-733.

Becker, Sascha O., Francisco J. Pino, and Jordi Vidal-Robert. 2021. "Freedom of the Press? Catholic Censorship during the Counter-Reformation." Working Paper.

Blasutto, Fabio, and David de la Croix. 2023. "Catholic Censorship and the Demise of Knowledge Production in Early Modern Italy." *The Economic Journal* 133 (656): 2899–2924.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2021. "Revisiting Event Study Designs: Robust and Efficient Estimation." Unpublished.

Braghieri, Luca, Ro'ee Levy, and Alexey Makarin. 2022. "Social Media and Mental Health." *American Economic Review* 112 (11): 3660–93.

Brandt, Loren, Debin Ma, and Thomas G. Rawski. 2014. "From Divergence to Convergence: Reevaluating the History Behind China's Economic Boom." *Journal of Economic Literature* 52 (1): 45-123.

Bristow, William. 2023. "Enlightenment." In *The Stanford Encyclopedia of Philosophy*, edited by Edward N. Zalta and Uri Nodelman, Fall 2023. Metaphysics Research Lab, Stanford University. https://plato.stanford.edu/archives/fall2023/entries/enlightenment/.

Britto, Diogo G. C., Paolo Pinotti, and Breno Sampaio. 2022. "The Effect of Job Loss and Unemployment Insurance on Crime in Brazil." *Econometrica* 90 (4): 1393–1423.

Cabello, Matias. 2023. "The Counter-Reformation, Science, and Long-Term Growth: A Black Legend?" *SSRN Electronic Journal*.

Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-differences with Multiple Time Periods." *Journal of Econometrics* 225 (2): 200–230.

Cantoni, Davide, and Noam Yuchtman. 2013. "The Political Economy of Educational Content and Development: Lessons from History." *Journal of Development Economics* 104: 233-244.

Chen, Juyuan. 2000. *Qingdai puxue yu zhongguo wenxue (Puxue in the Qing Dynasty and Chinese Literature)*. Nanchang, China: Baihuazhou Wenyi Press.

Chen, Yuyu, and David Y. Yang. 2019. "The Impact of Media Censorship: 1984 or Brave New World?" *American Economic Review* 109 (6): 2294–2332.

Chen, Zhiwu, and Chicheng Ma. 2022. "Prioritizing Stability over Productivity: How Confucianism Facilitated Population Growth in Preindustrial China." *Available at SSRN 3859796.*

Chen, Zuwu. 2012. *Qingdai xueshu yuanliu (The Origins and Development of Scholarship in the Qing Dynasty)*. Beijing: Beijing Normal University Press.

De Chaisemartin, Clément, and Xavier D'Haultfoeuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110 (9): 2964-96.

Dewitte, Edgard, Francesco Drago, Roberto Galbiati, and Giulio Zanella. 2023. "Science under Inquisition: The Allocation of Talent in Early Modern Europe." Working Paper.

Dincecco, Mark, and Gabriel Katz. 2016. "State Capacity and Long-Run Economic Performance." *The Economic Journal* 126 (590): 189–218.

Drelichman, Mauricio, Jordi Vidal-Robert, and Hans-Joachim Voth. 2021. "The Long-Run Effects of Religious Persecution: Evidence from the Spanish Inquisition." *Proceedings of the National Academy of Sciences* 118 (33): e2022881118.

Du, Shiran, and Qiupeng Jin. 2003. *Zhongguo kexue jishu shi: tongshi juan (History of Science and Technology in China: General History Volumes)*. Beijing: Science Press.

Edwards, Ronald A. 2005. "The Structure of Authority, Property Rights and Economic Stagnation: The Case of China (in Chinese)." *China Economic Quarterly* 4 (2): 541-562.

Fernández-Villaverde, Jesús, Mark Koyama, Youhong Lin, and Tuan-Hwee Sng. 2023. "The Fractured-Land Hypothesis*." *The Quarterly Journal of Economics* 138 (2): 1173–1231.

Ford, Guy. 1935. *Dictatorship in the Modern World*. Minneapolis, MN: University of Minnesota Press.

Fu, Zhengyuan. 1993. *Autocratic Tradition and Chinese Politics*. Cambridge, UK: Cambridge University Press.

Gross, Daniel P., and Bhaven N. Sampat. 2023. "America, Jump-Started: World War II R&D and the Takeoff of the US Innovation System." *American Economic Review* 113 (12): 3323–56.

Guo, Chengkang, and Tiejun Lin. 1990. *Qingchao wenziyu (The Literary Inquisition in the Qing Dynasty)* Beijing: Qunzhong Press.

Harvard University, Academia Sinica, and Peking University, China Biographical Database (CBDB) (January 1, 2023), https://projects.iq.harvard.edu/cbdb.

Hu, Zhen. 2013. "A Punishment for the Speech? ——'Regicide'and Literary Inquisition in Qing Dynasty (in Chinese)." *China Agricultural University Journal of Social Sciences Edition* 30 (4): 57–65.

Huang, Aiping. 2003. *Puxue yu qingdai shehui (Puxue and Qing Dynasty Society)*. Shijiazhuang: Hebei Renmin Press.

Huang, Ray. 1997. *Zibenzhuyi yu ershiyi shiji (Capitalism and the 21st Century)*. Beijing: SDX Joint Publishing Company.

Huang, Yasheng. 2023. *The Rise and the Fall of the EAST: Examination, Autocracy, Stability, and Technology in Chinese History and Today*. Yale University Press.

Ito, Shuntaro, Kenzo Sakamoto, Keiji Yamada, and Yoichiro Murakami. 1984. *Jianming shijie kexue jishu shi nianbiao (A Concise Chronology of World Science and Technology History)*. Translated by Zhenhuan Jiang and Guanxiong Ge. Harbin: Harbin Institute of Technology Press.

Jian, Bozan. 2006. *Zhongguo shi gangyao (An Outline of Chinese History)*. Beijing: Peking university Press.

Jin, Qiupeng. 1998. *Zhongguo kexue jishu shi: renwu juan (History of Science and Technology in China: Biographical Volumes)*. Beijing: Science Press.

Jin, Taijun. 1999. "On the Political Socialization Mechanism of Chinese Traditional Political Culture (in Chinese)." *CASS Journal of Political Science* 02: 49-61.

Jin, Xingyao. 2012. *Doushi wenzi re de huo: Qingdai wenziyu (Trouble Caused by Words: Literary Inquisitions of the Qing Dynasty)*. Beijing: The Palace Museum Press.

Jones, Eric. 2003. *The European Miracle: Environments, Economies and Geopolitics in the History of Europe and Asia*. 3rd ed. Cambridge: Cambridge University Press.

Jones, Peter M. 2017. *Industrial Enlightenment: Science, Technology and Culture in Birmingham and the West Midlands, 1760–1820.* Manchester, UK: Manchester University Press.

Koyama, Mark, and Melanie Meng Xue. 2015. "The literary inquisition: The persecution of intellectuals and human capital accumulation in China." Working Paper.

Kung, James Kai-sing, and Chicheng Ma. 2014a. “Autarky and the Rise and Fall of Piracy in Ming China.” *The Journal of Economic History* 74 (2): 509–34.

Kung, James Kai-sing, and Chicheng Ma. 2014b. “Can Cultural Norms Reduce Conflicts? Confucianism and Peasant Rebellions in Qing China.” *Journal of Development Economics* 111 (November): 132–49.

Landes, David, S. 2006. “Why Europe and the West? Why Not China?” *Journal of Economic Perspectives*, 20 (2): 3-22.

Lin, Justin Yifu. 1995. “The Needham Puzzle: Why the Industrial Revolution Did Not Originate in China.” *Economic Development and Cultural Change* 43 (2): 269-292.

Ma, Chicheng. 2021. “Knowledge Diffusion and Intellectual Change: When Chinese Literati Met European Jesuits.” *The Journal of Economic History* 81 (4): 1052–97.

Ma, Chicheng. 2022. “Classicism and Modern Growth: The Shadow of the Sages.” *Available at SSRN 4288048.*

Ma, Debin. 2008. “Economic Growth in the Lower Yangzi Region of China in 1911–1937: A Quantitative and Historical Analysis.” *The Journal of Economic History* 68 (2): 355–92.

McCloskey, Deirdre N. 2010. *Bourgeois Dignity: Why Economics can't Explain the Modern World*. Chicago: University of Chicago Press.

Mokyr, Joel. 1992. *The Lever of Riches: Technological Creativity and Economic Progress*. Oxford University Press.

Mokyr, Joel. 2002. *The Gifts of Athena: Historical Origins of the Knowledge Economy*. Princeton and Oxford: Princeton University Press.

Mokyr, Joel. 2005. “Mobility, Creativity, and Technological Development: David Hume, Immanuel Kant and the Economic Development of Europe.” in A von Günter, ed., *Kreativität. XX. Deutscher Kongress für Philosophie.* Berlin, pp. 1131-1161.

Mokyr, Joel. 2007. “The Market for Ideas and the Origins of Economic Growth in Eighteenth Century Europe.” *Tijdschrift voor Sociale en Economische Geschiedenis*, vol. 4, no. 1, pp. 3-38.

Mokyr, Joel. 2009. *The Enlightened Economy: An Economic History of Britain 1700–1850*. New Haven and London: Yale University Press.

Mokyr, Joel. 2017. *A Culture of Growth: The Origins of the Modern Economy*. Princeton NJ: Princeton University Press.

Needham, Joseph. 1981. *Science in Traditional China. Cambridge*. Massachusette：Harvard University Press.

Pearsall, Judy, Patrick Hanks, Catherine Soanes, and Angus Stevenson. 2007. *The New Oxford English-Chinese Dictionary*. 2nd ed. Shanghai: Shanghai Foreign Language Education Press.

Phelps, Edmund S. 2013. *Mass Flourishing: How Grassroots Innovation Created Jobs, Challenge, and Change.* Princeton, NJ: Princeton University Press.

Pomeranz, Kenneth. 2000. *The Great Divergence: China, Europe, and the Making of the Modern World Economy*. Princeton University Press.

Qian, Mu. 1997. *Zhongguo jin sanbai nian xueshu shi (The Academic History of China in the Last Three Hundred Years)*. Beijing: The Commercial Press.

Scheidel, Walter. 2019. *Escape from Rome: The Failure of Empire and the Road to Prosperity*. Princeton University Press.

Sun, Guangyan, and Jun Song. 2018. “Study of Literary Inquisition in Qing Dynasty Based on Data Statistics (in Chinese).” *Journal of Law Application (Judicial Case)*, no. 16: 127–37.

Sun, Liyang, and Sarah Abraham. 2021. “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects.” *Journal of Econometrics* 225 (2): 175–99.

Tilly, Charles. 1990. *Coercion, Capital, and European States, A.D. 990-1990*. First Edition. Cambridge, Mass., USA: Basil Blackwell.

Wager, Stefan, and Susan Athey. 2018. “Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests.” *Journal of the American Statistical Association* 113 (523): 1228–42.

Wakeman, Frederic E. 1985. *The Great Enterprise: The Manchu Reconstruction of Imperial Order in Seventeenth-Century China*, Vol. 1. Oakland, CA: University of California Press.

Wang, Fan-Sen. 2002. “Political Pressures on the Cultural Sphere in the Ch’ing Period.” *The Cambridge History of China. Ed. by William J. Peterson* 9. Part II: 606-648.

Wang, Yeh-chien. 1973. *Land taxation in Imperial China, 1750-1911*. Cambridge, MA: Harvard University Press.

Wanyan, Shaoyuan, ed. 2011. *Qingdai wenziyu dang (zengding ben) (Archives of the Literary Inquisition Cases (Updated Edition))*. Shanghai: Shanghai Bookstore Publishing House.

Weber, Max. 1915. *The Religion of China: Confucianism and Taoism*. Glencoe Ill: Free Press. Translated in 1951.

Xue, Melanie Meng. 2021. “Autocratic Rule and Social Capital: Evidence from Imperial China.” Working Paper.

Yan, Chongnian. 2016. *Kangxi di da zhuan (The Great Biography of Emperor Kangxi)*. Beijing: Zhonghua Book Company.

Yang, Qiankun. 1999. *Zhongguo gudai wenziyu (Ancient Chinese Literary Inquisitions)*. Shannxi: Shannxi Renmin Chubanshe.

Yang, Fengcheng. 1992. *Qian gu wenziyu — qingdai jishi (Eternal Literary Inquisition — A Record of the Qing Dynasty)*. Haikou: Nanhai Publishing Company.

Yu, Dahua. 1996. “A New View of the Literary Inquisition of the Qing Dynasty (in Chinese).” *Journal of Liaoning Normal University*, no. 1: 72–75.

Zhang, Ji, ed. 1934. *Qingdai wenziyu dang ji (Archives of the Literary Inquisitions Cases (Collected Volumes))*. Beijing: The Palace Museum.

Zhang, Shucai, and Jinghua Du. 1991. *Qingdai wenziyu an (Qing Literary Inquisition Cases)*. Beijing: The Palace Museum Press.

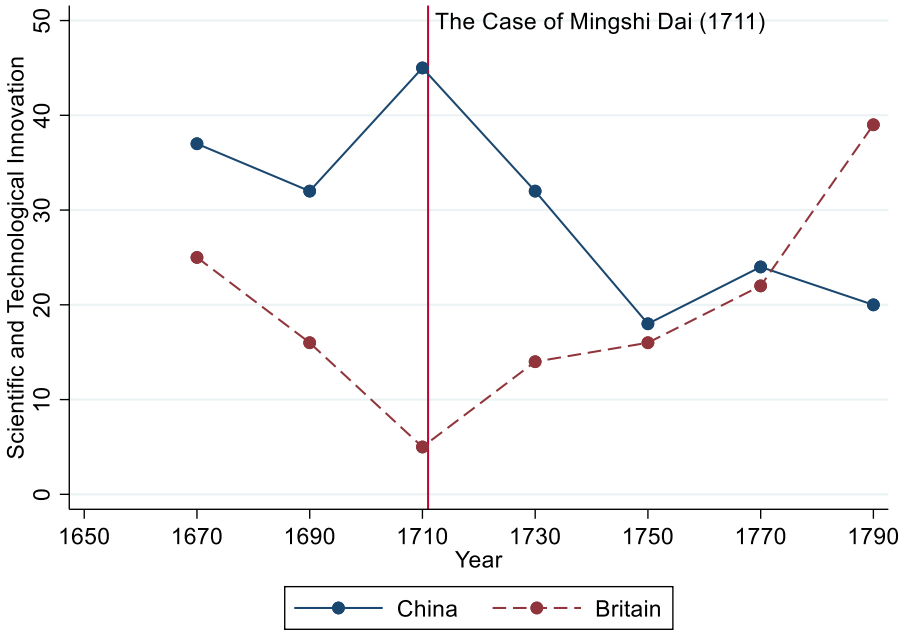
Zhang, Yuyan, and Cheng Gao. 2005. “Overseas Silver, Initial Institutions, and the Stagnation of the Orient: A Conjecture on Why China Missed Economic Takeoff in Late Ming Dynasty (in Chinese).” *China Economic Quarterly* 4(2): 491-518.

Zheng, Tianting. 1980. *Qingshi jian shu (A Brief History of the Qing Dynasty)*. Beijing: Zhonghua Book Company.

Zhi, Weiping. 1925. *Qingdai puxue dashi liezhuan (Biographies of Puxue Masters in the Qing Dynasty)*. Shanghai: Shanghai taidong tushuju.

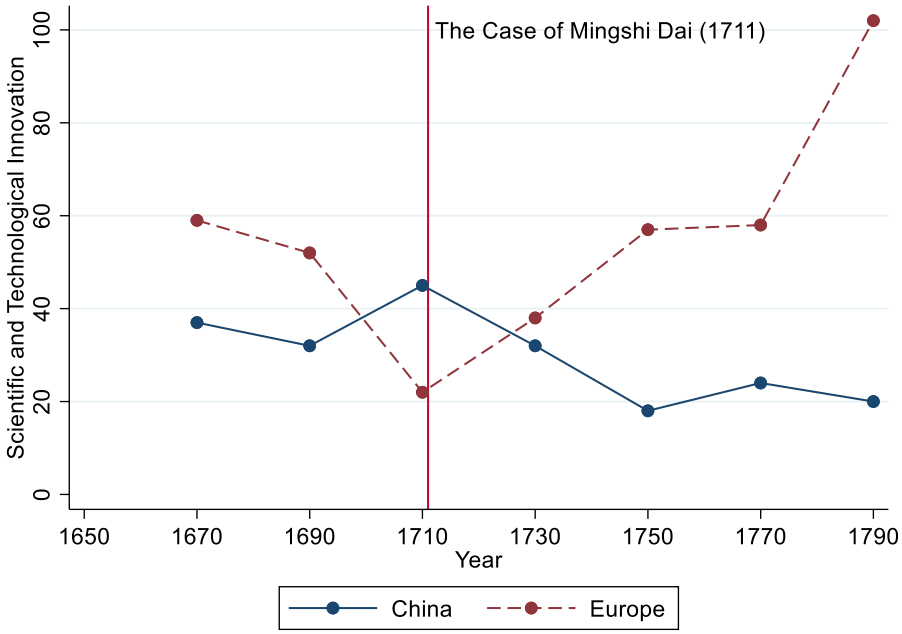
Zhu, Yang. 2013. “The Value of ‘Files of Literary Inquisition of Qing Dynasty’ as Historical Data (in Chinese).” *Journal of Harbin University* 34 (06): 98–101.

Figure 1: Number of Scientific and Technological Innovations in China and Britain (1670–1790)



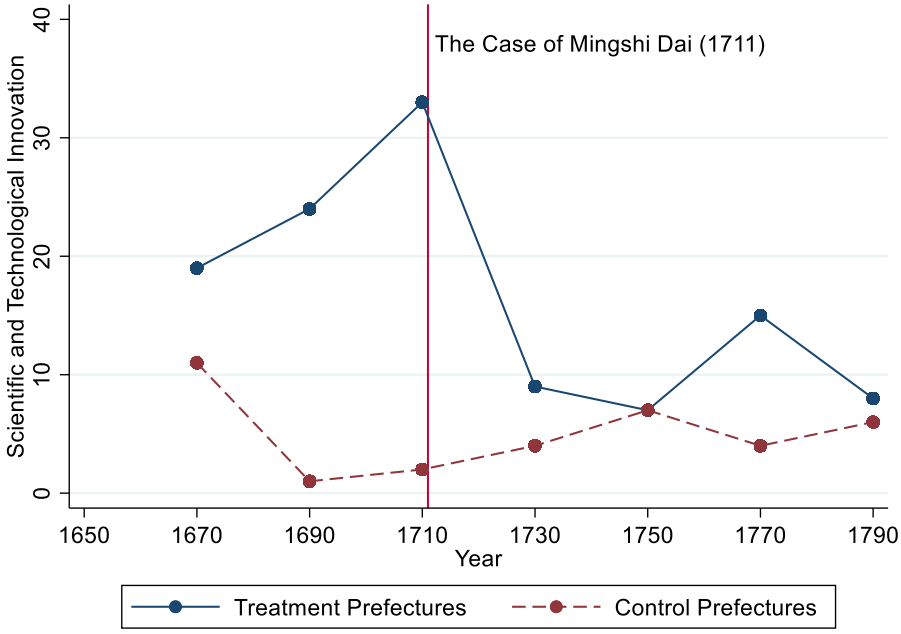
Notes: Data on scientific and technological innovations in China and Britain are from Ai and Song (2006) and Ito et al. (1984), respectively. The two books document important events of scientific and technological activities in Chinese and European history, respectively. We identify the events that reflect important advancements in S&T as innovations, such as publications of academic works on S&T and technological inventions. In general, these innovations include notable advancements in various disciplines, such as mathematics, physics, printing, agriculture, and medicine.

Figure 2: Number of Scientific and Technological Innovations in China and Europe (1670–1790)



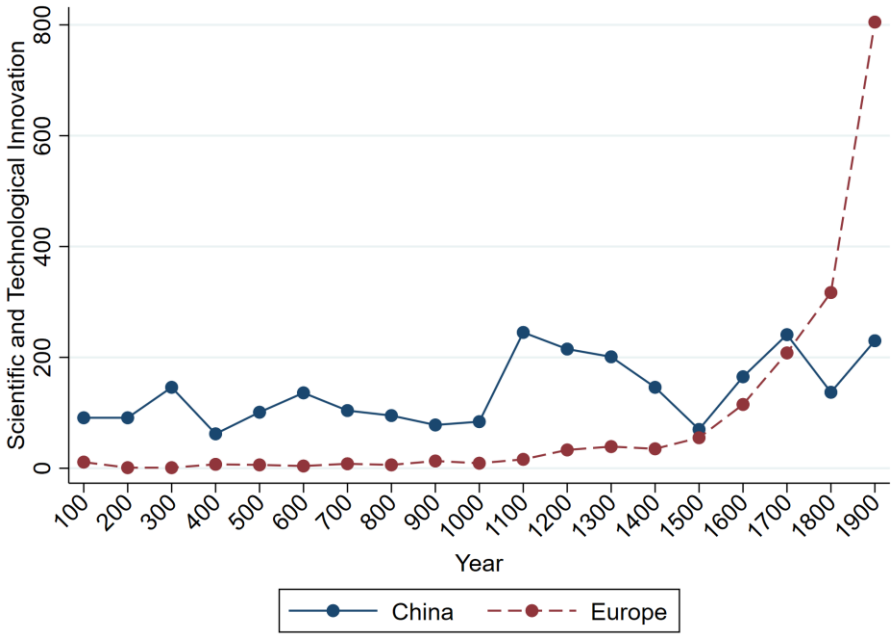
Notes: Similar to Figure 1, data on scientific and technological innovations in China and Europe are from Ai and Song (2006) and Ito et al. (1984), respectively.

Figure 3: Number of Scientific and Technological Innovations in China, Treatment and Control Prefectures (1670–1790)



Notes: Data on scientific and technological innovations in China are from Ai and Song (2006).

Figure 4: Number of Scientific and Technological Innovations in China and Europe (100–1900)



Notes: Similar to Figure 2, data on scientific and technological innovations in China and Europe are from Ai and Song (2006) and Ito et al. (1984), respectively.

Figure 5A: Number of Literary Inquisition Cases over Time (1700–1800)

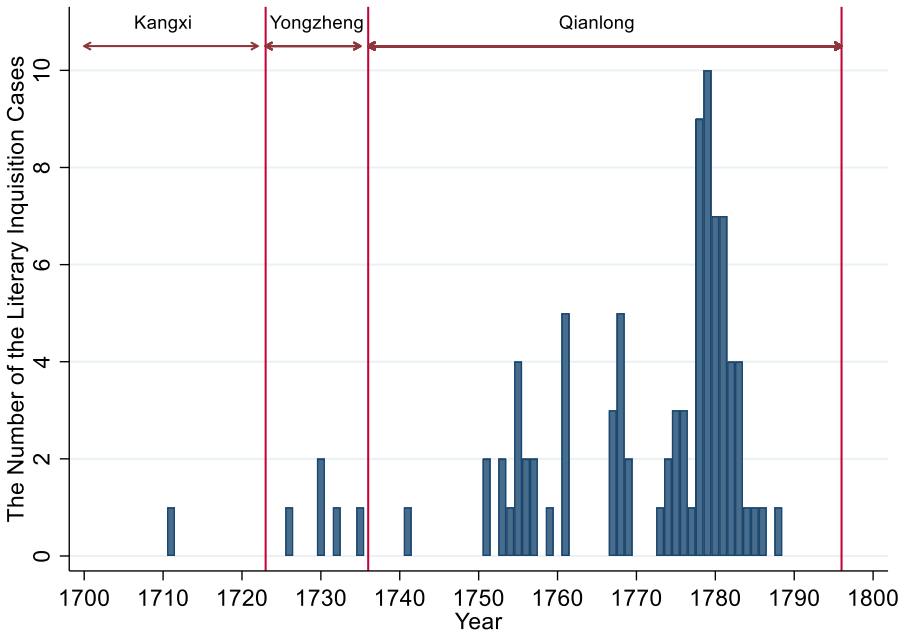
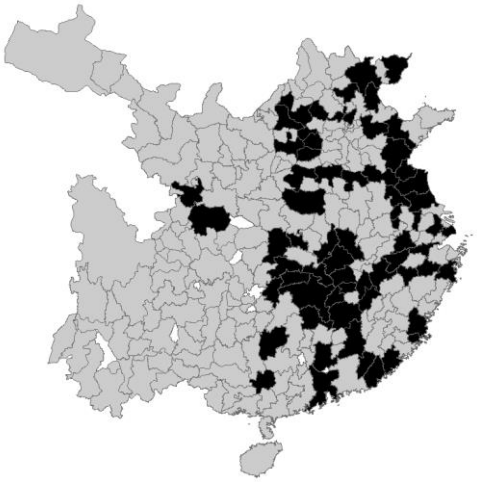


Figure 5B: Spatial Distribution of Prefectures Exposed to the Literary Inquisition (1700–1800)

(a) Kangxi Reign (1700–1722)

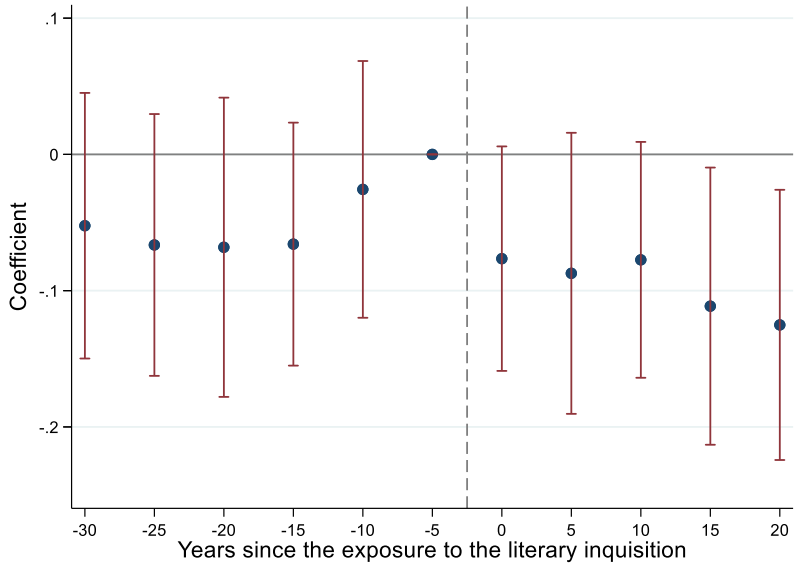
(b) Yongzheng Reign (1723–1735)

(c) Qianlong Reign (1736–1795)



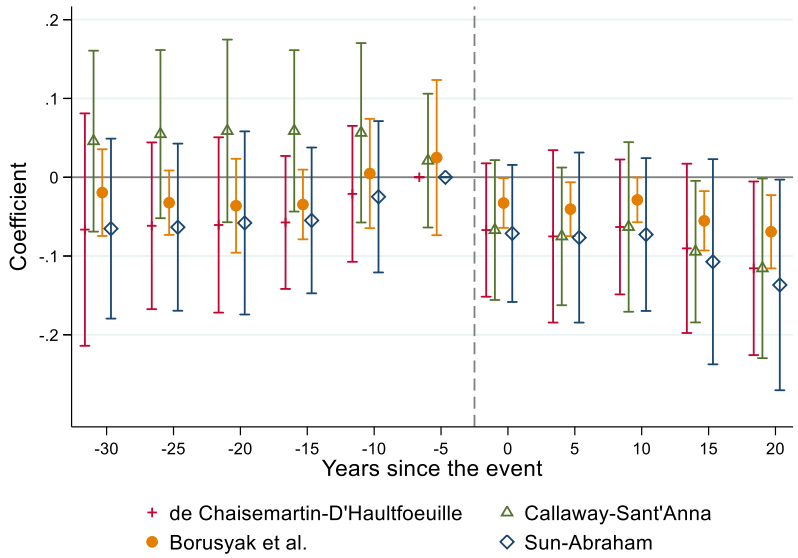
Notes: Figure 5B shows the prefectures exposed to the literary inquisition in the different periods in our sample. The dark areas represent prefectures that were first exposed to the literary inquisition in certain periods.

Figure 6A: Dynamic Effects of the Literary Inquisition on Scientific and Technological Innovation



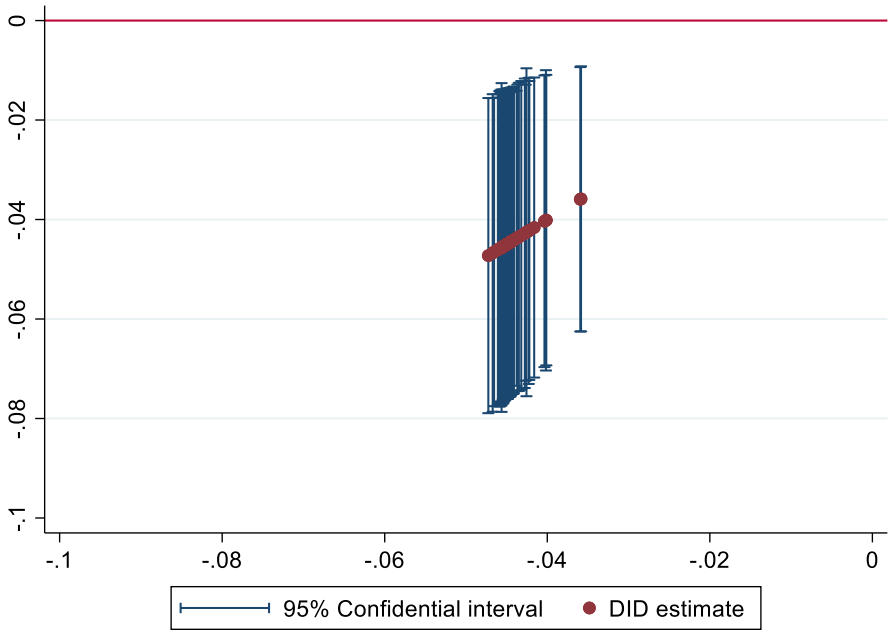
Notes: This figure visualizes the dynamic effects of the literary inquisition on innovation (OLS estimates obtained from the dynamic version of the TWFE model (Equation (2)), using the groups just before the literary inquisition (i.e., the groups five years prior to the literary inquisition) as the reference. Control variables include all baseline controls in Column (4) of Table 2. The bars represent 95% confidence intervals. Standard errors are clustered at the prefectural level.

Figure 6B: Dynamic Effects of the Literary Inquisition on Scientific and Technological Innovation: Staggered Treatment Correction



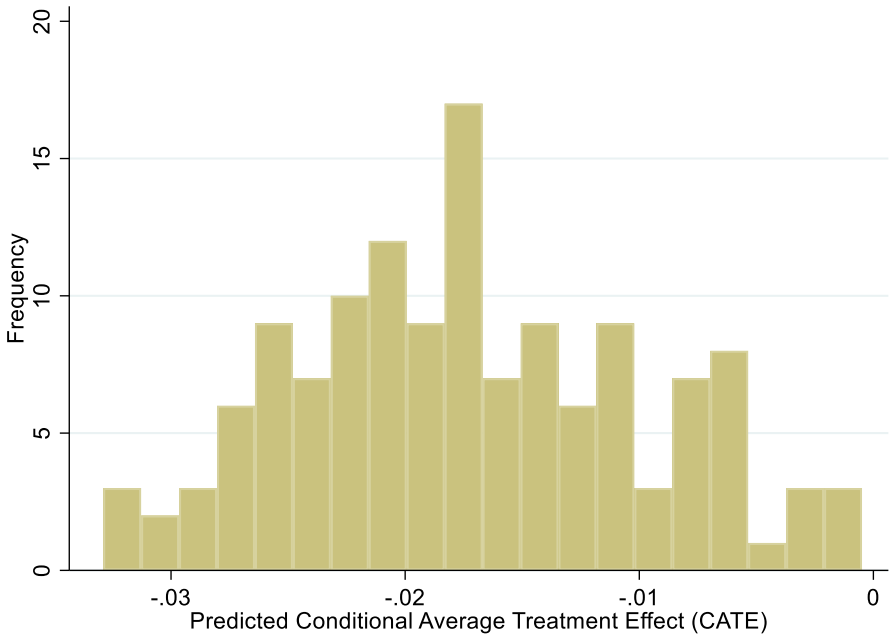
Notes: This figure presents the estimates of the dynamic effects of the literary inquisition on innovation, employing four alternative estimators. These include: Callaway and Sant'Anna (2021) (in green with triangle markers); Sun and Abraham (2021) (in blue with diamond markers); De Chaisemartin and d'Haultfoeuille (2020) (in red with cross markers); and Borusyak, Jaravel, and Spiess (2021) (in orange with circle markers). The outcome variable is the number of scientific and technological innovations (at the prefectural level), and control variables include all baseline controls in Column (4) of Table 2. The groups five years prior to the literary inquisition are used as the reference. The bars represent 95% confidence intervals. Standard errors are clustered at the prefectural level.

Figure 7: DID Estimates of the Effect of Literary Inquisition on Innovation, Omitting Individual Prefectures



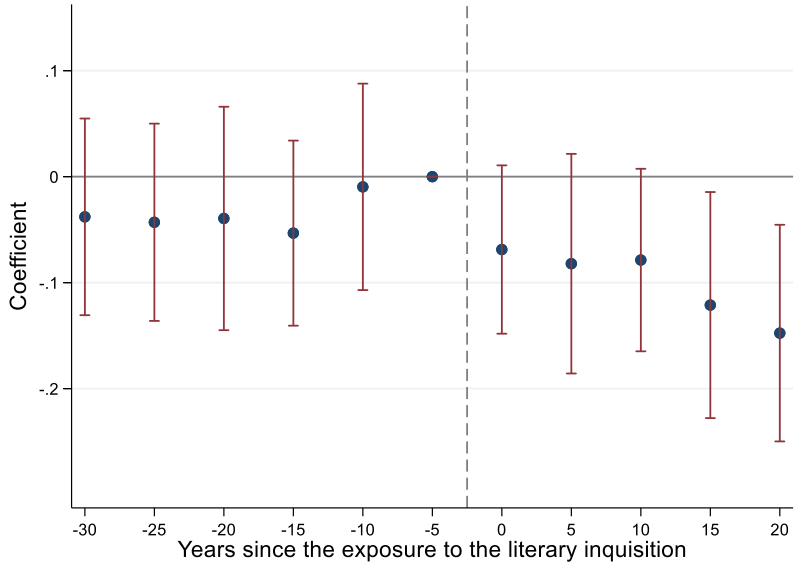
Notes: This figure shows the DID estimates of the effect of the literary inquisition on the number of S&T innovations, omitting individual prefectures from the estimation sample. Error bars represent a 95% confidence interval, computed from standard errors clustered at the prefectural level.

Figure 8: Distribution of the Predicted Conditional Average Treatment Effect



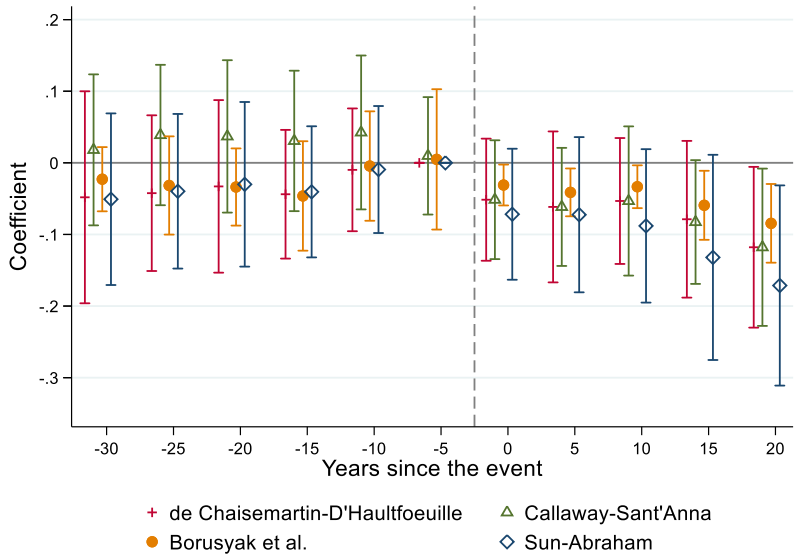
Notes: This figure shows the distribution of the predicted Conditional Average Treatment Effect (CATE). See the Appendix for the detailed procedure of obtaining the CATE.

Figure 9A: Dynamic Effects of the Literary Inquisition on Scientific and Technological Innovation (the Matched Sample)



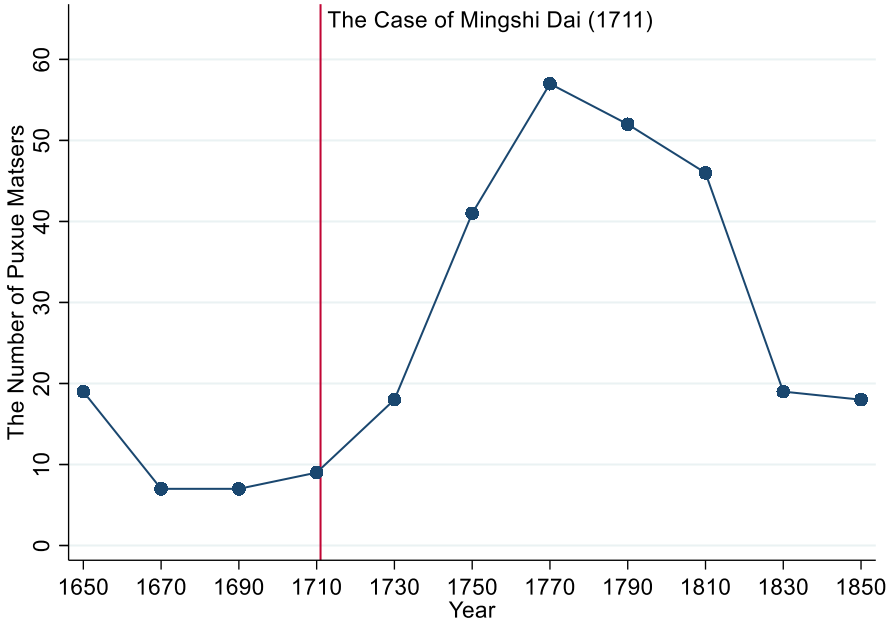
Notes: This figure visualizes the dynamic effects of the literary inquisition on innovation with the matched sample (OLS estimates obtained from the dynamic version of the TWFE model (Equation (2)), using the group five years prior to the literary inquisition as the reference. Control variables include all the baseline controls in Column (4) of Table 2. The bars represent 95% confidence intervals. Standard errors are clustered at the prefectural level.

Figure 9B: Dynamic Effects of the Literary Inquisition on Scientific and Technological Innovation: Staggered Treatment Correction (the Matched Sample)



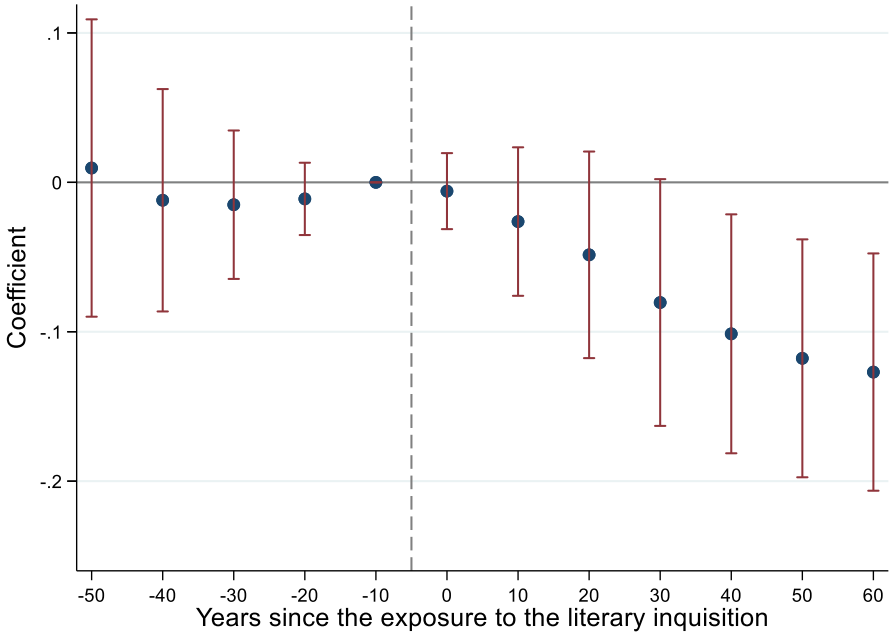
Notes: This figure presents the estimates of the dynamic effects of the literary inquisition on innovation for the matched sample, employing four alternative estimators. These include: Callaway and Sant'Anna (2021) (in green with triangle markers); Sun and Abraham (2021) (in blue with diamond markers); De Chaisemartin and d'Haultfoeuille (2020) (in red with cross markers); and Borusyak, Jaravel, and Spiess (2021) (in orange with circle markers). The outcome variable is the number of scientific and technological innovations (at the prefectural level), and control variables include all baseline controls in Column (4) of Table 2. The groups five years prior to the literary inquisition are used as the reference. The bars represent 95% confidence intervals. Standard errors are clustered at the prefectural level.

Figure 10: The Number of *Puxue* Masters in China (1650-1850)



Notes: This figure presents the number of *puxue* masters born in different periods. Each point represents the number of *puxue* masters born in the past 20 years. Data on *puxue* masters are from Zhi (1925), which records the biographical information of all *puxue* masters in the Qing Dynasty. Unfortunately, this book does not contain the information on the *puxue* masters' birth years. Thus, we further identify the *puxue* masters in another data set, i.e., Chinese Biographical Database (CBDB), and finally obtain a sample of 318 *puxue* masters with complete information.

Figure 11: Dynamic Effects of the Literary Inquisition on Land Tax Revenue



Notes: This figure visualizes the dynamic effects of the literary inquisition on land tax revenue (OLS estimates obtained from the dynamic version of the TWFE model (Equation (3)), using the groups 10 years prior to the literary inquisition as the reference. Control variables include all the controls in Table 5. The bars represent 95% confidence intervals. Standard errors are clustered at the prefectural level.

Table 1: Data Sources and Summary Statistics for the Main Variables

	Sources	Obs.	Mean	S.D.
Panel A: 1701-1800				
Outcome				
The number of scientific and technological innovations	1	4,960	0.019	0.185
Treatment				
Whether a prefecture was exposed to the literary inquisition	2	4,960	0.098	0.298
Controls				
Whether a prefecture is situated on the coast	4	4,960	0.137	0.344
Whether a prefecture includes one or more major rivers[1]	4	4,960	0.625	0.484
The number of *jinshi* in 1600–1710	5	4,960	42.617	67.199
Population density in 1630 (population/km^2)	5	4,960	83.287	118.513
Agricultural suitability for wetland rice	4	4,960	2.008	1.061
Agricultural suitability for foxmillet	4	4,960	2.912	1.312
Agricultural suitability for sweet potato	4	4,960	2.600	0.997
Language polarization index	4	4,960	0.162	0.299
Language fragmentation index	4	4,960	0.087	0.164
Panel B: 1680-1850				
Outcome				
Logarithm of land tax revenue	3	3,078	11.270	1.646
Treatment				
Whether a prefecture was exposed to the literary inquisition	2	3,078	0.188	0.391
Controls				
Whether a prefecture is situated on the coast	4	3,078	0.187	0.390
Whether a prefecture includes one or more major rivers	4	3,078	0.696	0.460
The number of *jinshi* in 1600–1710	5	3,078	61.766	89.696
Population density in 1630 (population /km^2)	5	3,078	92.803	131.981
Language polarization index	4	3,078	0.185	0.305
Language fragmentation index	4	3,078	0.099	0.170
Agricultural suitability for wetland rice	4	3,078	2.201	1.017
Agricultural suitability for foxmillet	4	3,078	2.796	1.298
Agricultural suitability for sweet potato	4	3,078	2.644	0.854
Land area (1000 km^2)	5	3,078	20.710	19.717
Basin HHI	4	3,078	0.619	0.226

Note: Major rivers are those ranked as first- and second-order streams in the Chinese river hierarchy.

Source: 1.Ai and Song (2003).
2.Wanyan (2011).
3.The General Records of all provinces at different times. See the Appendix for more details.
4.Bai and Jia (2016).
5.Ma (2021).

Table 2: DID Estimates of the Effect of the Literary Inquisition on Innovation

	Dependent Variable: # innovations			
	(1)	(2)	(3)	(4)
Literary Inquisition	–0.0454***	–0.0426***	–0.0441***	–0.0446***
	(0.0165)	(0.0155)	(0.0156)	(0.0157)
Prefecture FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Province FE × Time FE	Yes	Yes	Yes	Yes
Treat × After1790	Yes	Yes	Yes	Yes
Jinshi × Time FE	No	Yes	Yes	Yes
Pop density1630 × Time FE	No	Yes	Yes	Yes
Rice × Time FE	No	No	Yes	Yes
Foxmillet × Time FE	No	No	Yes	Yes
Sweet potato × Time FE	No	No	Yes	Yes
Language polarization × Time FE	No	No	No	Yes
Language fragmentation × Time FE	No	No	No	Yes
Coastal × Time FE	No	No	No	Yes
Main river × Time FE	No	No	No	Yes
Y mean	0.0187	0.0187	0.0187	0.0187
Observations	4960	4960	4960	4960
R-squared	0.103	0.128	0.133	0.158

Notes: This table examines the effect of the literary inquisition on the number of innovations in treatment prefectures. Specifically, it presents estimates of the coefficient β from Equation (1). In Column (1), we estimate Equation (1) without including controls (only control the prefecture fixed effects, time fixed effects, the interaction of province and time fixed effects, and the interaction term of the treatment dummy and post-1790 dummy). In Columns (2)–(4), we gradually include a series of interaction terms of the time fixed effects and the following prefectural characteristics in the regressions: *Jinshi* (number of *jinshi* in 1600–1710), Population density (the population density in 1630), agricultural suitability metrics for wetland rice, fox millet, and sweet potato, Language polarization (the language polarization index), Language fragmentation (the language fragmentation index), and two geographic dummies: Coast (whether a prefecture is situated on the coast), Main river (whether a prefecture includes at least one major river). Standard errors in parentheses are clustered at the prefectural level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: DID Estimates of the Effect of the Literary Inquisition on Innovation (Matched Sample)

	Dependent variable: # innovations			
	(1)	(2)	(3)	(4)
Literary Inquisition	–0.0554***	–0.0584***	–0.0616***	–0.0629***
	(0.0208)	(0.0222)	(0.0225)	(0.0234)
Prefecture FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Province FE × Time FE	Yes	Yes	Yes	Yes
Treat × After1790	Yes	Yes	Yes	Yes
Jinshi × Time FE	No	Yes	Yes	Yes
Pop density1630 × Time FE	No	Yes	Yes	Yes
Rice × Time FE	No	No	Yes	Yes
Foxmillet × Time FE	No	No	Yes	Yes
Sweet potato × Time FE	No	No	Yes	Yes
Language polarization × Time FE	No	No	No	Yes
Language fragmentation × Time FE	No	No	No	Yes
Coastal × Time FE	No	No	No	Yes
Main river × Time FE	No	No	No	Yes
Y mean	0.0288	0.0288	0.0288	0.0288
Observations	2220	2220	2220	2220
R-squared	0.141	0.179	0.193	0.237

Notes: This table examines the effect of the literary inquisition on the number of innovations in the treatment prefectures for the matched sample. Specifically, it presents estimates of the coefficient β from Equation (1) for the matched sample. Control variables are the same as those in Table 2. Standard errors in parentheses are clustered at the prefectural level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Spillover Effect of the Literary Inquisition on Innovation

	Dependent variable: # innovations			
	Full Sample		Matched Sample	
	Sub I	Sub II	Sub I	Sub II
	(1)	(2)	(3)	(4)
Literary Inquisition	–0.0437***	–0.0612**	–0.0643***	–0.0886**
	(0.0164)	(0.0264)	(0.0244)	(0.0385)
Baseline Controls	Yes	Yes	Yes	Yes
p-value		0.215		0.162
Y mean	0.0227	0.0291	0.0308	0.0399
Observations	3700	2680	2080	1480
R-squared	0.176	0.276	0.242	0.360

Notes: This table examines the spillover effect of the literary inquisition on innovation for the full and matched samples, respectively. Specifically, it presents estimates of the coefficient β from Equation (1) for the two sub-samples constructed from the full and matched samples. As discussed in Section 5.6, we construct two sub-samples to perform the DID analysis to investigate the potential spillover effect. Sub-sample I includes all treatment prefectures and closely located control prefectures within 60 km from the borders of the treatment prefectures. Sub-sample II includes all treatment prefectures and distant control prefectures over 60 km from the borders of the treatment prefectures. Baseline controls are the same as those in Column 4 of Table 2. Standard errors in parentheses are clustered at the prefectural level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Estimates of the Dynamic Effects of the Literary Inquisition on Land Tax Revenue

	Dependent variable: ln (land tax revenue)			
	Years since the event	Coefficients	Years since the event	Coefficients
	[–50, –40)	0.00959	[20, 30)	–0.0485
		(0.0508)		(0.0353)
	[–40, –30)	–0.0120	[30, 40)	–0.0804*
		(0.0380)		(0.0421)
	[–30, –20)	–0.0150	[40, 50)	–0.101**
		(0.0253)		(0.0408)
	[–20, –10)	–0.0111	[50, 60)	–0.118***
		(0.0123)		(0.0406)
	[0, 10)	–0.00588	[60, 70)	–0.127***
		(0.0130)		(0.0405)
	[10, 20)	–0.0262		
		(0.0254)		
Prefecture FE		Yes		Yes
Time FE		Yes		Yes
Province FE × Time FE		Yes		Yes
Tandingrumu		Yes		Yes
Land area × Time FE		Yes		Yes
Basin HHI × Time FE		Yes		Yes
Rice × Time FE		Yes		Yes
Fox millet × Time FE		Yes		Yes
Sweet potato × Time FE		Yes		Yes
Jinshi × Time FE		Yes		Yes
Pop density 1630 × Time FE		Yes		Yes
Language polarization × Time FE		Yes		Yes
Language fragmentation × Time FE		Yes		Yes
Coastal × Time FE		Yes		Yes
Main river × Time FE		Yes		Yes
Observations				3078
R-squared				0.601

Notes: This table presents the coefficients visualized in Figure 11. Except for the prefecture fixed effects, time fixed effects, and interaction of province and time fixed effects, we also control for a series of interaction terms of the time fixed effects and the following prefectural characteristics in the regressions: Land area, Basin HHI (basin fragmentation index), agricultural suitability metrics for wetland rice, fox millet, and sweet potato, *Jinshi* (number of *jinshi* in 1600–1710), Population density (population density in 1630), and two geographic dummies: Coast (whether a prefecture is situated on the coast), Main river (whether a prefecture includes at least one major river). Additionally, we control for the potential impact of the *Tandingrumu* policy by including a dummy variable $Tandingrumu_{pt}$ in the regression. Standard errors in parentheses are clustered at the prefectural level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Online Appendix for

Freedom of Speech, Spirit of Innovation, and Long-term Economic Development: Evidence from the Qing Dynasty of China

Benlai Wang[1], Xuebo Wang[2]

February 2024

[1] School of Economics, Shanghai University of Finance and Economics.
[2] School of Economics, Shanghai University of Finance and Economics. E-mail: wang.xuebo@mail.shufe.edu.cn.

A1. Data on Scientific and Technological Innovation

As shown in the main text, we use the data on scientific and technological innovation at the prefectural level in 1700–1800 to conduct our empirical analysis. The data are sourced from the book "History of Science and Technology in China (Chronology Volumes)" (Ai and Song, 2006), which documents important events in scientific and technological activities throughout Chinese history. We identify the events that reflect important advancements in science and technology (S&T) as innovations.

Figure A1 illustrates a sample of the records on scientific and technological events from which we extract the relevant information on innovations. This record includes essential information on the corresponding innovation, such as the time when the innovation was made (1705), the author of the innovation (Wending Mei), the title of the innovation (*ping san jiao ju yao*, a publication on mathematics), and a comprehensive description of the innovation itself. However, the record does not disclose the site where the innovation was made. In practice, we track the life of the innovator to determine his or her residence at the time of the innovation, attributing this location as the innovation site.

Our analysis includes 102 notable innovations in the 1700–1800 period. Among these innovations, approximately 80% (or 82 innovations) are scholarly publications on science and technology, encompassing fields such as mathematics, physics, astronomy, medicine, and geography. The remainder primarily comprises technological breakthroughs, encompassing enhancements in cultivation techniques, novel disease treatments, advancements in water engineering, and innovative solutions for road and bridge construction. Collectively, these innovations represent significant strides in S&T in China in this period.

In Figures 1 and 2 of the main text, we compare scientific and technological innovations in China and Europe (Britain) in the 1670–1790 period. Given that the data on innovations in China and Europe are sourced from two different books, people may doubt whether they are comparable. We now provide more information on the data on innovations in China and Europe to confirm that the data are reliable to a large extent.

As shown in the main text, the data on scientific and technological innovations in China and Europe are from Ai and Song (2006) and Ito et al. (1984), which document important events of

scientific and technological activities in Chinese and European history, respectively. We identify the events that reflect important advancements in S&T as innovations. Given that all important historical events that reflect significant advancements in S&T in China and Europe were recorded in the two books, our data on innovations in China and Europe in the 1650–1790 period largely reflect the S&T development trends in the two regions in this period.

Table A1 presents the compositions of scientific and technological innovations (by field) in China and Europe in the 1650–1790 period. We roughly divide these innovations into two categories: science and technology. Panel A shows that in Europe the shares of innovations on science and technology are 59% and 41%, respectively. In China the corresponding statistics are 75% and 25%, respectively. Panel A also presents these statistics for Europe and China before and after 1710. The shares of innovations on technology increased in both regions after 1710. While this increase in the share of technological innovations in Europe may reflect the surge of technological development before and during the Industrial Revolution, such an increase in China may be because that technological activities were less affected by the literary inquisition at the time.

Panels B and C further present the compositions of scientific and technological innovations by field in Europe and China, respectively. Specifically, we divide scientific innovations into the following categories based on the fields that they belong to: physics, biology, medicine, mathematics, chemistry, astronomy, geography, geology, and others. Similarly, we divide technological innovations into the following categories: mechanical engineering, chemical engineering, civil engineering, instrument manufacturing, ironmaking technology, materials engineering, military technology, agricultural technology, and others. Generally, most scientific innovations in Europe were on physics, biology, medicine, mathematics, and chemistry. By contrast, most scientific innovations in China were on medicine, mathematics, geography, and geology. When it comes to technological innovation, the dominant fields in Europe and China were mechanical engineering and agricultural technology, respectively.

Panels B and C of Table A1 also show that although scientific and technological innovations in China decreased sharply after the literary inquisition, innovations on some fields seem less affected by it. Specifically, the share of innovations on medicine in science and the share of innovations on

agricultural technology in technology increased significantly after 1710, indicating that medical and agricultural activities may be less affected by the literary inquisition. Although the literary inquisition may have had a significant negative impact on innovation in most scientific and technological fields, not all fields were equivalently susceptible to it.

Table A2 presents some examples of typical scientific and technological innovations in various fields in Europe and China in the 1650–1790 period. The comprehensive lists of all innovations in Europe and China in this period are available upon request.

People may still doubt that our measure of innovation may not be perfect and the obtained indicators of scientific and technological innovation in Europe and China may not be comparable at the time. We acknowledge that there may not be perfect measure of scientific and technological innovation hundreds of years ago. However, as discussed in the main text, we do not require that the number of innovations in a certain period in China or Europe accurately reflects their absolute level of science and technology at the time. In particular, if our definition and measure of innovation are constant across regions and over time, then our data on innovations for a certain period can largely reflect the development trends of science and technology in China and Europe in the period. Therefore, if the numbers of innovations are comparable across regions and over time, then it would not be a problem, particularly for our identifications of the impact of the literary inquisition on innovation within China.

A2. Data on the Literary Inquisition Cases

As noted in the main text, our primary data source on the literary inquisition is *Qingdai wenziyu dang (zengding ben)* (Qing Literary Inquisition Archives (Updated Edition)) (Wanyan, 2011). The book is a revised edition of *Qingdai wenziyu dang ji* (Qing Literary Inquisition Archives) (Zhang, 1934), compiled by historians using resources from the Qing Imperial Archives in the 1930s.

Zhu (2013) highlights three major advantages of the *Qingdai wenziyu dang (zengding ben)* as a data source on the literary inquisition:

1. Reliability of Qing Official Archives: The book sources its information from Qing official archives, including the Grand Council (*junji chu*)'s archives, palace memorials, and *Qing Shilu*. These official archives represent the most reliable source of historical records on the Qing literary inquisition.

2. Detailed Documentation: It provides exhaustive details on each case of literary inquisition, including the cause and course of the investigation, the emperor's response, the verdict, and information on the accused.
3. High Value as Primary Historical Material: The records in the book comprehensively capture the full scope of the literary inquisition cases, thereby providing invaluable first-hand historical documentation.

Xue (2021) uses *Qingdai wenziyu an* (Qing Literary Inquisition Cases) (Zhang and Du, 1991) as her data source on the literary inquisition. This book was published 20 years earlier than *Qingdai wenziyu dang (zengding ben)*, and it includes 65 cases from the *Qingdai wenziyu dang ji* and additional 21 cases from other sources such as intellectuals' notes. Records from different sources may not be comparable. In particular, intellectuals' notes on the literary inquisition may be subjective to some extent and thus not comparable to those official records. These facts cast shadow on the credibility of the records from this book. Additionally, the *Qingdai wenziyu dang (zengding ben)* documents ten cases not recorded in *Qingdai wenziyu an*, indicating its broader scope in historical documentation.

Reassuringly, the difference between the two data sources is relatively small, and our results are not sensitive to the choice of the data source on the literary inquisition. We conduct our empirical analysis using Xue's data on the literary inquisition and present the results in Panel A of Table A4. Both the estimates for the full and matched samples remain similar to those baseline results presented in Tables 2 and 3 in the main text. Additionally, we merged the two data sources to include all cases documented in both books, resulting in a final sample of 96 cases. Panel B of Table A4 presents the obtained estimates with this merged sample, which are also similar to the baseline estimates in the main text. Thus, our results remain robust across various specifications using alternative datasets on the literary inquisition.

A3. Data on Land Tax Revenue

Due to a lack of annual land tax revenue data for all prefectures in the Qing Dynasty, we had to construct a balanced panel data set on land tax revenue in which time intervals between prefectural observations are constant for our empirical analysis. We now demonstrate how to construct such a balanced panel data set on the land tax revenue at the prefectural level in the 1680–1850 period.

We first collect the data on the land tax revenue at the prefectural level in different years from both provincial and national General Records (*tong zhi*) in the Qing Dynasty. The provincial General Records chronicle significant provincial events during certain periods, including land tax revenue of all of their subordinate prefectures. Likewise, the national General Records document land tax revenue for all prefectures across the country in certain years. Table A3 presents all the General Records we employ to construct the panel data on prefectural land tax revenue in the Qing Dynasty. The provincial General Records provide prefectural land tax revenue for over 18 provinces in certain years in 1667–1739, while the national General Records provide similar information for all prefectures across the country in 1743, 1784, and 1820.

To construct a balanced panel data on land tax revenue at the prefectural level, we estimate annual land tax revenue for each prefecture using the average annual growth rate between two observations. For instance, we have data on land tax revenue of all prefectures in Jiangxi province in 1724 and 1743, from which we could calculate the average annual growth rate of the land tax revenue of these prefectures during this period. With an underlying assumption that the prefectural land tax revenue grew at a constant rate during this period, we predict the prefectures' annual land tax revenue in each year in 1724–1743. Evidently, such an exercise may not be completely accurate. However, our study aims to identify the long-term impacts of the literary inquisition on land tax revenue in affected regions, and thus such inaccurate estimation of the land tax revenue during a relatively short period may not be a problem.

Unfortunately, after 1820, the data on land tax revenue was available only at the provincial level. We therefore calculate the average annual growth rates of all provincial land tax revenue for a certain period. We then use these growth rates to approximate those of prefectures within each province. Such a practice carries an underlying assumption that the land tax revenue of all prefectures within the same province experienced the same growth rate in this period. For instance, we have the data on the land tax revenue of Zhejiang province in 1820 and 1841, and we can calculate the average annual growth rate of the land tax revenue in the province during this period. By assuming that land tax revenue of all prefectures in Zhejiang province grew at the same rate to that of the province, we can obtain the estimated annual land tax revenue of those prefectures in the 1820–1841 period.

We acknowledge that this is not a perfectly accurate practice. Particularly, the land tax revenue of treatment prefectures affected by the literary inquisition may experience a lower growth rate than that of control prefectures after 1820. However, this may not be a problem for our identification. Specifically, the literary inquisition could have a negative effect on the land tax revenue in treatment prefectures in the long term, and thus the average growth rates of land tax revenue of treatment prefectures could be lower than that of control prefectures of the same province. Therefore, if we assume that treatment and control prefectures experienced the same growth rate in land tax revenue after 1820, it would be more difficult for us to identify a negative effect of the literary inquisition on land tax revenue in treatment prefectures. Therefore, under this circumstance, the DID estimates would be a lower bound of the treatment effect of the literary inquisition on land tax revenue in treatment prefectures, and the obtained significantly negative estimates would present strong evidence of the treatment effects.

Following the above practice, we finally obtain a decadal panel data set of land tax revenue at the prefectural level in the 1680–1850 period.

A4. Robustness Checks

A4.1. Estimation Results at the Year Level

As mentioned in the main text, considering that in our sample the time intervals between observations are 5 years and a prefecture was exposed to the literary inquisition in a specific year, we need to determine its exact treatment time. It may not be precise to simply determine that a prefecture was first exposed to the literary inquisition in a period if it began to be affected by the literary inquisition from any year of this period. For instance, considering an observation of a prefecture during 1731–1735, if this prefecture was exposed to the literary inquisition in 1735, then it was affected by the event for only less than 1 year during this period. Thus, under this scenario, we determine that the prefecture was exposed to the literary inquisition from the next period (or 1736). In the main text, we define that a prefecture was exposed to the literary inquisition during a period if its exposure occurs in the first 4 years of the period.

People may still be concerned that such a practice is not precise enough. Reassuringly, our results are not sensitive to the choice of the treatment time. We obtain similar results by conducting the

analysis at the year level. We define a prefecture's treatment time as the year following its exposure to the literary inquisition. We conduct the event-study analysis with the yearly panel and report the estimates at five-year intervals, averaging yearly estimates from the preceding five years. Similar to Bailey and Goodman-Bacon (2015), we estimate the following event-study version of the TWFE model:

$$\#Innovation_{pt} = \sum_{\tau} \beta_{\tau} literary\ inquisition_p \times D_p^{\tau} + Treat_p \times After1790_t + \theta X_p \times \gamma_t$$

$$+\lambda_p + \gamma_t + \delta_{prov} \times \gamma_t + \epsilon_{pt}. \qquad (A1)$$

where $\#Innovation_{pt}$ is the number of scientific and technological innovations in prefecture p in year t; $literary\ inquisition_p$ is a dummy indicating whether prefecture p was ever exposed to the literary inquisition; D_p^{τ} is a dummy that equals to 1 if prefecture p is observed in event-year group τ. These groups are categorized based on the number of years y since prefecture p's first exposure to the literary inquisition, with different ranges: $y < -30$, $-30 \leqslant y < -25$, $-25 \leqslant y < -20$, $\cdots$, $-10 \leqslant y < -5$, $0 \leqslant y < 5$, $\cdots$, $20 \leqslant y < 25$ and $y \geqslant 25$ ($-5 \leqslant y < 0$ is omitted). Other variables are same to those in Equation (2) in the main text. β_{τ} is the parameter of interest, which aims to capture magnitudes and joint statistical significance of the prefectural event-year estimates (Bailey and Goodman-Bacon, 2015).

We estimate Equation (A1) for the full and matched samples and demonstrate the results in Panels (A) and (B) of Figure A2. Both figures look similar to those presented in the main text, confirming the robustness of our results.

Furthermore, we estimate Equation (1) in the main text using the yearly panel data and report the estimates of β in Table A5. Theoretically, other things equal, these yearly estimates should be one-fifth of those obtain at five-year intervals, as presented in the main text.

Panels A and B of Table A5, across Columns (1) to (4), consistently show that the DID estimates change little upon incrementally adding control variables in the regressions, demonstrating robustness. After controlling for all relevant variables, Column (4) of Panel A presents that the coefficient is approximately -0.009, which indicates that the literary inquisition reduces the yearly number of innovations in the treatment prefectures by approximately 0.009. Considering that the mean of the yearly number of innovations for all prefectures during 1700–1800 is approximately 0.004, this

represents a substantial negative impact. Specifically, the mean of the yearly number of innovations in the treatment prefectures after treatment is approximately 0.009. This estimate indicates that the corresponding value would be 0.018 if these prefectures were not exposed to the literary inquisition. That is, the literary inquisition reduced the yearly number of innovations in the treatment prefectures from the potential value of 0.018 to an actual value of 0.009, indicating a reduction of 50.0%, a magnitude close to our baseline results in Column (4) of Table 2 in the main text (50%). Similarly, Panel B of Table A5 indicates that the literary inquisition reduced the yearly number of innovations in the treatment prefectures by 0.012, or 57.1% in percentage, which is also similar in magnitude to the baseline estimate in Column (4) of Table 3 in the main text (57.8%). In summary, our results are still robust for analysis at the year level.

A4.2. Controlling for the Impacts of Other Historical Events

As mentioned in the Section 5 in the main text, other historical events in the 18th century may have also had important impacts on scientific and technological innovation at the time. We will discuss these events in details and control for their potential impacts on innovation in our DID framework.

A4.2.1 The Expulsion of the Jesuits

The first notable historical event is the expulsion of the Jesuits in the 1720s. Emperor Yongzheng launched a policy in 1723 to expel the Jesuits, which may also have negatively impacted scientific and technological activities in China at the time. Ma (2021) highlights that Jesuit scientists introduced European science to China from 1580 onward, resulting in a significant increase in scientific works within prefectures hosting Jesuit scientists. However, this knowledge diffusion in China facilitated by the Jesuits ceased in the 1720s, when the emperor began to expel them owing to the Chinese Rites Controversy with the Pope. Undoubtedly, the expulsion of the Jesuits may have also hindered knowledge diffusion and scientific and technological innovations at the time. We now control for the potential impact of this expulsion policy on innovation in our DID framework.

Ma (2021) categorizes the Jesuits in China into two groups. (1) Jesuit scientists engaged in scientific activities while preaching. (2) Jesuit priests exclusively conducted missionary work. Presumably, only the former would have fostered scientific advancements in the regions they influenced. Following Ma's approach, we include a series of control variables in Equation (1) in the

main text, comprising interaction terms between the variables of Jesuit scientists and a dummy variable representing the period of Jesuit expulsion (1721–1780). By doing so, we largely control for the potential impact of the Jesuit expulsion on innovation. In particular, variables associated with Jesuit scientists encompass their presence and number. The former indicates whether Jesuit scientists were present in a prefecture, while the latter is the aggregation of all Jesuit scientists for the decades of their presence in a prefecture preceding the Jesuit expulsion. To further control for the potential impact of the Jesuit expulsion, we also control for interaction terms between the variables of Jesuit priests and Jesuit expulsion dummy. The corresponding estimates of β in Equation (1) in the main text with these controls are presented in Table A6. Columns (1)–(4) of Table A6 show that the estimates change little compared with the baseline results in Table 2 as we add the aforementioned controls in the regressions. These results indicate that the Jesuit expulsion does not contaminate our DID estimates of the effects of the literary inquisition on innovation in the treatment prefectures.

A4.2.2 The Sea Ban Policy

The second significant event is the sea ban policy implemented by the Qing government after 1650, which may also have affected domestic scientific and technological activities. The Qing rulers began to strictly enforce a ban on maritime trade from 1655, and coastal regions were profoundly impacted by this policy. The ban was initiated to cut off communication between domestic residents and overseas anti-Qing forces and to consolidate the Qing regime. Three decades later, Emperor Kangxi abolished the sea ban policy when anti-Qing forces were defeated and the Qing regime was consolidated (Guo, 1984). However, the rapid development of international trade worried the Qing rulers that such uncontrolled maritime trade may threaten the Qing regime, leading them to reimpose the sea ban policy in the 18th century. In 1717–1727, the Qing government implemented the sea ban policy in Guangdong and Fujian provinces, banning residents there to go to sea and trade with other countries. In 1757, the Qing government began to ban all maritime trade throughout the country, and only kept Guangzhou port as a window to foreign countries. Consequently, maritime trade in 1757–1842 was strictly forbidden across the country except for Guangdong Province.

We control for the potential impact of the sea ban policy in the 18th century on innovation by adding corresponding controls in Equation (1) in the main text. Specifically, we control for interaction

terms between the dummies of the two periods of the sea ban policy (i.e., 1717–1727 and 1757–1800) and dummies of the corresponding affected regions in these periods to absorb the potential effects of the policy on innovation (i.e., "Coast I× Sea Ban (1717–1727)" and "Coast II× Sea Ban (1757–1800)"). The variable "Sea Ban (1717–1727)" is a dummy of 1717–1727 period, when the sea ban policy was enforced. "Coast I" is a dummy of coastal prefectures in Guangdong and Fujian provinces affected by this policy. This interaction term is supposed to capture the potential impact of the sea ban policy during 1717–1727 on innovation in the affected regions. Similarly, "Sea Ban (1757–1800)" is a dummy of the 1757–1800 period, when the other sea ban policy was in place. "Coast II" is a dummy of coastal prefectures (excluding Guangzhou) affected by this policy. This interaction term is intended to capture the potential impact of the post-1757 sea ban policy on innovation in the affected regions. Column (5) of Table A6 presents that our DID estimator remains similar to the baseline results after including the two interaction terms.

A4.2.3 The *Tandingrumu* Policy

The third historical event is the *Tandingrumu* policy, a significant tax reform implemented in the Yongzheng reign. This reform, transitioning from a poll tax to a property tax, fundamentally altered the tax structure of the Qing Dynasty. Such a change had the potential to impact the geographical distribution of the tax burden, which could, in turn, affect regional scientific and technological activities and land tax revenue.

In the early Qing period, the labor tax and land tax were levied separately. Such a tax system gave rise to many problems and exacerbated the tax burden on peasants (Wang, 2020). In response, the Qing government initiated the *Tandingrumu* policy, simplifying tax assessment by merging labor and land taxes into a single tax collected in silver (Zelin, 1984; Naquin and Rawski, 1987). This reform meant that the total tax owed by households became dependent solely on their cultivated land area (Wang, 2020). The policy was rolled out at the provincial level, and Table A7 demonstrates the specific timings across provinces (Wang, 2020).

To control for the potential impact of the *Tandingrumu* policy on innovation, we introduce an additional control variable in Equation (1) in the main text. Specifically, we construct a dummy variable $Tandingrumu_{pt}$, which equals 1 if prefecture *p* had implemented the policy by time *t*. This

variable aims to capture the potential impact of the *Tandingrumu* policy on regional innovation. Column (6) of Table A6 presents that our DID estimate changes little after we include this dummy variable in the regression. Such result suggests that the effect of the *Tandingrumu* policy on innovation does not significantly confound our analysis of the literary inquisition's impact.

A4.2.3 The Changes to the Civil Service Examination System

For more than 1300 years, China recruited its elites with a civil service examination system, and the exam system had profound societal influences, particularly on bureaucracy efficiency, elite circulation, talent allocation, and perceptions of social mobility (Weber, 1915; Chang, 1962; Qian, 1982; Elman, 2000; Bai and Jia, 2016). The Qing government implemented several reforms on the civil exam system in the 18th century, and such institutional changes could potentially impact regional innovation at the time. This section briefly discusses these reforms and evaluates the robustness of our DID estimates after controlling for their potential impacts on innovation.

A4.2.3.1 The Quota System Reform in 1724

The quota system, pivotal in the civil service examination, determined the number of successful candidates at each exam level, including prefecture and province levels (Bai and Jia, 2016). During the early Qing period, these quotas varied over time. In 1724, Emperor Yongzheng reformed the system, setting fixed quotas at the prefectural level, and this prefectural level quota persisted until 1851 (Bai and Jia, 2016). This reform could potentially affect regional innovation, particularly in prefectures with higher quotas.

To control for the potential impact of this reform on innovation, we introduce an additional control variable in Equation (1) in the main text. We construct an interaction variable $Quota_p \times Reform1724_t$, where $Quota_p$ is the assigned quota for prefecture p in 1724, and $Reform1724_t$ is a time dummy that equals 1 if $t > 1724$. This interaction term aims to absorb the potential impact of the quota system reform on innovation in affected regions. Column (7) of Table A6 presents that our DID estimate remains similar to the baseline estimate after including this interaction term in the regression, confirming the robustness of our results.

A4.2.3.2 Two Revisions to the Examination Content

The Qing government made two significant content revisions to the civil service examination in the 18th century. The first, in 1756, introduced a poetry composition requirement in eight-rhyme regulated verse. The second revision in 1787 mandated specialization in all Five Classics, as opposed to just one previously (Elman, 2013). These changes increased the exam's difficulty and may have pushed candidates to spend more time learning to write poetry and specializing in all of the *Five Classics*. This could potentially divert their efforts away from developing human capital for modern science, ultimately leading to a decline in S&T in Qing China. Prefectures with higher quotas, typically more developed and intellectually vibrant, might have been more affected.

We control for the potential impacts of these revisions by incorporating two additional interaction terms, $Quota_p \times keju1756_t$ and $Quota_p \times keju1783_t$, in Equation (1) in the main text. Similar to the above controls, $Quota_p$ is the assigned quota for prefecture p in 1724, and $keju1756_t$ and $keju1783_t$ are time dummies indicating years post-1756 and post-1783, respectively. These two interaction terms are intended to capture these revisions' potential impacts on regional innovation. Columns (8) and (9) of Table A6 present that our DID estimates change little after including the inclusion of these terms in the regressions, affirming the robustness of our results.

A4.3. Estimation Results for the Restricted Sample of the Literary Inquisition Cases

Qingdai wenziyu dang (zengding ben) documents a total of 90 investigated literary inquisition cases in the Qing Dynasty. The default categorization in this compilation considers any case examined by the authorities as a literary inquisition, regardless of the final sentence. This broad definition is generally reasonable. Specifically, if a case is officially filed for investigation, then all implicated individuals were imprisoned, waiting for the final sentences. Thus, the investigation itself was already a severe punishment for the suspects and has a deterrent effect on the public, justifying its classification as a literary inquisition.

Nonetheless, this categorization does have its limitations. Some cases concluded with acquittals, with the suspects being released. The classification of these instances as literary inquisitions is debatable. From the perspective of the final sentences, these suspects were not penalized, and the resulting deterrent effect on the public might be comparatively diminished.

To examine the robustness of our results, we apply more stringent criteria for defining a literary inquisition, focusing on cases involving legal penalties for offensive speech. Consequently, cases ending in acquittals are not considered literary inquisitions under this stricter definition. In this way, we obtain a restricted sample of 65 literary inquisition cases by including only those cases where the suspects were ultimately found guilty and subjected to severe punishment. These cases were undoubtedly literary inquisitions and should have exerted huge impacts on society at the time.

We examine whether our results remain robust for this restricted sample of the literary inquisition cases. Table A8 presents the estimation results using this restricted sample. Panels A and B of Table 8 report the estimates for the full and matched samples, respectively. These estimated coefficients are slightly larger than the baseline results shown in the main text, and all are significant. Such results indicate that our results are not sensitive to the criteria used for defining the literary inquisition.

A5. The Propensity Score Matching Approach

As discussed in the main text, we employ matching to reduce heterogeneity in the sample. After matching, there are no longer systematic differences between treatment and control prefectures prior to the treatment and the DID estimators are likely to capture the treatment effects of the literary inquisition on innovation.

In practice, we generate a propensity score for each prefecture by estimating:

$$Prob\big(Literary\ Inquisition_p = 1\big) = F(X_i) \qquad (A2)$$

where $Prob$ is the probability that a prefecture was exposed to the literary inquisition and X_i is the vector of covariates. We include all baseline controls in Column (4) of Table 2 in the main text in our set of covariates.

We adopt the single nearest-neighbor match strategy. This strategy requires us to select an appropriate caliper width to complete the matching, and such a selection needs to strike a balance between generating close matches for efficiency and avoiding the exclusion of too many observations (Lunt 2014). We select the proper caliper width on the basis of Cochran and Rubin (1973). Cochran and Rubin (1973) explore the rules of caliper width choices under different ratios of the variance of the propensity score in the treated group (σ_1^2) to that in the untreated group (σ_2^2). Their results suggest that given the same caliper width, the smaller σ_1^2/σ_2^2 ratio, the more bias can be reduced. If σ_1^2/σ_2^2

equals 2, a 0.2 standard deviations of the propensity score can effectively reduce bias by 98%. In our case, the σ_1^2/σ_2^2 ratio is smaller than 2 ($\sigma_1^2/\sigma_2^2 = 1.93$), and thus we select a caliper width of 0.2 standard deviations of the propensity score (0.04), which can effectively reduce bias by more than 98%.

Table A9 presents the balance of the geographical, economic, and human capital characteristics across the treatment and control prefectures, both before and after matching. Panel A of Table A9 clearly shows significant differences in these observable characteristics between the two groups before matching. However, as shown in Panel B of Table A9, we obtain a balanced sample with no observable differences between the treatment and control prefectures after matching. We now use this matched sample, in which the treatment and control prefectures are similar prior to the treatment, to estimate the effects of the literary inquisition on innovation in the treatment prefectures.

Figure A4 presents the distributions of the propensity scores for treatment and control prefectures before and after matching with a caliper width of 0.04. Evidently, the matching significantly improves the similarity between the treatment and control prefectures. Consistent with Table A9, treatment and control prefectures show no significant differences in observable characteristics after matching.

A6. Robustness Checks for the Estimates of the Spillover Effect

As discussed in the main text, when examining the potential spillover effect of the literary inquisition on innovation, we select a distance of 60 km as the cutoff between the close and distant control prefectures. We also stress that our results are not sensitive to the selection of this cutoff distance, and they remain robust if we select 30 km or 90 km as the cutoff distance. Panels A and B of Table A10 present the corresponding results if we select 30 km and 90 km as the cutoff distance, respectively. These results remain similar to the benchmark results presented in Table 4 in the main text, confirming the robustness of our results on the literary inquisition's spillover effect on innovation.

As discussed in the main text, we could alternatively define the close control prefectures as those adjacent to the treatment prefectures and the distant prefectures as those located far from the treatment prefectures' borders. Such distinction enhances the precision in detecting the potential spillover effects. Our examination of the data reveals that approximately 50% and 25% of control prefectures lie beyond 800 and 1000 km from the treatment prefectures' borders. Thus, we construct three sub-samples to perform the DID analysis: Subsample III, encompassing all treatment prefectures and their neighboring

control prefectures; Subsample IV, including treatment prefectures and control prefectures situated more than 800 km away; and Subsample V, comprising treatment prefectures and even more distant control prefectures, beyond 1000 km. Due to the limited number of control prefectures adjacent to the treatment ones in our matched sample, this analysis is confined to the full sample. The obtained results, presented in Table A11, reveal that the estimates from Subsamples IV and V are significantly larger (in absolute value) than that from Subsample III, with a *p*-value around 0.06. These results present strong evidence of the literary inquisition's spillover effect on innovation.

A7. Procedure for Obtaining the CATE Using Causal Forrest Method

As noted in the main text, the causal forest method is a recently developed supervised machine learning technique used for predicting heterogeneity in causal treatment effects (Athey and Imbens, 2016; Wager and Athey, 2018; Athey et al., 2019; Britto et al., 2022). The goal of the method is to estimate the Conditional Average Treatment Effect (CATE) for each prefecture. The CATE is defined as $E(Y_{1i} - Y_{0i}|X_i = x)$, where Y_{1i} and Y_{0i} denote the potential outcomes of interest for the *i*th prefecture when treated and untreated, and X is a vector of observable characteristics (Britto et al., 2022).

Athey et al. (2019) provide a guideline to the implementation of the causal forest. Given that our difference-in-differences setting is different from the common application based on randomized control trials (e.g., Davis and Heller, 2017; Bertrand et al., 2017), we follow Britto et al. (2022) and run the causal forest over first-differences. In this way, the treatment group indicator is orthogonal to the covariates, and thus the unconfoundness assumption in Wager and Athey (2018) is satisfied (Britto et al., 2022). Similar to Britto et al. (2022), we match each treatment prefecture with a control one and assign the treatment time of the treatment prefecture to the matched control prefecture as a placebo treatment time. The matching procedure is same to that in Section A5. We compute the innovation levels (measured by the average number of S&T innovations per five years during a period) for each prefecture both before and after the treatment. By calculating the difference between the post- and pre-treatment innovation levels within each prefecture, we generate the first-difference variable. This variable is intended to reflect the change in innovation levels resulting from the treatment and is employed as the outcome variable in the causal forest analysis.

As shown in Britto et al. (2022), the algorithm of the causal forest begins by building trees, each formed through data-driven splits in the sample that define distinct leaves. We set the minimum number of treatment and control observations allowed in a leaf to the default value (5). These leaves are then used to estimate the causal effect of a treatment, based on the specific characteristics (X) of each subgroup (Britto et al., 2022). The characteristics X included into the causal forest consist of all the control variables in Column (4) of Table 2 in the main text and the province-fixed-effects dummies. To avoid over fitting, we follow Britto et al. (2022) and adopt the "honest approach". Specifically, we randomly split the sample in two equal parts, with one part being used to define the sample splits (leaves) and the other part being used for estimating the predicted CATE. We repeat the procedure multiple times and ultimately build 2000 trees (the default value), with the final causal forest prediction being a weighted average over the predictions in each tree. Upon completion, the causal forest method yields the predicted Conditional Average Treatment Effect (CATE) for each prefecture. Figure 8 in the main text present the obtained results.

A8. More Robust DID Estimation Results

A8.1. More Results on the Effect of the Literary Inquisition on Innovation

As discussed in the main text, the TWFE model yields consistent estimates only under strong assumptions that treatment effects are homogeneous across treated groups and over time. We thus replicate our baseline results demonstrated in the main text using the robust estimators proposed by Borusyak et al. (2021), Callaway and Sant'Anna (2021), De Chaisemartin and d'Haultfoeuille (2020), and Sun and Abraham (2021). These estimators, which shut down the 2×2 difference in differences comparisons between newly-treated and already-treated units, are designed to be consistent even in the presence of heterogeneous treatment effects across treated units and over time.

Table A12 presents these alternative DID estimators on the effect of the literary inquisition on innovation for the full and matched samples, respectively. Panel A of Table A12 shows that the robust DID estimators for the full sample are generally larger than those baseline TWFE estimates presented in Table 2 in the main text. Although the robust estimators obtained from Callaway and Sant'Anna (2021) and De Chaisemartin and d'Haultfoeuille (2020) are not precisely estimated, the other

estimators are all significant at the 1% level, which present strong evidence that our baseline results in the main text are robust to a large extent.

Panel B of Table A12 shows that the robust DID estimators for the matched sample are similar to the corresponding baseline TWFE estimates presented in Table 3 in the main text. Once more, while the robust estimators obtained from Callaway and Sant'Anna (2021) and De Chaisemartin and d'Haultfoeuille (2020) are not precisely estimated, the other two robust estimators are precisely estimated and significant, which confirms the robustness of our baseline results presented in the main text.

A8.2. More Results on the Impact of the Literary Inquisition on Land Tax Revenue

In the main text, we estimate an event-study version of the TWFE model and present the results in Figure 11, which verifies that the common trends assumption for our DID strategy is likely to hold. However, the fully dynamic version of the TWFE model delivers consistent estimates only when the treatment effects are homogeneous across groups and over time (Sun and Abraham 2021). We thus further present the event-study figures generated by a set of recently developed robust estimators in the presence of treatment heterogeneity (De Chaisemartin and d'Haultfoeuille, 2020; Borusyak, Jaravel, and Spiess, 2021; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). Figure A5 demonstrates the obtained results. Figure A5 looks very similar to Figure 8 in the main text, which confirms the robustness of our results.

We now estimate the TWFE model to investigate the overall effects of the literary inquisition on the land tax revenue in the treatment prefectures. Specifically, we estimate the following equation:

$$\ln\left(LandTax_{pt}\right) = \beta Literary\ Inquisition_{pt} + \theta X_p \times \gamma_t + Tandingrumu_{pt} + \lambda_p + \gamma_t + \delta_{prov} \times \gamma_t + \epsilon_{pt} \qquad (A3)$$

This equation is similar to Equation (1) in the main text, except that here the dependent variable $ln\ (LandTax_{pt})$ is the logarithm of land tax revenue in prefecture *p* in time *t.*

Table A13 reports the estimates of β obtained from Equation A3. Columns (1) – (4) of Table A13 present that the estimates change little as we gradually add control variables in the regressions. Specifically, the estimates are around -0.08 and not significant. Given that the literary inquisition may only negatively affect the land tax revenue in the treatment prefectures in the long term, it is not difficult to explain such insignificant DID estimates which capture the overall effects of the literary inquisition on land tax.

Similar to Table A12, Table A14 presents more robust DID estimators of the impact of the literary inquisition on land tax revenue that allow for heterogenous treatment effects. Although one of these

estimators are not precisely estimated and thus insignificant, the other three estimators are negative and significant, and the estimates are between -0.05 and -0.09. These results indicate that the literary inquisition may have a modest negative effect on the land tax revenue in the affected regions.

References

Ai, Suzhen, and Zhenghai Song. 2006. *Zhongguo kexue jishu shi: nianbiao juan (Science and Technology in China: Chronological Volumes)*. Beijing: Science Press.

Athey, Susan, and Guido Imbens. 2016. "Recursive Partitioning for Heterogeneous Causal Effects." *Proceedings of the National Academy of Sciences* 113 (27): 7353–60.

Athey, Susan, Julie Tibshirani, and Stefan Wager. 2019. "Generalized Random Forests." *The Annals of Statistics* 47 (2): 1148–78.

Bai, Ying, and Ruixue Jia. 2016. "Elite Recruitment and Political Stability: The Impact of the Abolition of China's Civil Service Exam." *Econometrica* 84 (2): 677–733.

Bailey, Martha J., and Andrew Goodman-Bacon. 2015. "The War on Poverty's Experiment in Public Medicine: Community Health Centers and the Mortality of Older Americans." *American Economic Review* 105 (3): 1067–1104.

Bertrand, Marianne, Bruno Crépon, Alicia Marguerie, and Patrick Premand. 2017. "Contemporaneous and Post-Program Impacts of a Public Works Program: Evidence from Côte d'Ivoire." World Bank.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2021. "Revisiting Event Study Designs: Robust and Efficient Estimation." Unpublished.

Britto, Diogo G. C., Paolo Pinotti, and Breno Sampaio. 2022. "The Effect of Job Loss and Unemployment Insurance on Crime in Brazil." *Econometrica* 90 (4): 1393–1423.

Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-differencess with Multiple Time Periods." *Journal of Econometrics* 225 (2): 200–230.

Chang, Chung-li. 1962. *The Income of the Chinese Gentry*. Seattle, WA: University of Washington Press.

Cochran, William G., and Donald B. Rubin. 1973. "Controlling Bias in Observational Studies: A Review." *Sankhyā: The Indian Journal of Statistics, Series A (1961-2002)* 35, no. 4: 417–46.

Davis, Jonathan M. V., and Sara B. Heller. 2017. "Using Causal Forests to Predict Treatment Heterogeneity: An Application to Summer Jobs." *American Economic Review* 107 (5): 546–50.

De Chaisemartin, Clément, and Xavier D'Haultfoeuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110 (9): 2964-96.

Elman, Benjamin A. 2000. *A Cultural History of Civil Examinations in Late Imperial China*. Berkeley, CA: University of California Press.

Elman, Benjamin A. 2013. *Civil Examinations and Meritocracy in Late Imperial China*. Cambridge, Massachusetts London, England: Harvard University Press.

Guo, Yunjing. 1984. *Qingdai jingjishi jianbian 1644-1840 (A Brief Economic History of the Qing Dynasty 1644-1840)*. Zhengzhou: Henan Renmin Press.

Ito, Shuntaro, Kenzo Sakamoto, Keiji Yamada, and Yoichiro Murakami. 1984. *Jianming shijie kexue jishu shi nianbiao (A Concise Chronology of World Science and Technology History)*. Translated by Zhenhuan Jiang and Guanxiong Ge. Harbin: Harbin Institute of Technology Press.

Lunt, Mark. 2014. "Selecting an Appropriate Caliper Can Be Essential for Achieving Good Balance with Propensity Score Matching." *American Journal of Epidemiology* 179 (2): 226-235.

Ma, Chicheng. 2021. "Knowledge Diffusion and Intellectual Change: When Chinese Literati Met European Jesuits." *The Journal of Economic History* 81 (4): 1052–97.

Naquin, Susan, and Evelyn S. Rawski. 1987. *Chinese Society in the Eighteenth Century*. Yale University Press.

Qian, Mu. 1982. *Traditional Government in Imperial China: A Critical Analysis*. Hong Kong: Chinese University Press.

Sun, Liyang, and Sarah Abraham. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics* 225 (2): 175–99.

Wager, Stefan, and Susan Athey. 2018. "Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests." *Journal of the American Statistical Association* 113 (523): 1228–42.

Wang, Yuchen. 2020. "The Implications and Historical Insights of the Tandingrumu Policy (in Chinese)." *Taxation Research*, no. 02: 139–44.

Wanyan, Shaoyuan, ed. 2011. *Qingdai wenziyu dang (zengding ben) (Archives of the Literary Inquisition Cases (Updated Edition))*. Shanghai: Shanghai Bookstore Publishing House.

Weber, Max. 1915. *The Religion of China: Confucianism and Taoism*. Glencoe Ill: Free Press. Translated in 1951.

Xue, Melanie Meng. 2021. "Autocratic Rule and Social Capital: Evidence from Imperial China." Working Paper.

Zelin, Madeleine. 1984. *The Magistrate's Tael: Rationalizing Fiscal Reform in Eighteenth Century Ch'Ing China*. First Edition. Berkeley: University of California Press.

Zhang, Ji, ed. 1934. *Qingdai wenziyu dang ji (Archives of the Literary Inquisitions Cases (Collected Volumes))*. Beijing: The Palace Museum.

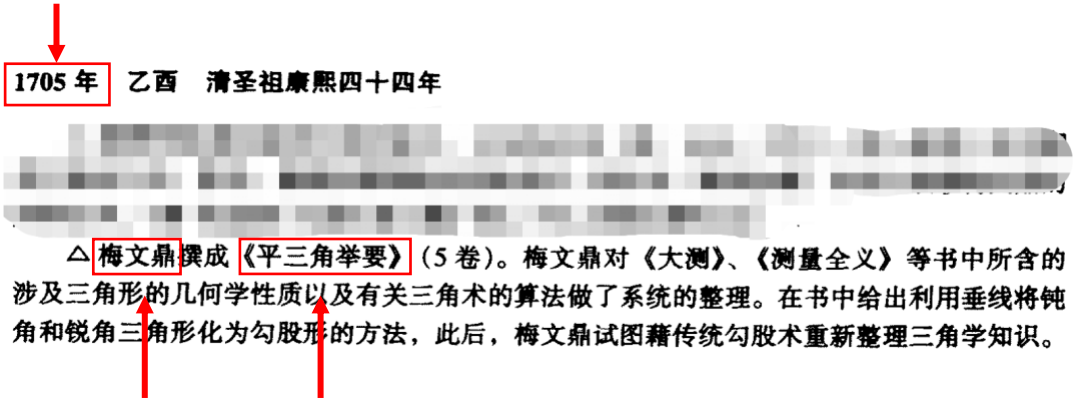
Zhang, Shucai, and Jinghua Du. 1991. *Qingdai wenziyu an (Qing Literary Inquisition Cases)*. Beijing: The Palace Museum.

Zhu, Yang. 2013. "The Value of 'Files of Literary Inquisition of Qing Dynasty' as Historical Data (in Chinese)." *Journal of Harbin University* 34 (06): 98–101.

B. Additional Figures and Tables

Figure A1: A Sample of the Records on Innovation in Ai and Song (2006)

Time when the innovation was made

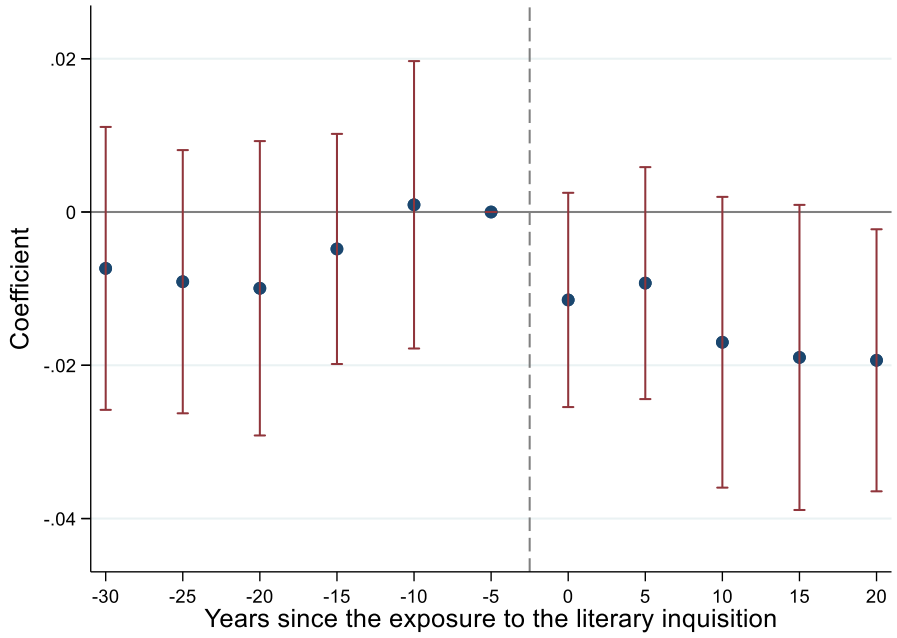

1705 年　乙酉　清圣祖康熙四十四年

△梅文鼎撰成《平三角举要》（5 卷）。梅文鼎对《大测》、《测量全义》等书中所含的涉及三角形的几何学性质以及有关三角术的算法做了系统的整理。在书中给出利用垂线将钝角和锐角三角形化为勾股形的方法，此后，梅文鼎试图藉传统勾股术重新整理三角学知识。

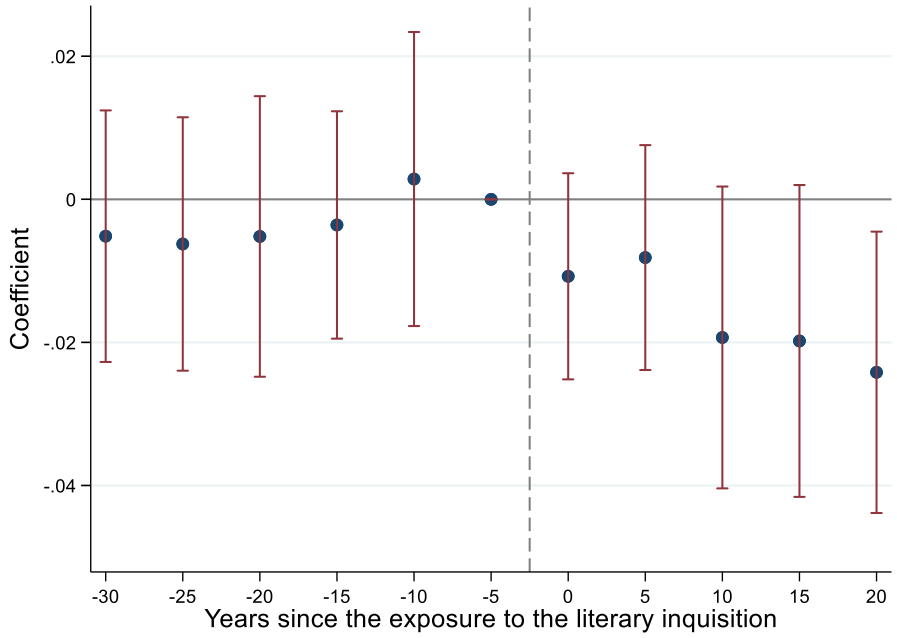
Author　　*Title of the innovation (publication on mathematics)*

Figure A2: The Dynamic Effects of the Literary Inquisition on Scientific and Technological Innovation (Yearly Panel)

(A) Full Sample



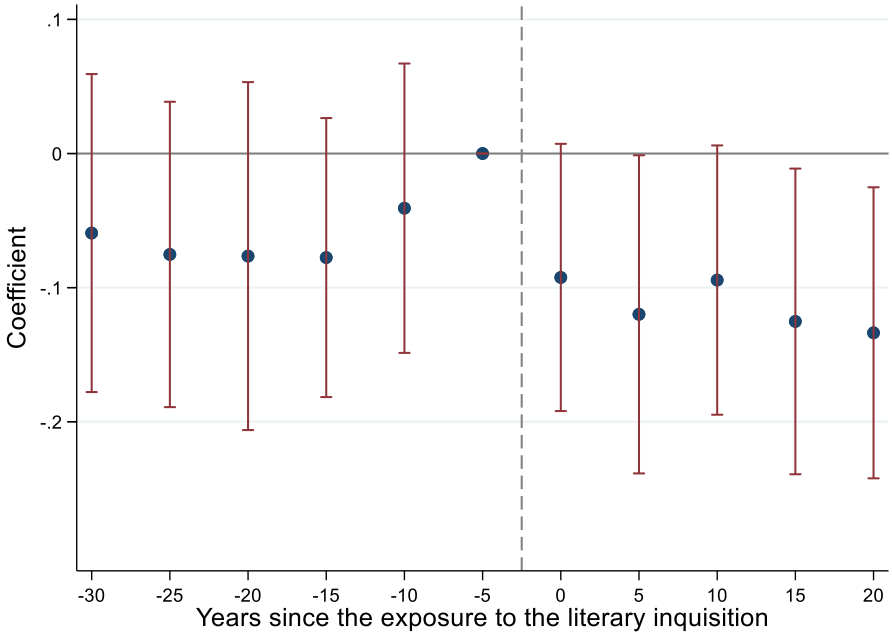
(B) Matched Sample



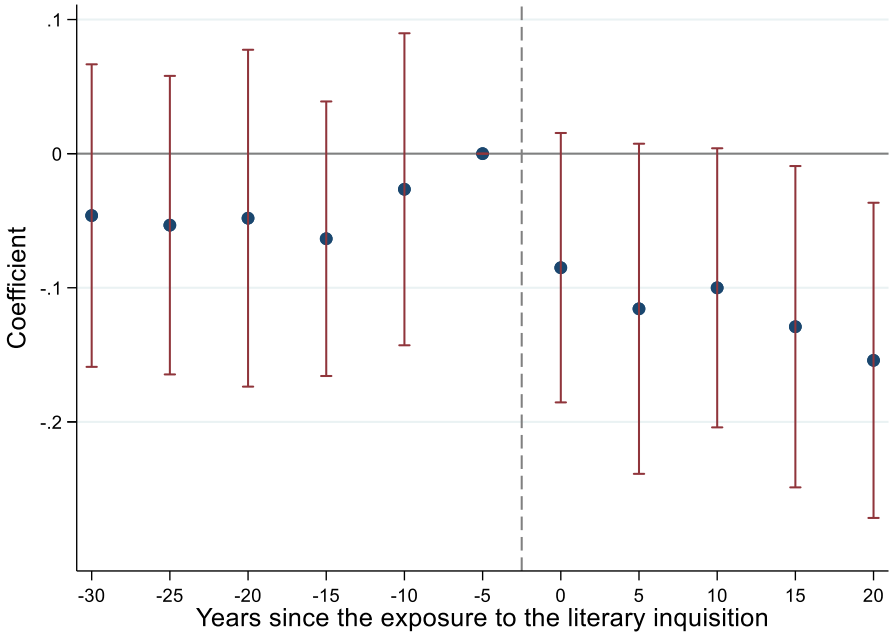
Notes: Panels (A) and (B) visualize the estimates of the dynamic effects of the literary inquisition on innovation for the full and matched samples, respectively. We obtain the OLS estimates from the dynamic version of the TWFE model (Equation (2)) with the yearly panel, using the groups just before the literary inquisition (i.e., the groups five years prior to the literary inquisition) as the reference. Control variables include all baseline controls in Column (4) of Table 2 in the main text. The bars represent 95% confidence intervals. Standard errors are clustered at the prefectural level.

Figure A3: The Dynamic Effects of the Literary Inquisition on Scientific and Technological Innovation (the Restricted Sample)

(A) Full Sample

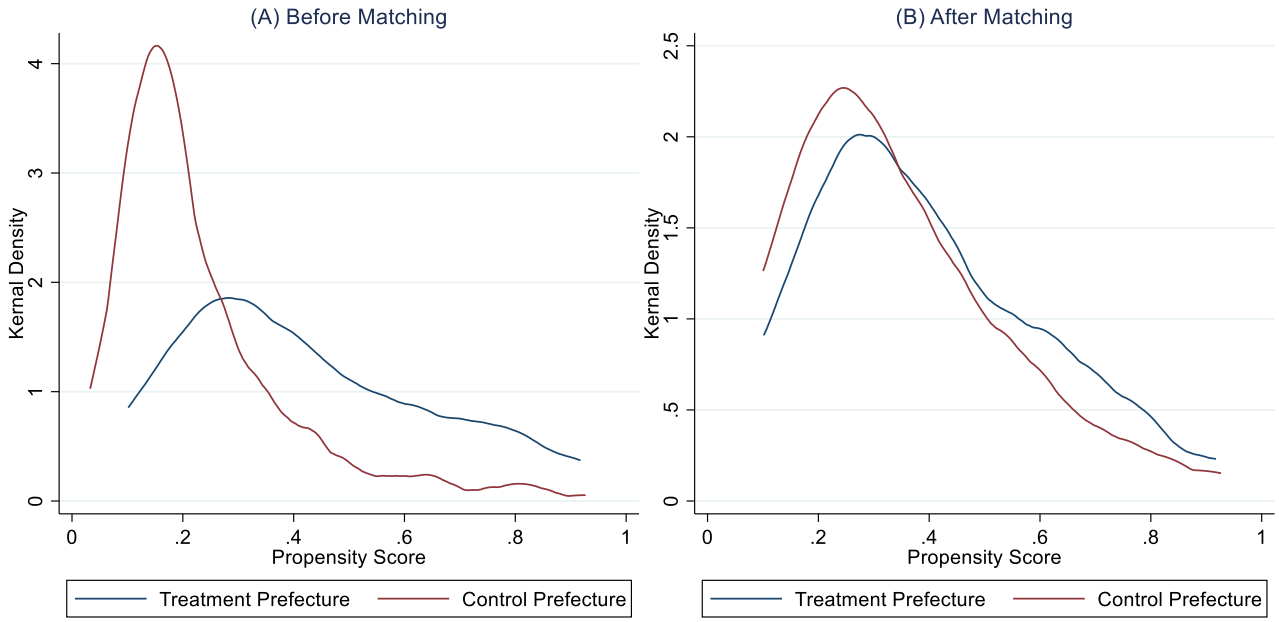


(B) Matched Sample



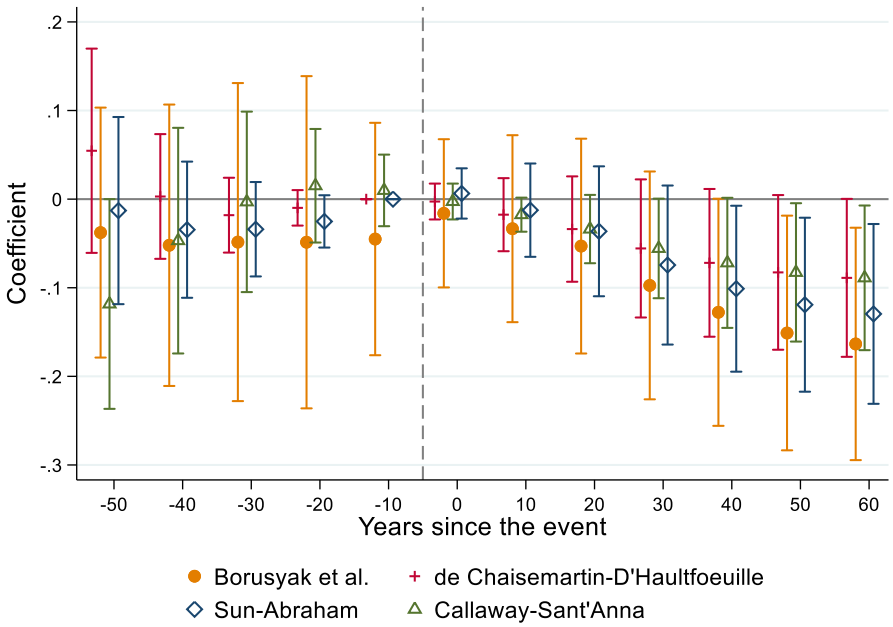
Notes: Panels (A) and (B) present the estimates of the dynamic effects of the literary inquisition on innovation for the full and matched samples, respectively. We obtain these OLS estimates from the dynamic version of the TWFE model (Equation (2)) with the restricted sample of literary inquisition cases where the suspects were found guilty and punished, using the groups just before the literary inquisition (i.e., the groups five years prior to the literary inquisition) as the reference. Control variables include all baseline controls in Column (4) of Table 2. The bars represent 95% confidence intervals. Standard errors are clustered at the prefectural level.

Figure A4: Propensity Score Matching: Distributions of the Propensity Scores Before and After Matching



Notes: Panels (A) and (B) show the propensity score distributions for the treatment and control prefectures, before and after matching, respectively.

Figure A5: The Dynamic Effects of the Literary Inquisition on Land Tax Revenue: Staggered Treatment Correction



Notes: This figure presents the estimates of the dynamic effects of the literary inquisition on land tax revenue, employing four alternative estimators. These include: Callaway and Sant'Anna (2021) (in green with triangle markers); Sun and Abraham (2021) (in blue with diamond markers); De Chaisemartin and d'Haultfoeuille (2020) (in red with cross markers); and Borusyak, Jaravel, and Spiess (2021) (in orange with circle markers). The outcome variable is the logarithm of land tax revenue and control variables include all the baseline controls in Column (4) of Table 5 in the main text. The groups 10 years prior to the literary inquisition are used as the reference. The bars represent 95 percent confidence intervals. Standard errors are clustered at the prefectural level.

Table A1: Compositions of Scientific and Technological Innovations in Europe and China in 1650–1790 (By Field)

	1650–1790		1651–1710		1711–1790	
	Europe	China	Europe	China	Europe	China
	Panel A: Scientific and Technological Innovations					
Total Number	349	153	119	90	230	63
Composition						
Science	58.70%	74.50%	69.70%	78.90%	53.00%	68.20%
Technology	41.30%	25.50%	30.30%	21.10%	47.00%	31.80%
	Panel B: Scientific Innovations					
Total Number	205	114	83	71	122	43
Composition						
Physics	31.20%	4.40%	32.50%	2.80%	30.30%	7.00%
Biology	22.40%	7.90%	18.10%	8.50%	25.40%	7.00%
Medicine	13.20%	24.60%	14.50%	15.50%	12.30%	39.50%
Mathematics	12.20%	22.80%	21.70%	26.80%	5.70%	16.30%
Chemistry	10.70%	0.90%	6.00%	0.00%	13.90%	2.30%
Astronomy	2.40%	5.30%	2.40%	26.80%	2.50%	9.30%
Geography	2.40%	13.20%	1.20%	12.70%	3.30%	14.00%
Geology	2.40%	20.20%	2.40%	5.60%	2.50%	4.70%
Others	2.90%	0.90%	1.20%	1.40%	4.10%	0.00%
	Panel C: Technological Innovations					
Total Number	144	39	36	19	108	20
Composition						
Mechanical engineering	48.6%	2.6%	66.7%	5.3%	42.6%	0.0%
Chemical engineering	9.7%	0.0%	0.0%	0.0%	13.0%	0.0%
Civil engineering	9.7%	17.9%	13.9%	21.1%	8.3%	15.0%
Instrument manufacturing	6.9%	7.7%	2.8%	15.8%	8.3%	0.0%
Ironmaking technology	6.3%	2.6%	2.8%	5.3%	7.4%	0.0%
Materials engineering	5.6%	10.3%	11.1%	0.0%	3.7%	20.0%
Military technology	2.8%	2.6%	2.8%	5.3%	2.8%	0.0%
Agricultural technology	2.8%	48.7%	0.0%	36.8%	3.7%	60.0%
Others	7.6%	7.7%	0.0%	10.5%	10.2%	5.0%

Table A2: Examples of Scientific and Technological Innovations in Europe and China in 1650–1790 (By Field)

	China	Europe
	Panel A: Science	
Mathematics	In 1672, Mei Wending presented the general solutions for a diverse range of linear equations in his work "*Fangcheng lun (Theory of Equations)*".	In 1686, Gottfried Wilhelm Leibniz proposed an influential integration algorithm in his paper "*New Method for the Greatest and the Least*".
Physics	In 1695, Liu Xianting documented the phenomenon of magnetic shielding in his notable work titled "*Guangyang zaji* (*The Miscellaneous Records of Guangyang)*".	In 1660, Robert Boyle proposed Boyle's Law in his work "*New Experiments Physico-Mechanicall, Touching the Spring of the Air and Its Effects*".
Astronomy	In 1673, Wang Xihuan proposed a model of the universe in his work "*Wuxing xingdu jie (Explanation of the Movements of the Five Planets)*".	In 1655, Christiaan Huygens discovered Saturn's moons and rings.
Chemistry	In 1765, Zhao Xuemin recorded organic acids, bases, and ammonia in his book "*Bencao gangmu shiyi (Supplement to the Compendium of Materia Medica)*".	In 1661, Robert Boyle proposed that the aim of chemical research is to know the nature of objects in his influential work "*The Sceptical Chymist*".
Medical science	In 1665, Qi Kun published the book "*Waike dacheng (Compendium of Surgery)*", which extensively discusses surgical conditions and their respective treatments.	In 1664, Thomas Willis published "*Cerebri Anatome*", a pioneering work on the nervous system that coined the term "neurology".
Biology	In 1683, Chen Dingguo published the book "*Li pu (Lychee Chart)*", which extensively studies lychee varieties.	In 1682, John Ray published the book "*Methodus Plantarum Nova*", which classifies plants into monocotyledons, dicotyledons, and other categories.
Geography	In 1662, Gu Yanwu compiled the book "*Zhao yu zhi (National Geography General)*", which covers various subjects such as settlement, geography, and landmarks.	In 1686, Edmund Halley studied the cause of trade winds and monsoons.
Geology	In 1664, Sun Tingquan published the book "*Yanshan zaji (Miscellaneous Records of Yanshan)*", which documents how to use rock formation to identify ore deposits.	In 1669, Nicolaus Steno postulated that various rocks are formed through the process of sedimentation in his work "*Preliminary Discourse to a Dissertation on a Solid Body Naturally Contained within a Solid*".
Others	In 1666, Xiong Bolong presented a critical analysis of traditional religious superstitions in his work titled "*Wuhe ji*", proposing that nature consists of entities without a conscious will.	In 1666, Robert Boyle discussed the mechanical universe theory in his work "*Origin of Forms and Qualities according to the Corpuscular Philosophy* ".

(*Continues*)

Table A2—*Continued*

	Panel B: Technology	
Mechanical engineering	In 1660, craftsmen in Nanjing and Shanghai began to imitate the striking clock.	In 1663, Isaac Newton designed a four-wheeled vehicle that utilized the propulsion of steam recoil for forward motion.
Civil engineering	In 1689, Jin Fu completed the book "*Zhihe fanglue (Flood Control Strategies)*", in which he summarized his experiences in flood control.	In 1679, V.Viviani implemented a flood control scheme for the Arno River in Florence.
Agricultural technology	In 1658, Zhang Lvxiang published the book "*Bu nong shu (Supplement to the Agricultural Treatise)*", which documents the agricultural technologies used in the Taihu Lake area.	In 1731, Jethro Tull designed a seed drill that can sow three ridges at once.
Materials engineering	In 1739, Tang Ying's work "*Tao ren xin yu (Poetry of the Ceramic Artisan)*" comprehensively recorded the technologies employed in the production of porcelain in Jingdezhen.	In 1662, T. Tilman made a crystalline lead glass.
Ironmaking technology	In 1696, a new technology was introduced to improve the size and construction of the iron smelting furnace in an ironworks in Guangdong Province.	In 1665, Dard Dudley published the book "*Metallum Martis*", which studied the use of coal in iron making.
Instrument manufacturing	In 1662, Sun Yunqiu made more than 70 kinds of optical instrument, including telescopes and magnifiers.	In 1714, Daniel Gabriel Fahrenheit invented mercury thermometer.
Chemical engineering	None	In 1736, Joshua Ward made sulfuric acid with sulfur and saltpeter for commercial manufacture.
Military technology	In 1687, Dai Zi invented a weapon "c*hongtian pao*" (which is similar to the mortar) for the Qing government.	In 1676, the French army used long grenades when attacking Algeria.
Others	In 1718, Xu Zhiding invented the "porcelain type", a typeface made from clay glazed and calcined for typesetting and printing.	In 1783, the Montgolfier brothers carried out a hot air balloon experiment, in which the hot air balloon carried the animals to a height of 500 meters and landed safely.

Table A3: Data Source of Land Tax Revenue

Region	Year Source	Author
Panel A: Provincial General Records		
Prefectures in Shannxi	1667 *Kangxi Shannxi Tongzhi*	Jia Hanfu et.al
Prefectures in Sichuan	1673 *Kangxi Sichuan Tongzhi*	Cai Yurong et.al
Prefectures in Shandong	1676 *Kangxi Shandong Tongzhi*	Zhao Xiangxing et.al
Prefectures in Zhejiang	1681 *Kangxi Zhejiang Tongzhi*	Zhao Shilin et.al
Prefectures in Shanxi	1682 *Kangxi Shanxi Tongzhi*	Ku Erkang et.al
Prefectures in Jiangxi	1682 *Kangxi Jiangxi Tongzhi*	Yu Chenglong et.al
Prefectures in Guangxi	1683 *Kangxi Guangxi Tongzhi*	Hao Yu et.al
Prefectures in Jiangsu and Anhiu	1683 *Kangxi Jiangnan Tongzhi*	Yu Chenglong et.al
Prefectures in Hubei and Hunan	1684 *Kangxi Huguang Tongzhi*	Xu Guoxiang et.al
Prefectures in Zhili	1685 *Yongzheng Jifu Tongzhi*	Li Wei et.al
Prefectures in Yunnan	1686 *Kangxi Yunnan Tongzhi*	Fan Chengxun et.al
Prefectures in Henan	1695 *Kangxi Henan Tongzhi*	Gu Qian et.al
Prefectures in Guizhou	1697 *Kangxi Guizhou Tongzhi*	Wei Jiqi et.al
Prefectures in Zhili	1724 *Yongzheng Jifu Tongzhi*	Li Wei et.al
Prefectures in Jiangxi	1724 *Yongzheng Jiangxi Tongzhi*	Xie Min et.al
Prefectures in Shannxi	1727 *Yongzheng Shannxi Tongzhi*	Liu Yuyi et.al
Prefectures in Sichuan	1728 *Yongzheng Sichuan Tongzhi*	Huang Tinggui
Prefectures in Henan	1728 *Yongzheng Henan Tongzhi*	Tian Wenjing et.al
Prefectures in Guangdong	1730 *Yongzheng Guangdong Tongzhi*	Hao Yulin et.al
Prefectures in Shanxi	1731 *Yongzheng Shanxi Tongzhi*	Jueluo Shilin et.al
Prefectures in Hubei and Hunan	1732 *Yongzheng Huguang Tongzhi*	Mai Zhu et.al
Prefectures in Yunnan	1732 *Qianlong Yunnan Tongzhi*	E Ertai et.al
Prefectures in Gaungxi	1733 *Yongzheng Guangxi Tongzhi*	Jin Gong et.al
Prefectures in Zhejiang	1733 *Yongzheng Zhejiang Tongzhi*	Li Wei et.al
Prefectures in Fujian	1734 *Qianlong Fujian Tongzhi*	Hao Yulin et.al
Prefectures in Shandong	1735 *Yongzheng Shandong Tongzhi*	Yue Jun et.al
Prefectures in Jiangsu and Anhui	1735 *Qianlong Jiangnan Tongzhi*	Zhao Hongen et.al
Prefectures in Gansu	1736 *Qianlong Gansu Tongzhi*	Xu Rong et.al
Prefectures in Guizhou	1739 *Qianlong Guizhou Tongzhi*	E Ertai et.al
Panel B: National General Records		
All prefectures	1743 *Kangxi Daqing Yitongzhi*	Xu Qianxue et.al
All prefectures	1784 *Qianlong Daqing Yitongzhi*	He Shen et.al
All prefectures	1820 *Jiaqing Chongxiu Yitongzhi*	Pan Xien et.al
Panel C: Land Tax Revenue at the Provincial Level		
All provinces	1841 *Shiqu Yuji*	Wang Qingyun
All provinces	1842 *Shiqu Yuji*	Wang Qingyun
All provinces	1845 *Shiqu Yuji*	Wang Qingyun
All provinces	1849 *Shiqu Yuji*	Wang Qingyun

Table A4: The Effect of the Literary Inquisition on Innovation (with Alternative Data Sources on the Literary Inquisition)

	Dependent variable: # Innovations			
	Panel A: Using Xue's Data		Panel B: Using Combined Sample	
	Full Sample	Matched Sample	Full Sample	Matched Sample
	(1)	(2)	(3)	(4)
Literary Inquisition	–0.0356**	–0.0455**	–0.0345***	–0.0481**
	(0.0148)	(0.0189)	(0.0132)	(0.0191)
Baseline Controls	Yes	Yes	Yes	Yes
Y mean	0.0187	0.0288	0.0187	0.0288
Observations	4960	2220	4960	2220
R-squared	0.157	0.236	0.157	0.236

Notes: This table presents the robustness of our DID estimates of the effect of the literary inquisition on the number of innovations with alternative data sources on the literary inquisition. Specifically, we estimate Equation (1) in the main text, using the data of Xue (2021) and a combined sample derived from both Xue's dataset and our own source on the literary inquisition. The results are reported in Panels A and B, respectively. Baseline controls are the same as those in Column 4 of Table 2 in the main text. Standard errors in parentheses are clustered at the prefectural level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: The Effect of the Literary Inquisition on Innovation (Using the Yearly Panel)

	Dependent Variable: # Innovations			
	(1)	(2)	(3)	(4)
	Panel A: Full Sample			
Literary Inquisition	–0.0088***	–0.0083***	–0.0085***	–0.0087***
	(0.0032)	(0.0029)	(0.0029)	(0.0029)
Y mean	0.0037	0.0037	0.0037	0.0037
Observations	24800	24800	24800	24800
R-squared	0.022	0.027	0.028	0.033
	Panel B: Matched Sample			
Literary Inquisition	–0.0103***	–0.0110***	–0.0115***	–0.0120***
	(0.0038)	(0.0039)	(0.0039)	(0.0040)
Y mean	0.0058	0.0058	0.0058	0.0058
Observations	11100	11100	11100	11100
R-squared	0.030	0.039	0.042	0.051
Prefecture FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Province FE × Time FE	Yes	Yes	Yes	Yes
Treat × After1790	Yes	Yes	Yes	Yes
Jinshi ×Time FE	No	Yes	Yes	Yes
Pop density1630 ×Time FE	No	Yes	Yes	Yes
Rice ×Time FE	No	No	Yes	Yes
Foxtail millet ×Time FE	No	No	Yes	Yes
Sweet potato ×Time FE	No	No	Yes	Yes
Language polarization ×Time FE	No	No	No	Yes
Language fragmentation ×Time FE	No	No	No	Yes
Coastal ×Time FE	No	No	No	Yes
Main river ×Time FE	No	No	No	Yes

Notes: This table examines the effect of the literary inquisition on the number of innovations in treatment prefectures using the yearly panel. Specifically, it presents estimates of the coefficient β from Equation (1) in the main text. Panels (A) and (B) present the results for the full and matched samples of prefectures, respectively. In Column (1), we estimate Equation (1) without including controls (only control the prefecture fixed effects, time fixed effects, the interaction of province and five-year fixed effects, and the interaction term of the treatment dummy and post-1790 dummy). In Columns (2)–(4), we gradually include a series of interaction terms of the five-year fixed effects and the following prefectural characteristics in the regressions: *Jinshi* (number of *jinshi* in 1600–1710), Population density (the population density in 1630), agricultural suitability metrics for wetland rice, fox millet, and sweet potato, Language polarization (the language polarization index), Language fragmentation (the language fragmentation index), and two geographic dummies: Coast (whether a prefecture is situated on the coast), Main river (whether a prefecture includes at least one major river). Standard errors in parentheses are clustered at the prefectural level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Robustness Checks: Controlling for the Potential Impacts of Other Historical Events

	Dependent variable: # Innovations								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Literary Inquisition	–0.0428***	–0.0449***	–0.0424***	–0.0443***	–0.0445***	–0.0446***	–0.0445***	–0.0428***	–0.0435***
	(0.0155)	(0.0156)	(0.0153)	(0.0156)	(0.0156)	(0.0157)	(0.0153)	(0.0153)	(0.0156)
Jesuit scientist number × Expulsion 1721–1780	Yes								
Jesuit scientist presence × Expulsion 1721–1780		Yes							
Jesuit priest number × Expulsion 1721–1780			Yes						
Jesuit priest presence× Expulsion 1721–1780				Yes					
Coast I ×Sea Ban 1717-1727					Yes				
Coast II ×Sea Ban 1757-1800					Yes				
Tandingrumu						Yes			
Quota × Reform 1724							Yes		
Quota × *keju* 1756								Yes	
Quota × *keju* 1787									Yes
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Y mean	0.0187	0.0187	0.0187	0.0187	0.0187	0.0187	0.0187	0.0187	0.0187
Observations	4960	4960	4960	4960	4960	4960	4960	4960	4960
R-squared	0.160	0.158	0.162	0.158	0.159	0.158	0.158	0.158	0.159

Notes: This table presents the DID estimates of the effect of the literary inquisition on the number of innovations in treatment prefectures after controlling for the potential impacts of other historical events at the time. Specifically, it presents estimates of the coefficient β from Equation (1) with more control variables. We control for the interaction terms between the variables of Jesuit scientists and a dummy of the period of Jesuit expulsion (1721–1780) to control for the potential effect of the Jesuit expulsion on innovation. Moreover, we control for the interaction terms between the dummies of the two periods of the sea ban polices (1717–1727 and 1757–1800) and dummies of the corresponding affected regions in the two periods to absorb the potential effect of the sea ban policies. Jesuit scientist (priest) presence is a dummy that equals 1 if a prefecture had Jesuit scientists (priests) between 1581 and 1720. Jesuit scientist (priest) number is the aggregation of all the Jesuit scientists (priests) for the decades of their presence in each prefecture between 1581 and 1720. Coast I and Coast II are two dummies that indicate the affected regions in the two periods of the sea ban polices (1717–1727 and 1757–1800, respectively). We also control the potential impacts of the *Tandingrumu* policy and changes to the civil exam system in Columns 6–9. Baseline controls are the same as those in Column 4 of Table 2 in the main text. Standard errors in parentheses are clustered at the prefectural level * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: The Implementation Time of the *Tandingrumu* Policy across Provinces

Time of implementation	Province
1716	Guangdong
1720	Sichuan
1724	Zhili, Fujian
1725	Shandong
1726	Yunnan, Shaanxi, Zhejiang, Gansu, Jiangsu, Anhui, Jiangxi, Hunan, Guangxi, Hubei
1731	Shanxi
1777	Guizhou

Notes: The data are sourced from Wang (2020).

Table A8: The Effect of the Literary Inquisition on Innovation (The Restricted Sample)

	Dependent Variable: # Innovations			
	(1)	(2)	(3)	(4)
	Panel A: Full Sample			
Literary Inquisition	–0.0492***	–0.0480***	–0.0500***	–0.0524***
	(0.0167)	(0.0151)	(0.0153)	(0.0157)
Y mean	0.0187	0.0187	0.0187	0.0187
Observations	4960	4960	4960	4960
R-squared	0.104	0.128	0.134	0.159
	Panel B: Matched Sample			
Literary Inquisition	–0.0560***	–0.0630***	–0.0686***	–0.0708***
	(0.0193)	(0.0218)	(0.0228)	(0.0244)
Y mean	0.0288	0.0288	0.0288	0.0288
Observations	2220	2220	2220	2220
R-squared	0.141	0.179	0.194	0.238
Prefecture FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Province FE ×Time FE	Yes	Yes	Yes	Yes
Treat × After1790	Yes	Yes	Yes	Yes
Jinshi ×Time FE	No	Yes	Yes	Yes
Pop density1630 ×Time FE	No	Yes	Yes	Yes
Rice ×Time FE	No	No	Yes	Yes
Foxtail millet ×Time FE	No	No	Yes	Yes
Sweet potato ×Time FE	No	No	Yes	Yes
Language polarization ×Time FE	No	No	No	Yes
Language fragmentation ×Time FE	No	No	No	Yes
Coastal ×Time FE	No	No	No	Yes
Main river ×Time FE	No	No	No	Yes

Notes: This table examines the effect of the literary inquisition on the number of innovations in treatment prefectures using the restricted sample of the literary inquisition cases where the suspects were found guilty and punished. Specifically, it presents estimates of the coefficient β from Equation (1) in the main text. In Column (1), we estimate Equation (1) without including controls (only control the prefecture fixed effects, time fixed effects, the interaction of province and time fixed effects, and the interaction term of the treatment dummy and the post-1790 dummy). In Columns (2)–(4), we gradually include a series of interaction terms of the five-year fixed effects and the following prefectural characteristics in the regressions: *Jinshi* (number of *jinshi* in 1600–1710), Population density (the population density in 1630), agricultural suitability metrics for wetland rice, fox millet, and sweet potato, Language polarization (the language polarization index), Language fragmentation (the language fragmentation index), and two geographic dummies: Coast (whether a prefecture is situated on the coast), Main river (whether a prefecture includes at least one major river). Standard errors in parentheses are clustered at the prefectural level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Balance of the Sample: Matching Covariates

	Panel A: Before Matching				
	Treatment Prefectures		Control Prefectures		
Variables	Obs.	Mean	Obs.	Mean	Diff in Mean
Coast	71	0.225	177	0.102	0.124**
Major river	71	0.634	177	0.621	0.012
Jinshi	71	82.986	177	26.424	56.562***
Population density1630	71	142.672	177	59.465	83.206***
Rice suitability	71	0.183	177	0.154	0.030
Foxtail millet suitability	71	0.097	177	0.083	0.014
Sweet potato suitability	71	2.391	177	1.855	0.536***
Language polarization index	71	2.980	177	2.884	0.096
Language fragmentation index	71	2.440	177	2.664	–0.225
	Panel B: After Matching				
	Treatment Prefectures		Control Prefectures		
Variables	Obs.	Mean	Obs.	Mean	Diff in Mean
Coast	67	0.194	44	0.205	–0.011
Major river	67	0.627	44	0.523	0.104
Jinshi	67	74.925	44	48.250	26.675
Population density1630	67	131.970	44	107.867	24.103
Rice suitability	67	0.194	44	0.178	0.016
Foxtail millet suitability	67	0.103	44	0.090	0.013
Sweet potato suitability	67	2.362	44	2.377	–0.015
Language polarization index	67	2.971	44	3.137	–0.166
Language fragmentation index	67	2.471	44	2.316	0.155

Notes: This table presents the differences between prefectures that were exposed to the literary inquisition and those that were not regarding their pre-treatment covariates. The covariates include two geographic dummies, namely, Coastal (whether a prefecture is situated on the coast), Main river (whether a prefecture includes at least one major river); and other prefectural characteristics: *Jinshi* (number of *jinshi* in 1600–1710), Population density (population density in 1630), agricultural suitability metrics for wetland rice, fox millet, and sweet potato, Language polarization (the language polarization index), Language fragmentation (the language fragmentation index).

Table A10: Robustness Checks: Spillover Effect of the Literary Inquisition on Innovation

	Dependent variable: # innovations			
	Full Sample		Matched Sample	
	Sub I	Sub II	Sub I	Sub II
	(1)	(2)	(3)	(4)
	Panel A: 30km			
Literary Inquisition	–0.0441***	–0.0571**	–0.0649***	–0.0865**
	(0.0165)	(0.0252)	(0.0243)	(0.0380)
Baseline Controls	Yes	Yes	Yes	Yes
p-value		0.339		0.220
Y mean	0.0239	0.0273	0.0314	0.0388
Observations	3520	2860	2040	1520
R-squared	0.178	0.269	0.243	0.356
	Panel B: 90km			
Literary Inquisition	–0.0434***	–0.0601**	–0.0641***	–0.0889**
	(0.0162)	(0.0273)	(0.0240)	(0.0396)
Baseline Controls	Yes	Yes	Yes	Yes
p-value		0.256		0.177
Y mean	0.0219	0.0307	0.0302	0.0410
Observations	3840	2540	2120	1440
R-squared	0.173	0.284	0.239	0.366

Notes: This table examines the robustness of the estimates of the spillover effect of the literary inquisition on innovation for the full and matched samples, respectively. Specifically, it presents estimates of the coefficient β from Equation (1) for the two sub-samples constructed from the full and matched samples. We construct two sub-samples to perform the DID analysis to investigate the potential spillover effect. Sub-sample I includes all treatment prefectures and closely located control prefectures within 30 km (Panel A) or 90 km (Panel B) from the treatment prefectures' borders. Sub-sample II includes all treatment prefectures and distant control prefectures over 30 km (Panel A) or 90 km (Panel B) from the the treatment prefectures' borders. Baseline controls are the same as those in Column 4 of Table 2 in the main text. Standard errors in parentheses are clustered at the prefectural level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Further Estimates of the Spillover Effect of the Literary Inquisition on Innovation

	Dependent variable: # innovations		
	0 km	800 km	1000 km
	Sub III	Sub IV	Sub V
	(1)	(2)	(3)
Literary Inquisition	–0.0444***	–0.0810**	–0.0818**
	(0.0167)	(0.0367)	(0.0368)
Baseline Controls	Yes	Yes	Yes
p-value		0.0643	0.0604
Y mean	0.0243	0.0466	0.0486
Observations	3460	1480	1420
R-squared	0.179	0.329	0.333

Notes: This table presents further estimates of the spillover effect of the literary inquisition on innovation for the full sample. Specifically, it presents estimates of the coefficient β from Equation (1) for the three sub-samples. Specifically, Sub-sample III includes all treatment prefectures and their adjacent control prefectures; Sub-sample IV comprises all treatment prefectures and distant (beyond 800 km) control prefectures; and Sub-sample V comprises all treatment prefectures and more distant (beyond 1000 km) control prefectures. Baseline controls are the same as those in Column 4 of Table 2 in the main text. Standard errors in parentheses are clustered at the prefectural level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Alternative DID Estimators of the Effect of the Literary Inquisition on Innovation

	Point Estimate	Standard Error	*p*-value	Lower Bound 95% Confidence Interval	Upper Bound 95% Confidence Interval
Panel A: Full Sample					
Borusyak-Jaravel-Spiess	–0.0377	0.0141	0.0074	–0.0653	–0.0101
De Chaisemartin-D'Haultfeuille	–0.0670	0.0432	0.1208	–0.1517	0.0177
Callaway-Sant'Anna	–0.0539	0.0476	0.2576	–0.1473	0.0394
Sun-Abraham	–0.0920	0.0133	0.0000	–0.1182	–0.0659
Panel B: Matched Sample					
Borusyak-Jaravel-Spiess	–0.0446	0.0168	0.0077	–0.0775	–0.0118
De Chaisemartin-D'Haultfeuille	–0.0515	0.0435	0.2367	–0.1367	0.0338
Callaway-Sant'Anna	–0.0576	0.0456	0.2061	–0.1470	0.0317
Sun-Abraham	–0.0534	0.0094	0.0000	–0.0719	–0.0350

Notes: This table presents the robustness of our baseline estimates in Tables 2 and 3 in the main text to use the alternative robust DID estimators introduced by Borusyak et al. (2021), Callaway and Sant'Anna (2021), De Chaisemartin and d'Haultfoeuille (2020), and Sun and Abraham (2021). We control the same variables in Column (4) of Table 2 in the main text in all regressions. See Borusyak et al. (2021), Callaway and Sant'Anna (2021), De Chaisemartin and d'Haultfoeuille (2020), and Sun and Abraham (2021) for the discussions about how they construct the estimators and why these estimators are robust to heterogenous treatment effects across treated units and over time. Standard errors are clustered at the prefectural level.

Table A13: The DID Estimates of the Effect of the Literary Inquisition on Land Tax Revenue in the Treatment Prefectures

	Dependent variable: ln (land tax revenue)			
	(1)	(2)	(3)	(4)
Literary Inquisition	–0.137**	–0.112**	–0.0890*	–0.0826
	(0.0545)	(0.0561)	(0.0534)	(0.0525)
Prefecture FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Province FE × Time FE	Yes	Yes	Yes	Yes
Tandingrumu	Yes	Yes	Yes	Yes
Land area × Time FE	Yes	Yes	Yes	Yes
Basin HHI × Time FE	Yes	Yes	Yes	Yes
Jinshi × Time FE	No	Yes	Yes	Yes
Pop density1630 × Time FE	No	Yes	Yes	Yes
Rice × Time FE	No	No	Yes	Yes
Fox millet × Time FE	No	No	Yes	Yes
Sweet potato × Time FE	No	No	Yes	Yes
Language polarization × Five FE	No	No	No	Yes
Language fragmentation × Five FE	No	No	No	Yes
Coastal × Time FE	No	No	No	Yes
Main river × Time FE	No	No	No	Yes
Observations	3078	3078	3078	3078
R-squared	0.571	0.580	0.596	0.602

Notes: This table presents the DID estimates of the effect of the literary inquisition on the land tax revenue in the treatment prefectures. Specifically, it presents estimates of the coefficient β from Equation (A2). Control variables are same to those in Table 5 in the main text. Standard errors in parentheses are clustered at the prefectural level.

Table A14: Alternative Difference-in-Differences Estimators (Land Tax Revenue)

	Point Estimate	Standard Error	*p*-value	Lower Bound 95% Confidence Interval	Upper Bound 95% Confidence Interval
Borusyak-Jaravel-Spiess	–0.0901	0.0543	0.0970	–0.1966	0.0163
De Chaisemartin-D'Haultfeuille	–0.0027	0.0103	0.7939	–0.0229	0.0176
Callaway-Sant'Anna	–0.0489	0.0283	0.0836	–0.1044	0.0065
Sun-Abraham	–0.0667	0.0344	0.0524	–0.1340	0.0007

Notes: This table presents robustness of our baseline estimates in Table A13 to use the alternative DID estimators introduced by Borusyak et al. (2021), Callaway and Sant'Anna (2021), De Chaisemartin and d'Haultfoeuille (2020), and Sun and Abraham (2021). We control the same variables in Column (4) of Table A13 in all regressions. Standard errors in parentheses are clustered at the prefectural level.